Agentic Agriculture: A Comprehensive Survey of AI Agents and Agentic AI in Precision Agriculture

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Abstract

This survey presents a first comprehensive analysis of the emerging paradigm of *Agentic Agriculture*, tracing the evolution from model-centric artificial intelligence to integrated, goal-driven agentic systems in precision agriculture. It begins with clearly defined research objectives and a structured literature synthesis methodology. The review covers foundational AI agents for core tasks, including sensing and monitoring, decision-support and advisory systems, and robotic field operations such as harvesting, spraying, and crop monitoring. Building on these, the survey examines multi-agent systems enabling distributed farm intelligence through IoT-agent architectures, swarm robotics, and multi-agent reinforcement learning for coordinated farm management and operational optimization. It then advances to Agentic AI, where autonomous agents extend these systems by integrating perception, reasoning, planning, and action into unified workflows. Applications include adaptive irrigation, crop monitoring, and resource allocation, supported by digital twins that evolve from static simulations to closed-loop, decision-driven optimization platforms. The survey further explores autonomous agricultural ecosystems with human–AI collaboration and coordinated robotics, including reinforcement learning–based irrigation and multi-robot harvesting. Finally, it identifies challenges in reliability, safety, data governance, ethics, and regulation. Overall, Agentic AI enables scalable, adaptive, and knowledge-driven agricultural systems, requiring robust integration of data-driven and process-based models for trustworthy deployment.

Keywords: AI Agents, Agentic AI, Precision Agriculture, Multi-Agent Systems, Autonomous Farming

1 Introduction

Precision and automated agriculture, originally conceptualized in the 1980s and 1990s Blackmore (1994); Srivastava et al. (1993), has evolved into one of the most transformative paradigms in modern farming Mgendi (2024); Karunathilake et al. (2023), driven by the convergence of sensing technologies, robotics, artificial intelligence (AI), and data-driven decision support systems Tantalaki et al. (2019); Nagireddy (2025); Jamal et al. (2025). Global agriculture faces unprecedented challenges including increasing demand for food, fiber an fuel, climate variability, labor shortages, and the need for sustainable management of scarce farming resources including water and nutrients, vos2019globalRahman et al. (2025); Vos and Bellù (2019). AI-enabled, precision and automated systems show promise to address thse challenges.

Traditional farming practices, which predominantly depend on manual observation Pulido and Bocco (2003); Misra and Ghosh (2024) , reactive management strategies, and uniform application of inputs are increasingly inadequate for addressing the growing complexity and challenges of modern agricultural industries Khatri et al. (2024); Sassenrath et al. (2008). Reactive management refers to decision-making approaches where actions are taken only after a problem has already occurred or

become visibly apparent Tantalaki et al. (2019); McCown (2002). For example, farmers may apply pesticides only after detecting visible signs of pest infestation Ye et al. (2025), irrigate fields after crops exhibit water stress symptoms Zhou et al. (2021), or add fertilizers once nutrient deficiencies manifest through leaf discoloration Polwaththa et al. (2024); Mikula et al. (2020); Laveglia et al. (2024). Similarly, disease control is often initiated only after infections have spread across plants, leading to delayed interventions and reduced effectiveness Roberts et al. (2021); Traversari et al. (2021).

Despite decades of research in precision agriculture focused on early detection and proactive management, real-world adoption remains constrained by challenges such as data heterogeneity, high deployment costs, limited scalability, and integration complexity, resulting in continued reliance on reactive practices that delay timely intervention under dynamic field conditions Juroszek and von Tiedemann (2011). In contrast, modern precision agriculture increasingly emphasizes predictive and data-driven strategies that enable early detection Aarif KO et al. (2025); Oyeboade and Olagoke-Komolafe (2023), timely intervention Arsenoaia et al. (2026), and continuous optimization of farm operations Akintuyi (2024); Prasanna et al. (2026). As a result, intelligent technologies capable of monitoring, analyzing, and autonomously responding to agricultural conditions are becoming central to the future of food production Aarif KO et al. (2025).

Over the past decade, AI has played a pivotal role in enabling data-driven agricultural systems Ugwu et al. (2025). Early applications focused on machine learning models for crop classification Ok et al. (2012); Agilandeeswari et al. (2022), yield prediction Nagini et al. (2016); Nyéki and Neményi (2022), pest and disease detection Yang (2020); Badage (2018), water stress Zhou et al. (2021) and soil analysis Malley et al. (2004). With the advancement of deep learning, computer vision models have enabled automated plant phenotyping, crop monitoring using UAV imagery, and robotic systems field operations such as harvesting and weeding Upadhyay et al. (2025); Qiao et al. (2022). These developments have significantly improved the efficiency and scalability of precision agriculture systems. However, most of these approaches remain largely model-centric, meaning they perform isolated prediction/actuation tasks without autonomous reasoning or coordinated decision-making.

More recently, the concept of *AI agents* has gained increasing attention across diverse application domains such as healthcare Huang (2025), finance Joshi (2025), transportation, and defense, as well as in enabling technologies including robotics and smart infrastructure Anjomshoae et al. (2019); Chan et al. (2025). AI agents are not a recent development but have been a foundational concept in artificial intelligence research since the 1990s, where they were formally defined as autonomous computational entities capable of perceiving their environment, reasoning about observed states, and executing actions to achieve specific objectives Franklin (1997). Early theoretical frameworks emphasized agents as systems situated within dynamic environments, capable of continuous interaction and goal-directed behavior, often modeled through perception–decision–action loops Castelfranchi (1998). These models laid the groundwork for understanding intelligence as an emergent property arising from continuous interaction between agents and their environments, rather than isolated computation. Subsequent formulations in classical AI literature further generalized agents as entities that map perceptual inputs to actions through rational decision-making processes, while increasingly incorporating learning and adaptation mechanisms that enable agents to improve performance over time based on experience and feedback, optimizing outcomes relative to predefined goals or utility functions Poole and Mackworth (2010).

Over time, the concept of AI agents has evolved to incorporate learning, adaptation, and probabilistic reasoning, enabling agents to operate under uncertainty and complex real-world conditions. Recent studies extend these principles by integrating modern machine learning and large-scale LLM data-driven models, enhancing the autonomy and scalability of agent-based systems across domains Maharramova et al. (2025). Thus, contemporary AI agents can be viewed as an evolution of long-established theoretical constructs, now empowered by advances in computation, data availability, and algorithmic advancement Hughes et al. (2025); Zuo (2025).

Beyond individual AI agents, the emergence of *Agentic AI* introduces a fundamentally new paradigm for autonomous intelligent systems, characterized by higher levels of autonomy, adaptability, and goal-directed behavior. Agentic AI refers to intelligent systems that are not only capable of perceiving and acting upon their environments but also of formulating objectives, reasoning over complex scenarios, planning multi-step actions, and dynamically utilizing external tools and/or knowledge sources to achieve desired outcomes Sapkota et al. (2025a); Abou Ali et al. (2025). Unlike traditional AI models that are typically designed for narrow, task-specific predictions, agentic systems operate as persistent entities that maintain contextual awareness, memory, and iterative decision-making

capabilities over extended time horizons. These systems integrate multiple components including large language models, planning modules, memory architectures, and tool interfaces into cohesive frameworks that enable autonomous orchestration of complex workflows Alva and Pandey (2026).

A defining characteristic of Agentic AI is its ability to perform sequential reasoning and adaptive planning, allowing it to decompose high-level goals into executable sub-tasks while continuously refining its strategy based on environmental feedback. This capability enables applications such as autonomous problem solving, dynamic resource allocation, and real-time system optimization across diverse domains and environments Bandi et al. (2025). Furthermore, agentic systems are inherently collaborative, often operating within multi-agent ecosystems where multiple specialized agents interact, coordinate, and share information to solve complex problems that exceed the capabilities of individual agents Jaggavarapu (2025). This collaborative intelligence is particularly relevant in domains such as agriculture, where distributed sensing, multi-robot coordination, and decision-making must occur simultaneously across spatially heterogeneous environments.

Recent advancements in Agentic AI also emphasize the importance of trust, risk management, and governance in Agentic AI systems, leading to the integration of frameworks such as Trust, Risk, and Security Management (TRiSM) to ensure reliable and responsible deployment and adoption Raza et al. (2026). Consequently, Agentic AI represents a shift from static, model-centric intelligence toward dynamic, system-level intelligence, where autonomous agents continuously perceive, reason, act, and learn in complex real-world environments Dwivedi et al. (2026); Yellanki (2025).

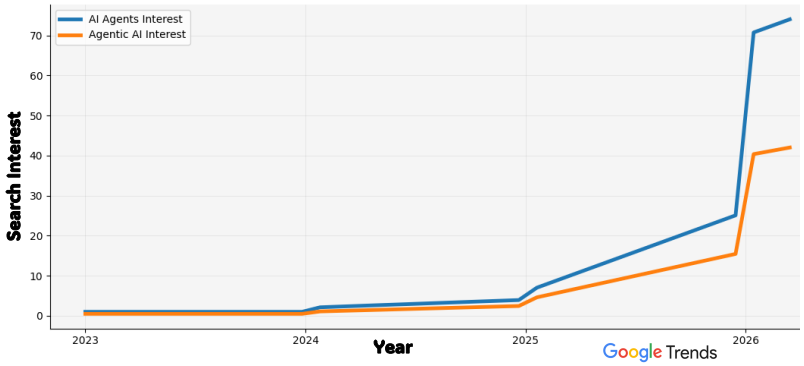


Fig. 1 Temporal evolution of normalized search interest for "AI Agents" and "Agentic AI" from post-ChatGPT emergence to March 23, 2026 (2:30 PM NY time), highlighting accelerating adoption and rising prominence of agentic paradigms in contemporary AI discourse.

1.1 Motivation

Despite these rapid interest in AI Agents and Agentic AI as shown in Figure 1, the literature on AI-driven precision agriculture remains fragmented across several domains including robotics, computer vision, machine learning, and farm management systems. While many surveys review deep learning in agriculture Kamilaris and Prenafeta-Boldú (2018); Santos et al. (2019); Ren et al. (2020); Albahar (2023); Altalak et al. (2022); Waqas et al. (2025); Qin et al. (2025) or agricultural robotics Prajapati et al. (2023); De Silva et al. (2024); Vougioukas (2019); Oliveira et al. (2021), to date, no comprehensive studies have systematically surveyed and reviewed the emergence and integration of AI agents and Agentic AI within precision agriculture. In particular, there is a lack of comprehensive synthesis that examines how autonomous agents, multi-agent collaboration, and agentic decision architectures are transforming agricultural production systems.

Our survey addresses this gap by providing a comprehensive assessment of AI agents and Agentic AI applications in precision agriculture. The paper reviews the evolution of intelligent agricultural systems from early sensing agents and decision-support systems to collaborative multi-agent platforms and fully autonomous farming architectures. The survey also highlights emerging research directions including agentic robotics, digital twin farms, and self-adaptive agricultural ecosystems.

1.2 Objective and Research Questions

The primary objective of this survey is to provide a comprehensive and structured synthesis of the emerging role of AI agents and Agentic AI in precision agriculture. While significant progress has been

made in agricultural AI, existing studies largely focus on isolated machine learning models or domain-specific applications, lacking a unified perspective on autonomous, goal-driven, and collaborative intelligence. This survey aims to bridge this gap by systematically analyzing the evolution from foundational AI agents to multi-agent systems and, more recently, to Agentic AI frameworks that enable autonomous decision-making and coordinated farm operations. Furthermore, the survey seeks to identify key technological trends, architectural paradigms, and open challenges that define the transition toward fully autonomous agricultural ecosystems.

To guide this analysis, we formulate four research questions (RQs) that structure the literature review and highlight critical dimensions of agentic intelligence in agriculture.

- **RQ1: How have AI agents been developed and applied in precision agriculture for foundational tasks?** This question investigates the evolution of individual AI agents in agricultural systems, focusing on foundational tasks including environmental sensing, state interpretation, decision support, and autonomous field execution. It examines sensing agents for perception, advisory agents for agronomic decision-making, and robotic agents for operations such as harvesting, spraying, and monitoring. The objective is to synthesize how early agent-based systems enabled data-driven and site-specific management in precision agriculture.
- **RQ2: How do multi-agent systems enable collaborative agricultural intelligence and distributed farm optimization?** This question examines the transition from single-agent systems to collaborative multi-agent frameworks. It focuses on distributed farm intelligence through IoT-agent architectures, swarm robotics and coordinated multi-robot systems, and multi-agent reinforcement learning approaches for various applications in farming such as optimizing irrigation, resource allocation, and crop management. The objective with this question, therefore, is to understand how coordination and communication among agents enhance scalability and efficiency in precision agriculture by supporting site-specific operations, optimizing resource allocation, and enabling adaptive management across spatially and temporally variable field conditions.
- **RQ3: How does Agentic AI enable autonomous agricultural decision-making and adaptive farm management?** This question explores the emergence of Agentic AI as a paradigm shift toward goal-oriented and reasoning-capable systems and their application in agriculture. It analyzes autonomous agricultural agents capable of planning and decision-making, agentic systems for crop health and resource management, and digital twin–based farm management platforms, among other applications. The objective with this question, therefore, is to characterize how agentic architectures integrate perception, reasoning, and action into unified autonomous workflows in agriculure.
- **RQ4: What are the future directions and challenges toward fully autonomous agentic agriculture systems?** This question addresses the long-term vision of autonomous farming ecosystems enabled by Agentic AI. It includes human–AI collaborative agriculture and agentic robotic operations as well as challenges related to safety, reliability, trust, ethics, and governance of Agentic AI as it applies to precision agriculture. With this synthesis, we will identify research gaps and define pathways for responsible and scalable deployment of agentic agricultural systems.

1.3 Literature Search and Synthesis

This survey adopts a structured methodology for literature search and synthesis, as illustrated in Fig. 2. The workflow is designed to systematically capture recent advancements (2020–2026) in AI agents and Agentic AI within precision agriculture, ensuring both breadth and relevance. As shown in Fig. 2, the process consists of three key stages: literature search and filtering, review design with research question formulation, and thematic synthesis of selected studies.

In Stage 1, a comprehensive keyword-driven search was conducted across major academic databases, including Scopus, Web of Science, IEEE Xplore, and ScienceDirect, using domain-specific and SEO-optimized terms such as “AI agents,” “Agentic AI,” and “multi-agent agriculture.” The initial pool of publications was refined through duplicate removal, relevance screening, and exclusion of non-agricultural or non-agentic studies. Stage 2 focuses on defining the scope of the review by formulating four core research questions and establishing inclusion criteria based on autonomy level, agent-based intelligence, collaboration capability, and empirical contribution. These criteria ensure that only technically relevant and methodologically rigorous studies are included.

Finally, Stage 3 organizes the selected literature into four major thematic categories, corresponding to the evolution from foundational AI agents to fully agentic agricultural ecosystems. As depicted

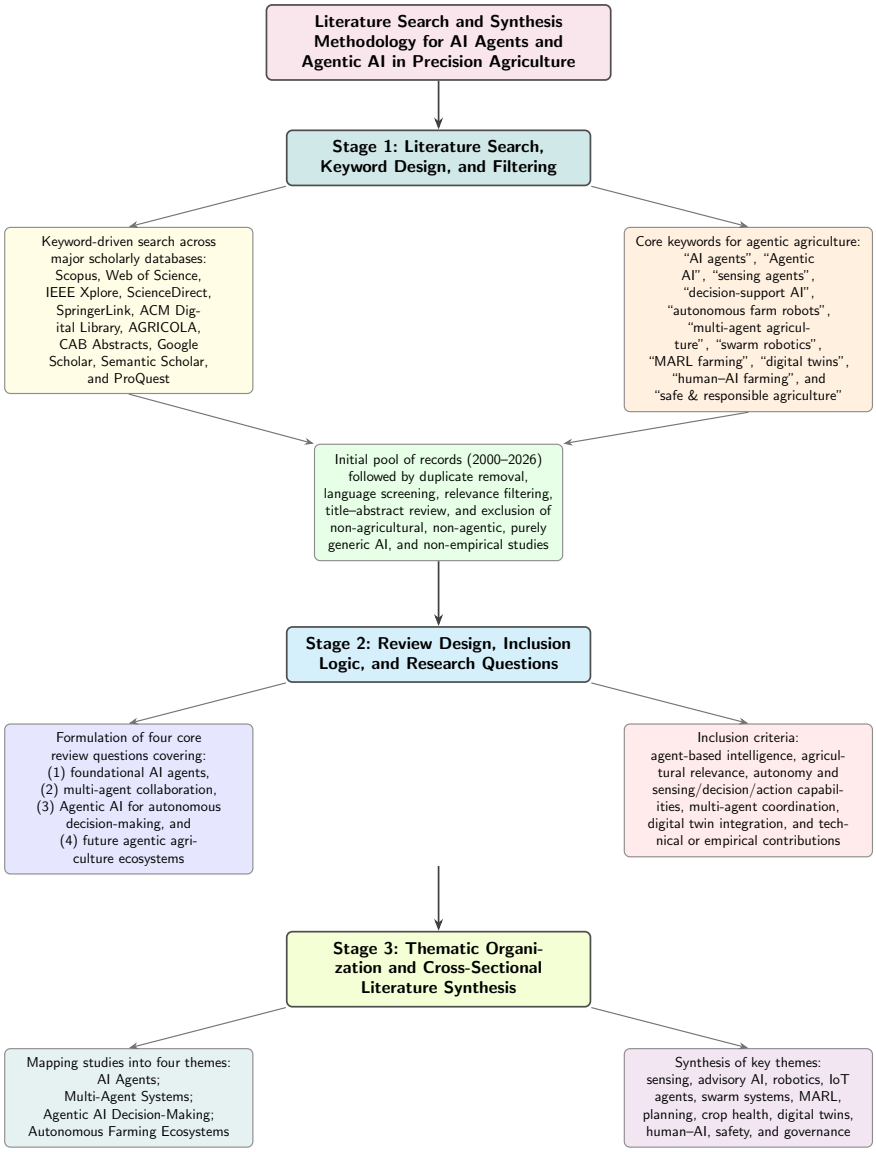


Fig. 2 Schematic representation of the systematic literature search, screening, and synthesis pipeline employed in this survey. The workflow progresses from literature retrieval and filtering to review design and inclusion logic, followed by thematic organization and cross-sectional synthesis for analyzing AI agents and Agentic AI in precision agriculture.

in Fig. 2, this stage also synthesizes cross-cutting themes such as sensing, decision support, robotics, multi-agent coordination, and digital twin integration into a coherent analytical framework.

2 AI Agents in Precision Agriculture: Foundations and Early Applications

AI agents are often perceived as a recent development associated with modern large-scale AI systems such as LLMs Akerkar (2026); however, as summarized in Fig. 3, their conceptual foundations and practical applications in agriculture date back several decades. Following early pre-2000 efforts centered on rule-based expert systems and agricultural simulation models Shaffer and Brodahl (1998), research in the 2000s and early 2010s advanced toward agent-based systems, distributed artificial intelligence, and decision-support frameworks, enabling more dynamic and decentralized solutions for complex challenges such as resource allocation, pest and nutrient management, crop-load optimization, and environmental monitoring Perini and Susi (2004); Ishaaya et al. (2001); Zhu et al. (2007). These systems leveraged the notion of intelligent agents to represent stakeholders, environmental processes, and operational units within agricultural systems, enabling decentralized decision-making and adaptive system behavior.

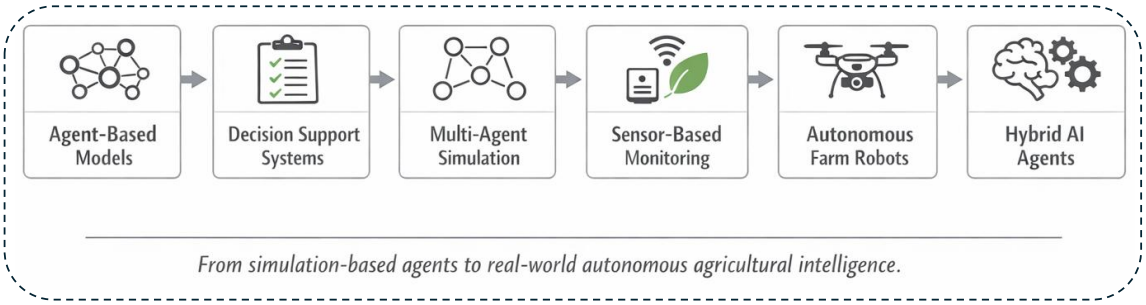


Fig. 3 Conceptual evolution of AI agents in precision agriculture, illustrating the sequential progression from foundational agent-based models to advanced multi-agent and hybrid intelligent systems.

As illustrated in Fig. 3, the evolution of AI agents in agriculture follows a clear progression from early agent-based models and decision-support systems toward more complex multi-agent simulations and sensor-driven monitoring frameworks. These early developments established the foundational paradigm of agents as computational entities capable of perception, reasoning, and action within dynamic and uncertain farming environments. Consequently, rather than being a recent trend, AI agents represent a mature and continuously evolving framework that has long contributed to the advancement of precision agriculture and laid the groundwork for modern agentic AI systems.

Initial applications of AI agents in agriculture primarily focused on decision-support systems and distributed information processing Pavlovic et al. (2008); Cortès et al. (2000). For instance, agent-based decision frameworks were developed for integrated agricultural production and policy analysis, where agents represented stakeholders, crops, or environmental processes Perini and Susi (2004); Berger (2001); Happe et al. (2006); Zhu et al. (2009). Similarly, multi-agent simulations were used to model agricultural ecosystems, enabling the study of resource use, farmer behavior, and policy impacts Bousquet and Le Page (2004); Parker et al. (2003); Barnaud et al. (2008). These approaches demonstrated the ability of agent-based systems to capture the complexity and heterogeneity of agricultural systems, which are inherently spatial, temporal, and socio-technical in nature.

With the advancement of sensing technologies and computational capabilities, AI agents were increasingly integrated into real-world agricultural systems during the early 2010s. Sensor-based agent systems were deployed to monitor farming environmental conditions, including soil moisture, and microclimate variables, enabling real-time data acquisition and decision-making FUKATSU et al. (2006). Concurrently, agent-based architectures were incorporated into robotic platforms, where agents functioned as decision-making modules controlling autonomous field robots and precision agriculture machinery Jepsen et al. (2006). These systems combined perception, control, and planning capabilities, marking a transition from purely simulation-based models to embodied intelligent agents operating in physical agricultural environments.

In addition to sensing and robotics, AI agents were also utilized in agricultural logistics, product quality control, and supply chain management Zhu et al. (2010). Agent-based frameworks were proposed for agricultural product traceability, enabling distributed monitoring and verification of product quality across different stages of the supply chain Ruiz-Garcia et al. (2010); Zhu et al. (2010). Similarly, web-based multi-agent systems facilitated knowledge sharing and communication among stakeholders in agricultural communities, highlighting the role of agents in enabling collaborative agricultural intelligence Jayarathna and Hettige (2013).

Furthermore, the integration of AI agents with machine learning and hybrid AI techniques expanded their capabilities beyond rule-based decision-making Wen (2007); Sarma et al. (2010). Early studies explored the combination of neural networks, expert systems, and agent-based architectures to enhance predictive accuracy and adaptability in agricultural applications Panigrahi and Ting (2012); Wen (2007). This hybridization allowed agents to operate under uncertainty and learn from data, paving the way for more advanced intelligent systems in precision agriculture.

Overall, the evolution of AI agents in agriculture reflects a clear and sustained progression from early theoretical models and simulation-based frameworks to practical, real-world systems integrating sensing, decision-making, and actuation capabilities, as evidenced by the studies summarized in Table 1. The wide range of applications reported in Table 1, spanning agricultural policy modeling, land-use simulation, sensor-based monitoring, and intelligent control systems, demonstrates that AI agents have been actively utilized in agricultural systems for more than two decades. These developments confirm that AI agents are not a recent trend, but rather a mature and continuously

evolving paradigm that has significantly contributed to the advancement of precision agricultural systems. Building upon these foundational systems, the recent emergence of Agentic AI extends this paradigm toward more advanced capabilities, including autonomous reasoning, goal-driven planning, and large-scale coordination in modern agricultural ecosystems.

Table 1 Summary of research studies demonstrating the long-standing use of AI agent-based systems in agriculture from 2001 to 2025. This table highlights how AI agents have been extensively applied across diverse agricultural domains, including policy modeling, land-use simulation, sensing systems, irrigation management, livestock systems, and supply chain optimization.

Author (Year)	Application Area	Agent-Based Contribution
Berger (2001) Berger (2001)	Agricultural policy and resource modeling	Introduced multi-agent systems to simulate farmer behavior, resource use, and policy impacts in agricultural systems
Sengupta and Bennett (2003) Sengupta and Sieber (2007)	Spatial decision support	Proposed agent-based spatial decision systems for agricultural watershed management and planning
Happe (2004) Happe (2004)	Agricultural policy simulation	Developed AgriPoliS, an agent-based model capturing structural changes and decision-making in agricultural production systems
Barreteau et al. (2004) Barreteau et al. (2004)	Irrigation and water systems	Utilized multi-agent simulations to model farmer behavior and irrigation system viability in agricultural regions
Janssen (2005) Janssen (2005)	Land-use and ecological agriculture	Applied agent-based modeling to simulate agricultural land-use decisions and environmental interactions
Castella et al. (2005) Castella et al. (2005)	Agricultural transition modeling	Developed agent-based simulation of agrarian systems capturing socio-economic and environmental dynamics
Happe et al. (2006) Happe et al. (2006)	Farm structural dynamics	Extended agent-based simulation for agricultural policy analysis with heterogeneous farm agents and dynamic interactions
Fukatsu et al. (2006) Fukatsu et al. (2006)	Sensor-based agriculture	Implemented distributed agent-based sensor networks for real-time environmental monitoring in precision agriculture
Macmillan and Huang (2008) Macmillan and Huang (2008)	Agricultural system simulation	Modeled agricultural production systems using agents representing households and land-use decisions
Schreinemachers and Berger (2009) Berger and Schreinemachers (2009)	Agri-environmental policy and land-use modeling	Developed agent-based land-use models (MP-MAS) to support collaborative learning and policy analysis in agricultural systems
Schreinemachers et al. (2010) Schreinemachers et al. (2010)	Agricultural innovation adoption	Applied agent-based modeling for ex-ante assessment of tree crop innovations, analyzing adoption dynamics under economic and environmental constraints
Zhu et al. (2010) Zhu et al. (2010)	Product quality control	Designed an agent-based system for agricultural product quality monitoring and distributed decision-making
Pomar et al. (2011) Pomar et al. (2011)	Precision livestock feeding systems	Developed an agent-based simulation framework for intelligent coordination of livestock feeding processes and resource optimization
Schreinemachers and Berger (2011) Schreinemachers and Berger (2011)	Human–environment agricultural systems	Proposed MP-MAS framework for modeling farmer decision-making and environmental interactions using multi-agent systems
Niazi and Hussain (2011) Niazi and Hussain (2011)	Agricultural agent-based systems	Applied scientometric analysis to highlight the role of intelligent agents in agricultural research and distributed AI systems
Berger and Troost (2014) Berger and Troost (2014)	Climate adaptation in agriculture	Utilized agent-based modeling to simulate farmer responses to climate change and policy interventions
Dong et al. (2014) Dong et al. (2014)	Sensor-based agricultural monitoring	Developed mobile agent-based systems for efficient data collection and intelligent environmental monitoring
Yamashita and Hoshino (2018) Yamashita and Hoshino (2018)	Agricultural land-use management	Designed an agent-based model to simulate land preservation decisions and rural agricultural planning
Anbari et al. (2021) Anbari et al. (2021)	Irrigation and groundwater systems	Modeled socio-ecological agricultural systems using agent-based simulation of stakeholder interactions and water use
Wang et al. (2020) Wang et al. (2020)	Greenhouse monitoring and control	Proposed an IoT-enabled agent-based system for real-time monitoring and automated environmental control in agriculture
Ravaioli et al. (2023) Ravaioli et al. (2023)	Agricultural land-use modeling	Introduced a data-driven agent-based framework integrating remote sensing and micro-data for agricultural decision modeling
Ionescu et al. (2024) Ionescu et al. (2024)	AI-enabled agent-based modeling	Demonstrated integration of AI techniques with agent-based systems for improved decision-making in sustainable agriculture
Alotibi (2025) Alotibi (2025)	Smart agriculture adoption	Developed a socio-technical agent-based model to simulate adoption dynamics of intelligent agricultural technologies
Shadkam and Irannezhad (2025) Shadkam and Irannezhad (2025)	Agricultural supply chain systems	Proposed an agent-based intelligent framework for optimizing agricultural supply chains and digital agriculture workflows

2.1 Intelligent Sensing and Monitoring Agents

Intelligent sensing and monitoring agents represent one of the earliest and most impactful applications of AI agents in precision agriculture Murad et al. (2026). These agents are designed to continuously perceive agricultural environments through heterogeneous sensing modalities, analyze large-scale data streams, and provide actionable insights for farm management Dhanaselvam et al. (2026); Miller et al. (2025). Unlike traditional monitoring systems that rely on periodic manual observations Hamadani et al. (2021), AI-based sensing agents operate in a persistent, autonomous manner, enabling real-time situational awareness across spatially distributed agricultural systems Chen et al. (2024); Murad et al. (2026). As precision agriculture evolved in the early 2010s Zaman (2023); Tey et al. (2024), the integration of IoT sensors, remote sensing platforms, and machine learning models led to the

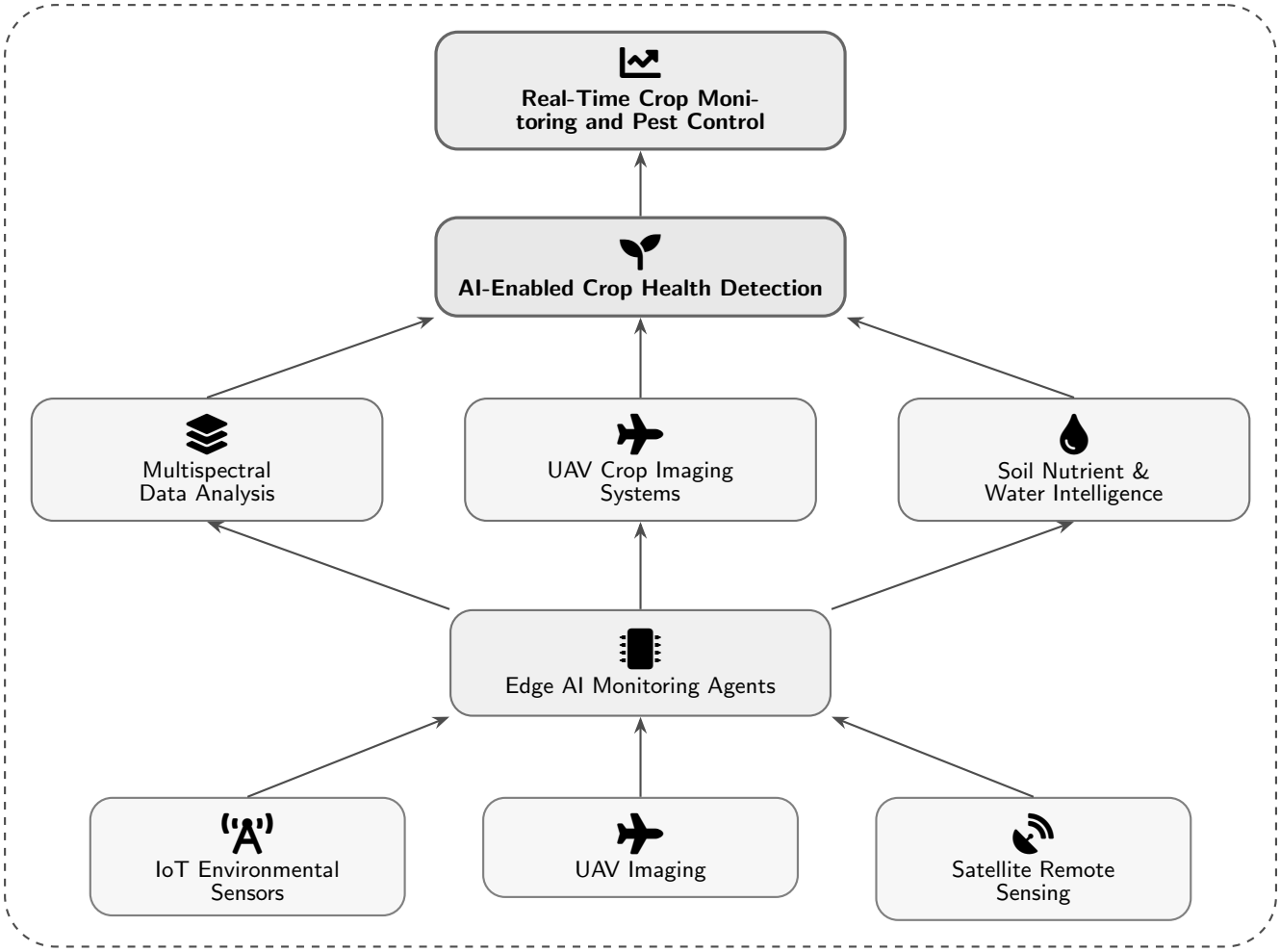


Fig. 4 Architecture of intelligent sensing and monitoring agents for precision agriculture, showing a clear bottom-up flow from heterogeneous sensing sources through edge AI fusion and parallel analytical modules to AI-enabled crop health detection and real-time crop monitoring.

emergence of intelligent agents capable of transforming raw environmental data into meaningful agronomic intelligence SS et al. (2024); Sharma and Shivandu (2024b).

As depicted in Figure 4, fundamental component of intelligent sensing agents is the use of IoT-based environmental monitoring systems Miller et al. (2025); Álvarez-López et al. (2026). These systems deploy distributed sensor networks across agricultural fields to collect data such as soil moisture, temperature, humidity, pH levels, and nutrient concentrations Ramani et al. (2025). AI agents embedded within these networks process sensor data either at the edge or through cloud-based architectures Ficili et al. (2025); Sarma et al. (2022), enabling dynamic decision-making for applications such as irrigation scheduling EMARA et al. (2026); Zhang et al. (2025); Hu and Tsang (2026), fertilization, pest control Yang et al. (2025); Bal et al. (2026) and environmental control Liu et al. (2025); Uma et al. (2026). For example, soil moisture sensing agents can trigger irrigation when moisture levels fall below threshold values, although more robust decisions are achieved by integrating weather forecasts, plant water status indicators (e.g., stomatal conductance) Zhang (2026); Vishwakarma (2025), and crop phenological stages, while temperature and humidity agents capture microclimate variations that influence crop growth Hu and Tsang (2026). The integration of wireless communication protocols and edge computing further enhances the responsiveness of these systems, allowing agents to operate in real-time without reliance on continuous cloud connectivity Tariq et al. (2025); Srivastav and Das (2025).

In addition to ground-based sensing, UAV-based crop monitoring platforms have significantly expanded the capabilities of AI agents in agriculture Sapkota et al. (2025b); Assiri et al. (2025). Unmanned aerial vehicles equipped with RGB, multispectral, and thermal cameras enable high-resolution imaging of agricultural fields, capturing spatial variability that is not observable through

ground sensors alone Olson and Anderson (2021). AI agents process these aerial datasets using computer vision and deep learning techniques to identify crop stress, disease symptoms, and pest infestations KP and Gowrishankar (2025); Ali et al. (2024); Zhu et al. (2024). For instance, convolutional neural networks (CNNs) have been widely used to detect early signs of plant diseases by analyzing leaf texture, color variations, and canopy structure Tugrul et al. (2022). Similarly, UAV-based agents can generate vegetation indices such as the Normalized Difference Vegetation Index (NDVI), which serves as a proxy for plant health and biomass estimation Seralathan and Edward (2025); Wu et al. (2026). These capabilities allow farmers to implement site-specific interventions, reducing input costs and improving crop productivity.

Satellite remote sensing further complements UAV-based systems by providing large-scale, temporal monitoring of agricultural landscapes Phang et al. (2023); Xing et al. (2025). Satellite-based AI agents analyze multi-temporal imagery to track crop growth patterns, assess drought conditions, and monitor land-use changes Feng et al. (2026); Gao et al. (2026). The integration of machine learning with satellite data enables predictive analytics Kumar and Sankar (2025b), where agents can forecast yield, detect anomalies, and identify long-term environmental trends Miller et al. (2025); Ishaq et al. (2025). For example, time-series analysis of satellite imagery can reveal deviations in crop development, allowing early detection of stress conditions before they become visually apparent Ramaiah et al. (2025). This scalability makes satellite-based agents particularly valuable for monitoring large agricultural regions and supporting policy-level decision-making.

Another critical application of intelligent sensing agents as illustrated in Figure 4 lies in crop health monitoring using machine learning. AI agents leverage supervised and unsupervised learning techniques to analyze complex datasets derived from image data, time-series measurements from sensors (e.g., soil moisture, temperature, and humidity), and environmental variables Bandi et al. (2025). Deep learning models, particularly CNNs and transformer-based architectures, have demonstrated high accuracy in detecting plant diseases Zeng et al. (2025); Thakur et al. (2021), nutrient deficiencies Banerjee et al. (2024); Nfor et al. (2025), and pest infestations Fang et al. (2024); Zhang et al. (2025). For example, agents trained on labeled image datasets can classify diseases with high precision KP and Gowrishankar (2025). Beyond classification, segmentation models enable pixel-level identification of affected regions, supporting precise intervention strategies Pinheiro and Collobert (2015). Furthermore, multimodal AI agents that integrate image data with environmental parameters can improve diagnostic accuracy by capturing the interplay between climate conditions and crop health Yang et al. (2025).

Soil nutrient and moisture monitoring systems, when integrated with AI-driven perception, decision-making, and actuation capabilities, function as intelligent agents that play a crucial role in optimizing resource use and supporting sustainable agricultural practices Khaliq et al. (2025).. These agents utilize sensor data combined with predictive models to estimate various farming parameters such as nutrient availability, soil fertility, and water requirements Ahmad et al. (2025); Mallik et al. (2025). For instance, machine learning models can predict nitrogen levels based on soil properties, leaf reflectances, and historical data Liu et al. (2022), enabling targeted fertilization. Similarly, moisture sensing agents can optimize irrigation schedules by accounting for evapotranspiration rates, weather forecasts, and crop growth stages Mallik et al. (2025); Hu and Tsang (2026). Such systems reduce water wastage and prevent over-fertilization, contributing to both economic and environmental sustainability of farming industries.

Another key advancement in intelligent sensing systems is the incorporation of edge computing architectures. In traditional cloud-based systems, sensor data must be transmitted to centralized servers for processing, which introduces latency and dependency on network connectivity Khriji et al. (2022); Shukla et al. (2023). In contrast, edge-based AI agents perform data processing locally on embedded devices, enabling faster decision-making and reduced communication overhead Santoso and Surya (2024a); El Jarroudi et al. (2024). For example, edge-deployed vision agents on UAVs or ground robots can process images in real-time to detect anomalies and trigger immediate actions Wang et al. (2025); McEnroe et al. (2022); Tariq et al. (2025). This is particularly important in remote agricultural settings where connectivity is limited. Edge intelligence also enhances system robustness and scalability, allowing distributed agents to operate independently while still contributing to a coordinated farm management system Yu et al. (2025).

The integration of these sensing modalities-IoT sensors, UAV imaging, and satellite data, and machine learning technologies forms a comprehensive monitoring ecosystem in which AI agents act as the central intelligence layer Aristeidis et al. (2026). These agents not only process and analyze data but also interact with other agents and systems to enable coordinated decision-making Papadopoulos

Table 2 AI agent-based sensing and monitoring systems in precision agriculture studied over the last 20 years

Study	Sensing Type	Agent Function
Fukatsu (2006) Fukatsu et al. (2006)	IoT sensors	Distributed environmental monitoring agents
Zhu (2010) Zhu et al. (2010)	Quality control	Product monitoring and traceability agents
Pomar (2011) Pomar et al. (2011)	Livestock sensors	Feeding and behavior monitoring agents
Schreinemachers (2011) Schreinemachers and Berger (2011)	Environmental data	Human–environment interaction monitoring agents
Dong (2014) Dong et al. (2014)	Wireless sensors	Mobile agent-based environmental sensing
Yamashita (2018) Yamashita and Hoshino (2018)	Land-use sensing	Agricultural land preservation monitoring agents
Wang (2020) Wang et al. (2020)	IoT in greenhouse	Real-time control and monitoring agents
Anbari (2021) Anbari et al. (2021)	Water systems	Irrigation and groundwater monitoring agents
Ravaioli (2023) Ravaioli et al. (2023)	Satellite & Ground Remote sensing + data	Data-driven land-use monitoring agents
Wei (2023) Wei et al. (2023)	IoT sensor networks	AI-based environmental monitoring agents for real-time agricultural safety and climate sensing
Indira (2023) Indira et al. (2023)	IoT field sensing	Distributed sensor-based agents for continuous monitoring of temperature, humidity, and soil conditions
Pierre (2023) Pierre et al. (2023)	Weather sensing systems	AI agents for real-time environmental monitoring and climate-aware agricultural resource optimization
Dusadeerungsikul (2024) Dusadeerungsikul and Nof (2024)	Precision monitoring algorithms	AI-based responsive sensing agents for crop-level monitoring and adaptive agricultural management
Obeidat (2024) Obeidat et al. (2024)	Smart farming sensors	Multi-agent communication systems for coordinated environmental monitoring and data-driven agriculture
Han (2024) Han et al. (2024)	Remote sensing	AI-enabled sensing agents for climate and crop monitoring using satellite and multispectral data
Fuentes-Peñailillo (2024) Fuentes-Peñailillo et al. (2024)	IoT + remote sensing	Integrated sensing agents combining underground sensors and remote sensing for crop monitoring
Akintuyi (2024) Akintuyi (2024)	Smart agriculture sensors	AI-driven sensing agents for real-time monitoring and adaptive farm condition assessment
Naqvi (2025) Naqvi et al. (2025)	IoT sensor networks	AI-enhanced sensing agents for real-time crop monitoring and autonomous environmental control

et al. (2025). For instance, a sensing agent detecting crop stress may communicate with an irrigation agent to adjust water delivery or with a robotic agent to apply targeted treatments Patel et al. (2025); Bayar et al. (2025). This interconnected architecture reflects the transition from isolated sensing systems to intelligent, agent-driven agricultural ecosystems.

2.2 Decision Support and Advisory Agents

Decision support and advisory agents constitute a critical class of AI agents in precision agriculture Posadas et al. (2023), focusing on transforming raw agricultural data into actionable recommendations for farmers and other agricultural stakeholders including government agencies and policy makers Wu and Zhong (2025). Unlike sensing agents that primarily collect and analyze environmental data, decision-support agents operate at a higher level of abstraction, integrating multi-source data streams with agronomic knowledge and predictive models to guide farm management decisions Pavlovic et al. (2008); Obeidat et al. (2024); Cantonjos and Biswas (2026). Rooted in early pre-2000 developments such as agricultural expert systems and knowledge-based advisory tools McKinion and Lemmon (1985); Rafea (2009), decision-support agents were further formalized in the early 2000s through rule-based and expert-driven frameworks Ishaaya et al. (2001); Happe (2004); Romera

et al. (2004), and have since evolved into advanced data-driven systems capable of adaptive and context-aware recommendations Jha et al. (2019).

In this context, decision-support agents can be viewed as a logical extension of intelligent sensing agents, where the outputs of perception-driven systems are further processed to enable higher-level reasoning and actionable decision-making. As illustrated in Fig. 5, these agents operate within a layered framework that integrates heterogeneous sensing data with predictive models and agronomic knowledge bases to generate optimized recommendations. This paradigm shifts the role of AI from passive monitoring to active decision support, enabling adaptive, feedback-driven agricultural management.

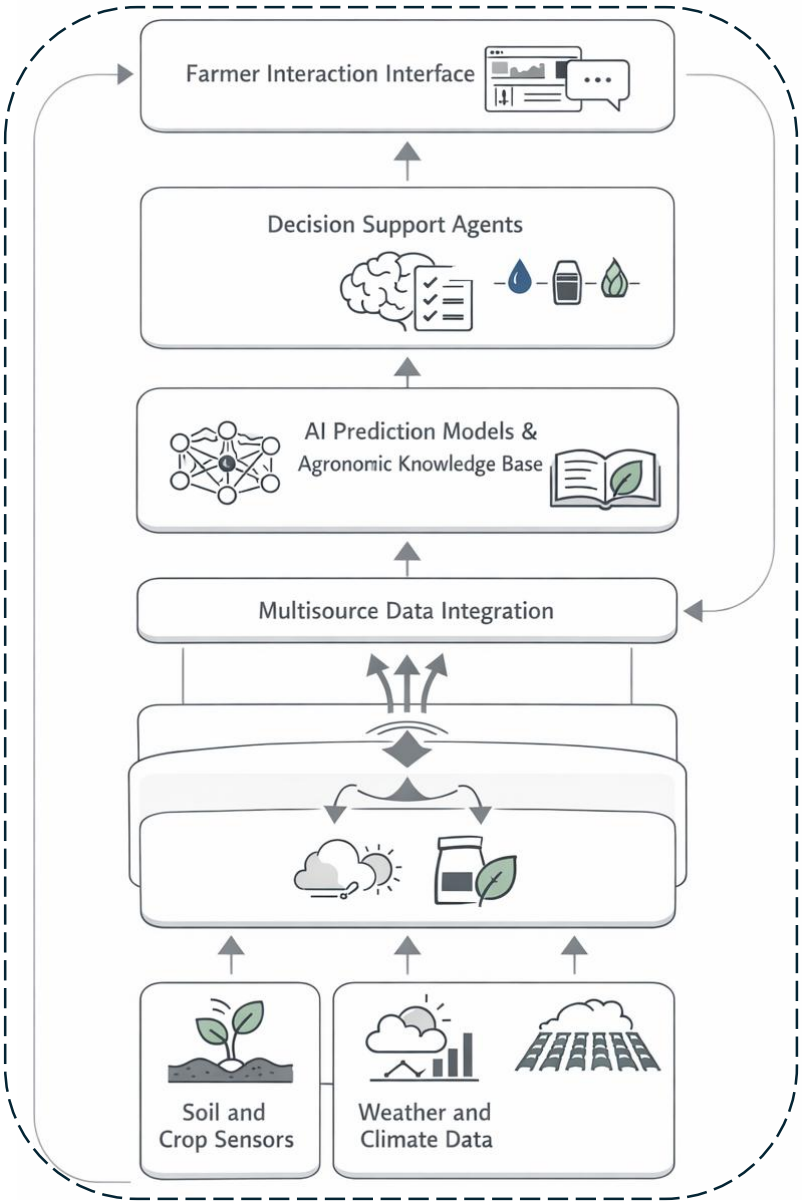


Fig. 5 Layered architecture of AI-based decision support agents in precision agriculture, illustrating the transformation of heterogeneous sensing data into actionable farm recommendations. Soil, crop, and climate data are integrated through multi-source fusion and processed by AI prediction models combined with agronomic knowledge bases. Decision-support agents generate optimized recommendations for irrigation, fertilization, and crop planning, among other applications, which are delivered through user interaction interfaces. Feedback loops enable continuous learning and adaptive decision-making for efficient and data-driven agricultural management.

At the core of these systems are AI-powered farm advisory platforms that integrate machine learning models with domain-specific knowledge bases Priya et al. (2025); Miller et al. (2024). These agents ingest heterogeneous data sources, including soil properties Wadoux (2025), weather forecasts

Vishwakarma (2025), crop growth stages Srinivasu et al. (2025), and historical yield records, to generate recommendations for irrigation, fertilization, pest management, and crop selection Bayar et al. (2025); Khaliq et al. (2025), among other applications. Beyond crop production, emerging advisory systems in animal agriculture incorporate livestock-specific data streams such as animal behavior, physiological indicators, feeding patterns, and environmental conditions within barns or grazing systems Neethirajan (2026); Dao et al. (2025), enabling decision support for health monitoring, precision feeding, and welfare management Zhang et al. (2025); Akintan et al. (2025). Early advisory systems relied on rule-based logic derived from agronomic and physiological expertise Shaffer and Brodahl (1998); Gabriel Filho et al. (2011), where decision rules were manually encoded into the system for both crop and livestock management contexts Moore et al. (2014). However, modern advisory agents increasingly incorporate statistical learning and data-driven models that enable continuous improvement based on observed outcomes, integrating multi-modal data across both plant and animal systems to provide adaptive, context-aware, and scalable decision support Ugwu et al. (2025); Sapkota et al. (2025)..

One of the most prominent applications of advisory agents is intelligent irrigation scheduling Jimenez et al. (2020). Water management is a critical challenge in agriculture, particularly in regions facing water scarcity Qadir et al. (2003). AI agents designed for irrigation optimization integrate multi-source data, including soil moisture measurements, weather forecasts, and evapotranspiration models, with crop water requirements commonly estimated using crop coefficient (Kc)-based approaches, while increasingly incorporating plant water stress indicators from both contact sensors (e.g., stomatal conductance, leaf water potential) and non-contact sensing methods (e.g., thermal and hyperspectral imaging) to determine optimal irrigation timing and quantity Kaushik and Singh (2025a); Mishra et al. (2026).. These agents can dynamically adjust irrigation schedules in response to changing environmental conditions, reducing water waste while maintaining crop health. For example, reinforcement learning-based irrigation agents have been proposed to learn optimal policies through interaction with environmental conditions, balancing water usage and crop yield objectives Zhang et al. (2025).

Crop recommendation systems represent another important application of decision-support agents Swati et al. (2026); Senapaty et al. (2024). These systems analyze soil characteristics, climate conditions, and historical cultivation data, while also considering the availability of key inputs such as water, labor, and other resources, to recommend suitable crops for specific regions or seasons Shams et al. (2024). AI agents in this context often leverage classification and clustering algorithms to identify crop suitability patterns Akkem et al. (2024), while also incorporating external factors such as market demand and price trends Tzachor et al. (2022); Ali et al. (2025). By providing data-driven recommendations, these agents assist farmers in selecting crops that maximize yield and profitability while minimizing environmental impact Titirmare et al. (2024); Kumari et al. (2025).

Yield prediction agents further enhance decision-making by forecasting crop production based on environmental conditions and management practices as well as crop growth models Hernández Hernández et al. (2025). These agents integrate time-series data from sensors and remote sensing platforms with machine learning models such as regression algorithms, neural networks, and ensemble methods Kazanskiy et al. (2025). Accurate yield prediction enables farmers to plan harvesting schedules, allocate resources efficiently, and optimize supply chain operations Celis et al. (2024); Dhal and Kar (2024). Moreover, predictive analytics allows early identification of potential yield losses due to stress factors or potential overcropping that may compromise produce quality, enabling proactive interventions SS et al. (2024).

Agricultural expert systems, which represent some of the earliest forms of AI agents in agriculture McKinion and Lemmon (1985); Rafea (2009); Holzinger et al. (2024), continue to play a role in modern advisory platforms Aryan et al. (2025). These systems encode expert knowledge in the form of rules, ontologies, and decision trees, providing explainable recommendations to users Shams et al. (2024). In contemporary implementations, expert systems are often combined with machine learning models to create hybrid advisory agents that benefit from both knowledge-driven and data-driven approaches He et al. (2026); Priya et al. (2025).

A notable recent advancement in advisory systems is the integration of conversational interfaces powered by large language models (LLMs) Reddy and Reddy (2025); Sapkota et al. (2025). These interfaces enable farmers to interact with AI agents using natural language, improving accessibility while extending system capabilities beyond traditional rule-based or narrowly trained models. In contrast to conventional agents that rely on limited, developer-curated datasets, LLM-based systems

can leverage and reason over vast corpora of published literature, agronomic knowledge, and domain-specific information sources to generate more informed and context-aware recommendations Sapkota et al. (2025). For instance, farmers can query advisory agents about irrigation scheduling, disease management, or fertilizer application strategies and receive responses that synthesize both data-driven insights and broader knowledge sources Murad et al. (2026). As a result, such systems more effectively bridge the gap between complex AI models and practical agricultural decision-making.

Table 3 summarizes key developments in decision-support AI agents in agriculture over the past two decades. The table shows progression from rule-based approaches to advanced AI-driven platforms integrating sensing, prediction, and reasoning for data-driven farm management. Notably, emerging large language models (LLMs) and vision-language models (VLMs) present a particularly promising direction for advisory systems, as they enable the fusion of multimodal data with extensive knowledge sources to deliver more comprehensive and context-aware recommendations. As a result, future advisory agents are likely to become more autonomous, adaptive, and user-centric, further strengthening their role in precision agriculture.

2.3 Autonomous Robotics and Field Operation Agents

Autonomous robotics and field operation agents represent the physical embodiment of AI agents in agriculture, where intelligent decision-making is directly coupled with actuation capabilities Gonzalez-de Santos et al. (2020); Murad et al. (2026). These agents are integrated into robotic platforms such as autonomous tractors, harvesting robots, UAVs, and precision spraying systems, enabling automated execution of agricultural tasks Sharma and Shivandu (2024b); Jia et al. (2025). The development of such systems marks a significant transition from passive decision-support systems to active, agent-driven agricultural operations.

Robotic fruit, flower, and vegetable harvesting systems, for instance, are among the most challenging and impactful applications of autonomous agents in agriculture Oliveira et al. (2021). Harvesting tasks require precise perception, manipulation, and control, as crops vary in shape, size, and location within complex environments Ji et al. (2023). AI agents embedded in harvesting robots utilize computer vision and deep learning to detect target objects such as fruits, estimate their ripeness/readiness, and determine optimal harvsting strategies Zhou et al. (2022); Tang et al. (2020). For example, convolutional neural networks (CNNs) and Transfermer-based models are used to identify fruits in crop canopy images, while depth sensors and 3D reconstruction techniques provide spatial information for robotic manipulation Wang et al. (2022). These agents must also account for occlusions, lighting variations, and dynamic environmental conditions, making fruit, flower and vegetable harvesting a highly complex task Zhao et al. (2016); Neo et al. (2022).

Autonomous tractors and field robots extend the capabilities of AI agents to large-scale field operations such as planting, tillage, and fertilizer application Jia et al. (2025). These systems integrate GPS, LiDAR, and vision-based perception to robustly navigate agricultural fields and perform tasks with minimal human intervention. AI agents control navigation, path planning, and task execution, ensuring efficient coverage and avoiding obstacles Yang et al. (2023); Aremu et al. (2026). For instance, path planning algorithms enable tractors to optimize field traversal patterns, reducing operational time and fuel consumption Rimada (2025); Gao et al. (2026).

UAV-based crop scouting agents provide a complementary approach to ground-based robots by offering aerial perspectives of agricultural fields Kannan et al. (2025). These agents autonomously navigate predefined or dynamically adjusted flight paths, capture high-resolution images, and analyze data in real-time or offline. UAV agents are particularly useful for monitoring large areas, detecting anomalies, and guiding targeted interventions Velusamy et al. (2021); Ahmad et al. (2025). For example, UAV-based agents can identify regions affected by pests or diseases and communicate this information to ground robots for localized treatment Niaz et al. (2025); Ali et al. (2024).

UAVs are also deployed for targeted field operations in constrained environments (e.g., sloped terrains or small-scale farms), where perception-driven agents enable localized interventions such as site-specific chemical application. Beyond aerial systems, autonomous ground agents are widely utilized in precision spraying and robotic weed management, where perception outputs are directly coupled with actuation to enable selective treatment of weeds, pests, or diseased plants Meshram et al. (2022); Balaska et al. (2023). These systems reduce chemical usage and environmental impact by operating at fine spatial resolution and responding dynamically to field variability Fountas et al. (2022); Lochan et al. (2024); Vijayakumar et al. (2025). More broadly, such capabilities are not

Table 3 Representative studies on AI agent-based decision support and advisory systems in agriculture

Study	Application	Agent Function
Karetsos (2002) Karetsos et al. (2002)	E-agribusiness	Intelligent business agents for market interaction and agricultural information exchange
Parrott (2003) Parrott et al. (2003)	Dairy production	Multi-agent decision support for knowledge sharing and production management
Perini (2004) Perini and Susi (2004)	Farm DSS	Agent-based decision frameworks for integrated agricultural production planning
Kamel (2004) Kamel (2004)	Knowledge-based systems	Agent-based agricultural knowledge systems for distributed problem solving and advisory support
Ziervogel (2005) Ziervogel et al. (2005)	Climate advisory	Social simulation agents for assessing farmer responses to seasonal climate forecasts
Happe (2006) Happe et al. (2006)	Policy analysis	Agent-based decision support for agricultural policy simulation and farm structural dynamics
Schreinemachers and Berger (2006) Schreinemachers and Berger (2006)	Land-use decisions	Multi-agent representations of farmer decision-making for land-use planning and adaptation
Matthews (2006) Matthews (2006)	Landscape management	Agent-based integration of human decision-making with biophysical agricultural simulations
Liu and Zhu (2008) Liu and Zhu (2008)	Distributed crop management	Cooperative agent-based simulation for distributed crop decision support and management
Yue (2008) Yue et al. (2008)	Agricultural expert systems	Agent-based inspection and advisory modules for breeding and crop management decisions
Adikari (2010) Adikari (2010)	Farmer advisory systems	Dynamic multi-agent advisory framework for farmer-centered agricultural recommendations
Valbuena (2010) Valbuena et al. (2010)	Regional land-use planning	Agent-based modeling of heterogeneous farmer decisions for land-use change assessment
Ng et al. (2011) Ng et al. (2011)	Watershed management	Agent-based decision models linking farmer behavior with water quality and agricultural policy
Berger and Troost (2014) Berger and Troost (2014)	Climate adaptation	Multi-agent decision support for evaluating agricultural adaptation and mitigation strategies
Singh and Sharma (2015) Singh and Sharma (2015)	Agricultural information systems	Intelligent agent-based personalized advisory and knowledge retrieval for farmers
Pérez-Salazar et al. (2019) Pérez-Salazar et al. (2019)	Supply chain decision support	Agent-based decision system for optimizing agricultural production scheduling and logistics
Khemakhem et al. (2020) Khemakhem et al. (2020)	Agricultural DSS optimization	Intelligent agent-based decision support for optimizing agricultural production and logistics processes
Badillo-Márquez (2021) Badillo-Márquez et al. (2021)	Climate vulnerability	Agent-based model-driven decision support for assessing sugarcane vulnerability under climate change
Cavaliere and Senatore (2021) Cavaliere and Senatore (2021)	Agroforestry systems	Knowledge-enhanced multi-agent models for decision support in agroforestry management
Sun (2022) Sun et al. (2022)	Agri-environmental adoption	Agent-based policy decision support for technology adoption under environmental incentives
Vinyals (2023) Vinyals et al. (2023)	Innovation diffusion	Agent-based simulation for AI-assisted policy design and digital tool adoption in agriculture
Lopez (2023) Lopez-Jimenez et al. (2023)	Smart irrigation monitoring	Agent-based sensor placement and state estimation for optimizing irrigation through cost-aware network design and Kalman filter-based monitoring of heterogeneous field conditions
Tripathi (2023) Tripathi et al. (2023)	Supply chain advisory	Farmer-centric decision support for market dynamics in volatile agricultural supply chains
Kale and Rao (2025) Kale and Rao (2025)	Fertilizer recommendation	Multi-agent climate-resilient decision support for adaptive fertilizer recommendations
Pargo (2025) Pargo et al. (2025)	Smart irrigation systems	Hybrid agent-based and system dynamics framework for IoT-enabled irrigation, enabling cooperative control of soil moisture through rule-based agents and optimized water and energy management
Carbo (2026) Carbo et al. (2026)	Irrigation district management	Agent-based simulation of farmer decision-making and water distribution, incorporating BDI-based reasoning, interaction protocols, and pressure head estimation for decentralized and autonomous irrigation management

limited to spraying or weed control but extend across multiple agricultural operations, including irrigation, nutrient management, disease and insect control, and harvesting, where perception, reasoning, and action are tightly integrated within agent-driven frameworks. What distinguishes these systems from conventional precision agriculture approaches is the explicit formulation of closed-loop perception–decision–action pipelines, where AI agents continuously interpret multimodal sensory inputs

and autonomously adapt actions in response to changing environmental conditions. This integration is enabled by advanced perception systems that combine data from cameras, LiDAR, GPS, and other sensors through fusion techniques, allowing agents to robustly detect targets, estimate spatial context, and execute context-aware actions Xiang et al. (2023); Polanía et al. (2026); Ogbu (2023); Khan et al. (2025).

Edge computing plays a crucial role in enabling near real-time operation of robotic agents by performing data processing locally on the robot, thereby reducing communication latency and dependency on cloud infrastructure, although system responsiveness may still be constrained by computational and sensing delays in complex field environments Khan (2025); Waqar et al. (2024). In addition, distributed architectures enable multiple robots to coordinate and share information, forming collaborative multi-agent systems capable of executing tasks such as harvesting, precision spraying, smart irrigation and water management operations Ota (2006); Rasheed et al. (2022).

As summarized in Table 4, AI agent-based robotics in agriculture has evolved from early multi-agent prototyping and cooperative control frameworks to advanced systems supporting fleet coordination, autonomous navigation, and diverse task-specific operations, including harvesting, irrigation, and pest management. The studies listed in Table 4 highlight that AI agents function as the core intelligence layer, enabling robots to perceive environments, coordinate with other agents, and adapt to dynamic field conditions. Conceptually, this paradigm can be viewed as both an extension and a realization of the decision-support frameworks discussed in the previous subsection, where recommendations are no longer limited to human interpretation but are directly translated into autonomous actions. As illustrated in Fig. 6, perception, decision-making, and actuation are tightly integrated within a closed-loop architecture, in which multimodal sensing informs agent reasoning, generates executable commands, and continuously updates system behavior through feedback. Together, Table 4 and Fig. 6 demonstrate that AI agents enable the transition from advisory decision support to embodied, autonomous operations, forming the foundation for scalable multi-robot agricultural ecosystems.

3 Multi-Agent Systems and Collaborative Agricultural Intelligence

While individual AI agents enable task-specific intelligence in precision agriculture, many real-world agricultural challenges inherently require coordination across multiple processes, spatial scales, and operational units Srinivasu et al. (2025); Sapkota et al. (2025b). Tasks such as irrigation scheduling, crop monitoring, robotic harvesting, and resource allocation are interdependent and distributed across large farm environments Adinarayana et al. (2024). Consequently, multi-agent systems have emerged as a essential paradigm for enabling *collaborative agricultural intelligence*, where multiple agents interact, coordinate, and jointly optimize system-level objectives Tweedale et al. (2007); Kaye-Blake et al. (2010).

As modern agricultural systems are transitioning from isolated decision-making modules to interconnected ecosystems of sensing, reasoning, and actuation agents Murad et al. (2026), these agents operate across heterogeneous infrastructures including IoT sensors, UAVs, ground robots, and cloud-based platforms forming distributed intelligence networks capable of adaptive, real-time decision-making Vijay et al. (2025); Dazzi (2025). This system-level coordination is conceptually aligned with the closed-loop perception–decision–action pipeline illustrated in Fig. 7, where feedback-driven interactions enable robust and scalable agricultural automation.

3.1 Distributed Farm Intelligence and IoT-Agent Architectures

As illustrated in Figure 7, distributed farm intelligence represents the foundational layer of multi-agent agricultural systems, where multiple sensing and decision agents operate across farm infrastructure. These architectures integrate IoT sensor networks, edge computing, and cloud-based platforms into hierarchical agent-based systems capable of continuous monitoring and adaptive decision-making Radhakrishnan and Prasanth (2025); Vimalajeewa et al. (2021).

At the sensing level, distributed IoT networks act as perception agents, continuously collecting environmental and crop-related data such as soil moisture, temperature, humidity, nutrient levels, and canopy conditions Miller et al. (2025). These sensors are spatially distributed across fields, enabling high-resolution monitoring of heterogeneous agricultural environments. Unlike traditional

Table 4 Representative AI agent-based robotic systems in agricultural operations

Study	Application	Agent Function
Arguenon et al. (2006) Arguenon et al. (2006)	Agricultural robot prototyping	Multi-agent coordination of agricultural robot behaviors and interactions for field operations
Nielsen et al. (2006) Nielsen et al. (2006)	Cooperative field robots	AI agents for role allocation, cooperation, and navigation among agricultural robots
Yue (2008) Yue et al. (2008)	Agricultural expert robotics	Agent-based coordination modules supporting robotic inspection and cooperative agricultural tasks
Nebot et al. (2011) Nebot Roglá et al. (2008)	Agricultural robot teams	HLA-based distributed agent architecture for coordination of real and simulated agricultural robots
Pentjuss et al. (2011) Pentjuss et al. (2011)	Precision field operations	Multi-agent planning and control framework for robotic execution in precision agriculture
Lavendelis et al. (2012) Lavendelis et al. (2012)	Multi-robot task allocation	Agent-based robotic architecture for task allocation, coordination, and management in agricultural settings
Steckel et al. (2013) Steckel et al. (2013)	Mobile agricultural processes	Agent-based technologies for coordinating mobile processes in non-deterministic agricultural environments
El Yasmine et al. (2014) El Yasmine et al. (2014)	Autonomous logistcs	Multi-agent coordination frameworks supporting autonomous logistics and robotic interaction in agri-food systems
Kuandykov et al. (2015) Kuandykov et al. (2015)	Anti-locust protection	Multi-agent robotic coordination for territory surveillance and agricultural pest response
Guo et al. (2016) Guo et al. (2016)	Agricultural supply robotics	Agent-based negotiation mechanisms supporting autonomous coordination across operational agricultural chains
Chowhan et al. (2017) Chowhan et al. (2017)	Autonomous pollination drones	Intelligent agent-based control for autonomous drone-pollinator coordination in greenhouse agriculture
Oprea (2018) Oprea (2018)	Multi-robot agriculture	Agent-based modeling of multi-robot systems for coordination in agricultural foraging and field tasks
Nair et al. (2019) Nair (2019)	Greenhouse telerobotics	Agent-based human–robot coordination for greenhouse monitoring and disease-management robotics
Lujak et al. (2020) Lujak et al. (2020)	Agri-robot fleets	Multi-agent coordination strategies for collaborative operation of agricultural robot fleets
Jimenez et al. (2020) Jimenez et al. (2020)	Irrigation robotics	Cyber-physical intelligent agents for autonomous irrigation scheduling and control in horticultural crops
Eiffert et al. (2022) Eiffert et al. (2022)	Ground robot autonomy	Agent-aware path planning for long-term autonomous ground robot operation in agriculture
Roberts-Elliott et al. (2022) Roberts-Elliott et al. (2022)	Soil compaction mapping	Agent-based simulation of multi-robot coordination for precision soil monitoring tasks
Gutierrez Cejudo et al. (2023) Gutierrez Cejudo et al. (2023)	Agrirobot digital twins	Multi-agent architecture for dynamic simulation and coordination of agricultural robot fleets
Chreim et al. (2023) Chreim et al. (2023)	Mushroom harvesting supervision	AI-agent-based supervision of robotic harvesting systems within farm-scale system-of-systems settings
Gutierrez-Cejudo et al. (2023) Gutiérrez-Cejudo et al. (2023)	Dynamic agri-robot coordination	Agent-based agrirobot architecture and simulator for coordinated fleet actuation and task execution
Griepentrog and Stein (2024) Griepentrog and Stein (2024)	Sustainable crop robotics	Agent-based architectural concepts for evaluating robotic systems in sustainable crop production
Essakkat et al. (2025) Essakkat et al. (2025)	AI-based weeding robots	Agent-based adoption modeling for deployment and operational use of robotic weeding systems on farms
Anchekov (2025) Anchekov and Kurashev (2025)	Intelligent agricultural systems	Software architecture for agent-based modeling of interactions among robots, plants, and farm entities

centralized monitoring systems, IoT-based architectures allow real-time, localized data acquisition at scale Obeidat et al. (2024).

In this context, federated learning has emerged as a promising approach for leveraging distributed sensing networks while preserving data locality and privacy. Instead of transmitting raw data to

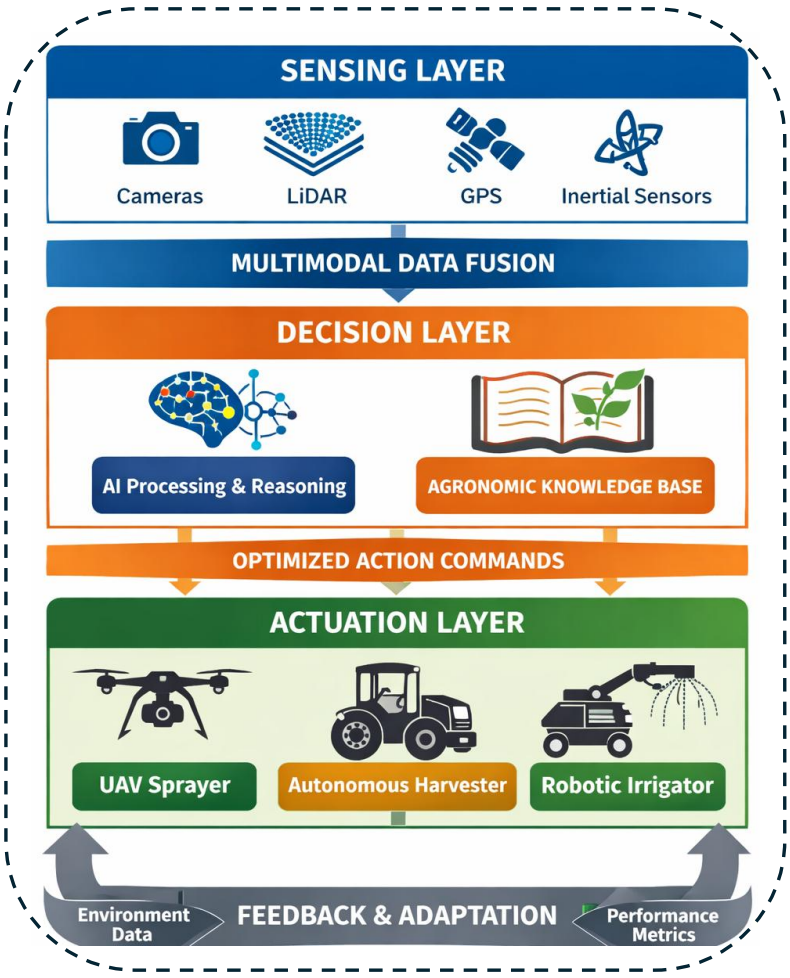


Fig. 6 System-level architecture of agent-driven autonomous agricultural systems, illustrating a closed-loop integration of perception, decision-making, and actuation. Multimodal sensing and data fusion provide environmental inputs to AI agents, which perform object detection, reasoning, and planning. These outputs are translated into robotic actuation for tasks such as precision spraying, harvesting, and irrigation using ground and/or aerial robots. Feedback mechanisms continuously update the system, enabling adaptive and robust operation in dynamic field environments.

centralized servers, local nodes collaboratively train shared models by exchanging model updates, enabling scalable and communication-efficient learning across heterogeneous field conditions McMahan et al. (2017); Park et al. (2021). This paradigm is particularly beneficial in agricultural settings where data variability, bandwidth constraints, and privacy concerns are significant. By integrating federated learning with IoT-based sensing architectures, perception agents can continuously improve model performance while maintaining decentralized and robust operation across large-scale farming systems Gupta et al. (2025).

To address latency and bandwidth constraints associated with centralized processing, edge computing plays a crucial role in enabling agricultural intelligence Kalyani and Collier (2021). Edge agents deployed on embedded devices or field gateways process sensor data locally, performing tasks such as anomaly detection, crop stress identification, and preliminary decision-making Tariq et al. (2025). This enables rapid responses to environmental changes, such as triggering irrigation or issuing alerts, without relying on continuous cloud connectivity.

At a higher level, cloud-based farm management systems aggregate data from multiple edge agents and perform global optimization using advanced AI models Tariq et al. (2025); Kannan (2025). These cloud agents integrate historical data, weather forecasts, and predictive analytics to generate farm-wide recommendations for irrigation scheduling, fertilization, and other crop management tasks Roy et al. (2025); Olayinka et al. (2026). The interaction between edge and cloud agents creates a hierarchical decision-making structure that balances local responsiveness with global optimization Liu et al. (2026).

Data flow within these architectures follows a continuous feedback loop. Sensor agents provide real-time data to edge agents, which process and forward relevant information to cloud agents Kalyani

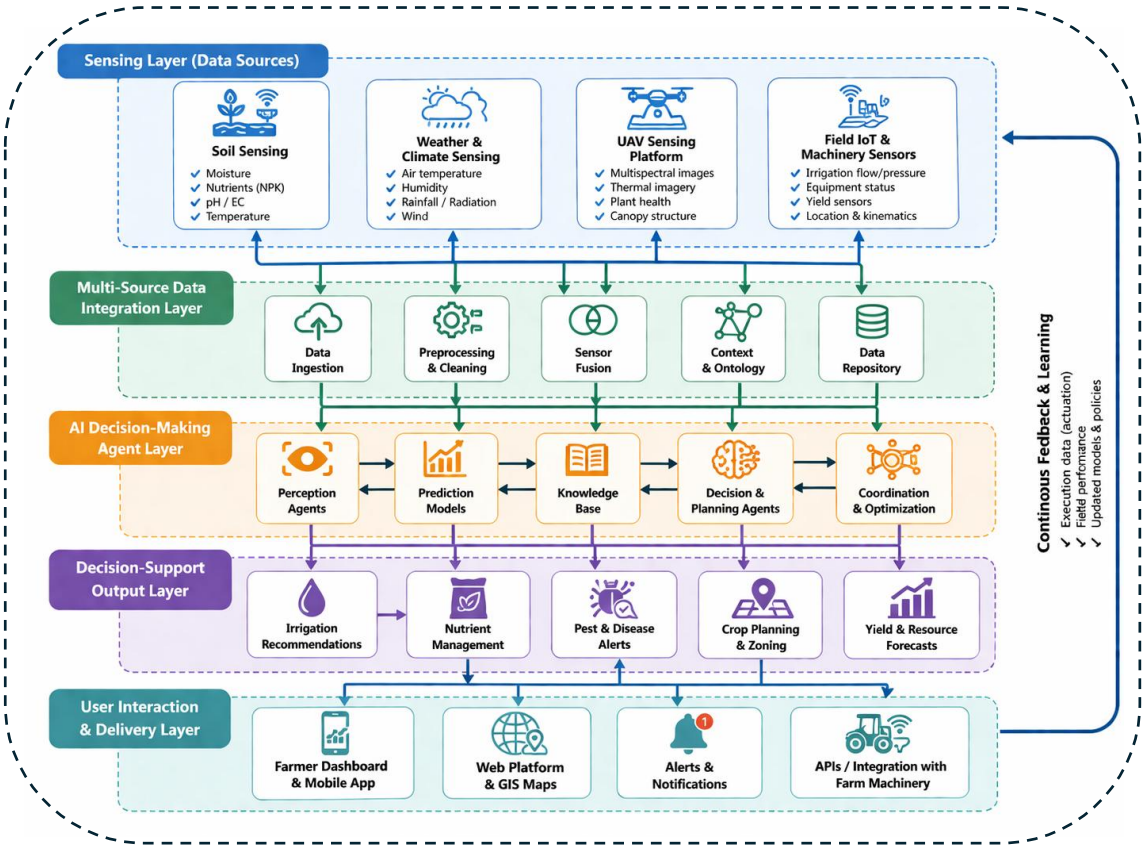


Fig. 7 Layered architecture of an AI agent-driven system for precision agriculture, showing the integration of heterogeneous sensing platforms (soil and crop sensors, weather stations, UAV-based imaging, and IoT devices) with multi-source data fusion, AI-based decision-making, and service delivery layers. The framework enables prediction, planning, and optimization for applications such as irrigation, nutrient management, pest control, and crop planning, while continuous feedback from operational and environmental outcomes supports adaptive and data-driven decision-making.

and Collier (2021). In turn, cloud-level decisions are communicated back to edge agents and actuation systems, enabling adaptive control of farm operations Olatinwo et al. (2025). This bidirectional flow ensures that agricultural systems remain responsive to both immediate conditions and long-term trends leading to more proactive, precise farm management compared to conventional reactive management practices.

Furthermore, hierarchical agent architectures allow functional specialization, where sensing agents, decision agents, and actuation agents operate as modular components within a unified system Tapia et al. (2013); Li et al. (2025). This modularity enhances scalability, interoperability, and robustness, enabling seamless integration of new technologies such as UAV-based monitoring and robotic systems Singh et al. (2021). As a result, distributed IoT-based agent architectures form the backbone of smart farming systems, enabling data-driven and adaptive agricultural intelligence Heideker et al. (2020).

3.2 Swarm Robotics and Multi-Robot Agricultural Systems

Beyond distributed sensing, as explained in Figure 8, multi-agent collaboration extends to physical robotic systems, where multiple robots coordinate their actions to perform complex agricultural tasks. Swarm robotics and multi-robot systems represent a key advancement in this domain, enabling decentralized coordination among autonomous agents operating in dynamic environments Alqudsi and Makaraci (2025a); Nguyen (2024).

Swarm UAV systems as depicted in Figure 8 have become increasingly important for large-scale crop monitoring. Multiple aerial agents operate collaboratively to capture high-resolution imagery, covering extensive agricultural areas more efficiently than single UAV systems Ming et al. (2023). These agents share information about field conditions, enabling coordinated data collection and real-time anomaly detection Qu et al. (2022); Assiri et al. (2025). For example, UAV swarms can identify

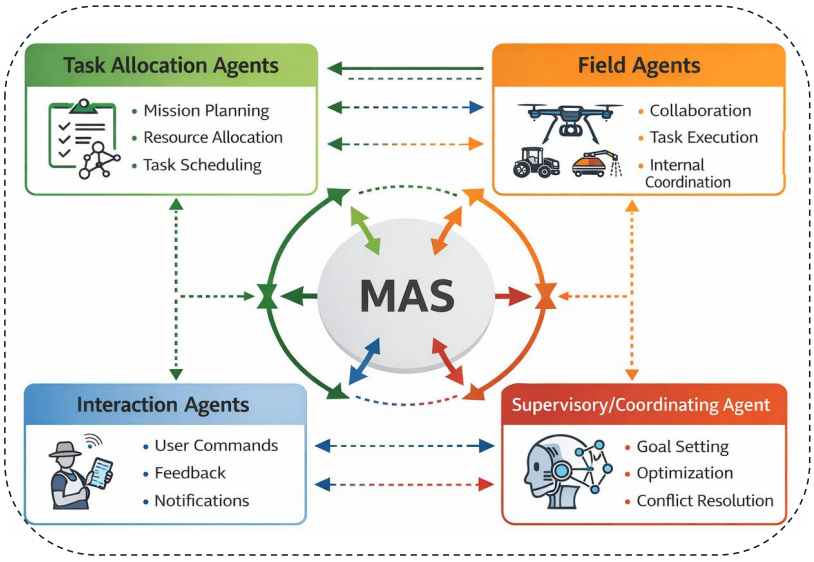


Fig. 8 Multi-agent system architecture for precision agriculture, illustrating coordinated interaction among task allocation, field, interaction, and supervisory agents. Bidirectional communication enables decentralized decision-making, resource optimization, and efficient task execution. The framework supports collaborative autonomy, integrating human inputs, system-level coordination, and real-time feedback to achieve efficient and adaptive agricultural operations.

regions affected by disease or water stress and dynamically adjust their flight paths to focus on areas of interest Ali et al. (2024); Ming et al. (2023).

Similarly, cooperative harvesting robots demonstrate the potential of multi-agent coordination in labor-intensive tasks Jia et al. (2025). In such systems, multiple robotic agents collaborate to detect, localize, and harvest crops Lytridis et al. (2021). Task allocation mechanisms allow robots to distribute workloads efficiently, reducing redundancy and improving overall system performance Afrin et al. (2021); Onteddu et al. (2025). For instance, one agent may specialize in perception (fruit detection) Dutta et al. (2025); Srinivasu et al. (2025), while others focus on manipulation and collection Javed et al. (2025); Kandamali et al. (2025); however, equally important is the dynamic allocation of multiple agents performing the same task, such as deploying a greater number of manipulators to regions with higher fruit density or reallocating robots to more productive areas of the field, thereby improving harvesting efficiency and overall system throughput.

Robot fleet coordination is another critical aspect of multi-agent agricultural systems Gutierrez Cejudo et al. (2023). Autonomous tractors, sprayers, and ground robots must coordinate their movements and tasks to avoid collisions, optimize coverage, and minimize operational time Lujak et al. (2020). Decentralized coordination strategies enable robots to make local decisions based on shared information, rather than relying on a central controller Katikaridis et al. (2025); Hernández et al. (2025). This decentralized coordination approach improves system robustness and scalability, particularly in large and complex farm environments.

Distributed robotic sensing further enhances the capabilities of multi-robot systems Tejada et al. (2025). Robots equipped with heterogeneous sensors such as cameras, LiDAR, and environmental sensors collect complementary data, which is shared among agents to improve situational awareness Wijayathunga et al. (2023). For example, UAVs may provide aerial imagery, while ground robots collect close-range data, enabling a multi-scale understanding of crop conditions Zhang and Zhu (2023).

Swarm intelligence principles underpin these systems, where collective behavior emerges from local interactions among agents Beni (2020). Decentralized control strategies allow robots to coordinate without explicit central supervision, adapting dynamically to environmental changes Bossens et al. (2022). This decentralized coordination paradigm is particularly valuable in agriculture, where field conditions such as crop variability, terrain heterogeneity, and dynamic weather patterns are inherently complex and unpredictable.

3.3 Multi-Agent Reinforcement Learning for Farm Optimization

While coordination mechanisms enable collaboration among agents, learning-based approaches further enhance their ability to adapt and optimize decisions over time. As illustrated in Figure 9,

Multi-Agent Reinforcement Learning (MARL) has emerged as a powerful framework for enabling agents to learn optimal policies through interaction with agricultural environments Boubin et al. (2022); Marwah et al. (2023).

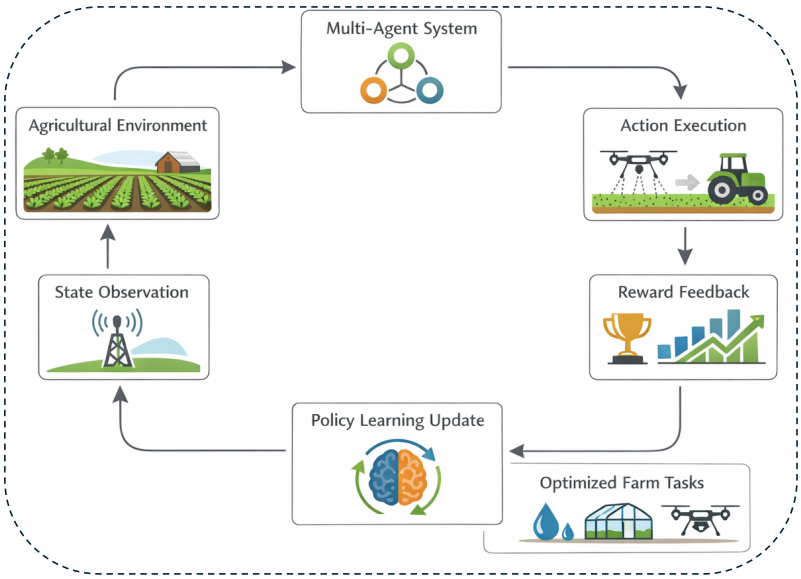


Fig. 9 Showing a multi-agent reinforcement learning framework for agricultural systems, where multiple agents interact with a dynamic environment to optimize farm operations. Agents observe environmental states, execute actions, and receive reward feedback, forming a closed learning loop. Policy updates enable continuous adaptation and coordination among agents. The framework supports parallel decision-making and real-time optimization, ultimately improving agricultural tasks such as irrigation management, greenhouse control, and autonomous UAV-based monitoring through data-driven, feedback-guided learning.

For example, in irrigation management, reinforcement learning agents optimize water allocation by learning policies through reward-driven feedback based on outcomes such as yield, water-use efficiency, and plant stress responses, while incorporating environmental conditions, crop requirements, and weather forecasts. In multi-agent settings, experiences from different field regions can be aggregated to support collective learning, enabling more robust and adaptive irrigation strategies under spatial variability Jia and Pei (2025). By interacting with the environment, these agents continuously refine their policies to balance water usage and crop yield Hung and Yang (2021). Multi-agent formulations allow different zones of a farm to be managed independently while maintaining overall system efficiency Rangarajan et al. (2026); Assiri et al. (2025).

Multi-agent crop management systems extend this concept to broader agricultural operations Villarrubia et al. (2017). Agents representing different components such as crops, soil, and weather systems interact to optimize decisions related to fertilization, pest control, and resource allocation González-Briones et al. (2019); Chevalier et al. (2016). These systems capture the complex interdependencies within agricultural ecosystems, enabling more holistic and adaptive decision-making Villarrubia et al. (2017).

Autonomous greenhouse optimization is another important example of how MARL could be applied to precision agriculture. In controlled environments, multiple agents manage variables such as temperature, humidity, lighting, and CO_2 levels Ajagekar et al. (2024). Reinforcement learning enables these agents to learn optimal control strategies that maximize crop growth while minimizing energy consumption Mansour et al. (2025); Hu and You (2022). Multi-agent coordination ensures that different environmental factors are balanced effectively.

Energy-efficient UAV routing further demonstrates the potential of MARL in agricultural systems Assiri et al. (2025); Ye et al. (2023). UAV agents learn optimal flight paths that minimize energy consumption while maximizing coverage and data quality Betalo et al. (2023). Multi-agent coordination allows multiple UAVs to distribute tasks efficiently, avoiding redundancy and improving operational efficiency Abro et al. (2024).

A key advantage of MARL is its ability to adapt policies over time through continuous interaction with dynamic environments Gronauer and Diepold (2022). Unlike rule-based systems, learning-based agents can respond to changing conditions, uncertainties, and complex interactions among system

components Wang et al. (2017); Albrecht et al. (2024). This adaptability is critical in agriculture, where environmental variability and uncertainty are inherent.

Moreover, MARL enables decentralized learning, where agents learn independently while sharing information with others which reduces the need for centralized control and enhances scalability Ahmed et al. (2025). However, challenges such as coordination complexity, convergence stability, and data requirements remain active areas of research Wong et al. (2023).

4 Agentic AI for Autonomous Agricultural Decision-Making

Agentic AI represents a paradigm shift beyond traditional AI agents by introducing capabilities for autonomous reasoning, goal-driven planning, tool utilization, and long-term decision-making Sapkota et al. (2025a). While classical AI agents in agriculture have primarily focused on reactive or narrowly scoped tasks such as sensing, monitoring, or rule-based decision support Raman et al. (2024); Babenko (2026), agentic AI systems operate with a higher degree of autonomy, enabling end-to-end orchestration of complex agricultural workflows Tariq et al. (2025); Kumar et al. (2026). These systems are capable of interpreting high-level objectives, decomposing them into actionable sub-tasks, dynamically adapting to environmental conditions, and coordinating multiple tools and agents over extended temporal horizons Bandi et al. (2025).

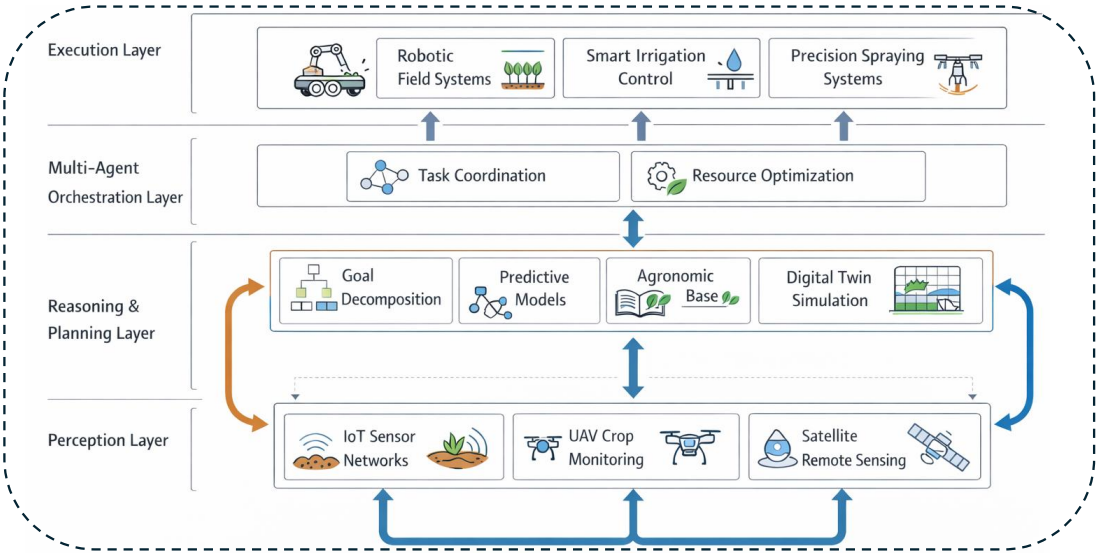


Fig. 10 Illustrating a layered Agentic AI architecture for autonomous agriculture, integrating perception, reasoning, orchestration, and execution modules. Multi-modal sensing (IoT-, UAV-, satellite-based) feeds into planning components including predictive models, agronomic knowledge, and digital twin simulations. A multi-agent orchestration layer coordinates tasks and optimizes resources, enabling adaptive decision-making. The execution layer deploys robotic systems, irrigation, and precision spraying. Closed-loop feedback and digital twin interaction ensure continuous learning, real-time adaptation, and efficient farm-scale autonomous operations.

In agricultural contexts, as depicted in Figure 10, this transition is particularly significant due to the inherently dynamic, uncertain, and multi-scale nature of farming systems. Crop growth, environmental variability, resource constraints, and interdependent farm operations such as the coordination of irrigation, fertilization, labor, and machinery require continuous adaptation and long-term planning Hanson et al. (2007); Verma et al. (2025). Agentic AI systems address these challenges by embedding reasoning and planning capabilities directly within the decision-making loop Vishwakarma (2025); Gupta (2026), effectively transforming agricultural automation from reactive control systems into proactive, goal-oriented intelligence frameworks Hosseini and Seilani (2025). This evolution builds upon the multi-agent coordination frameworks discussed in the previous section, extending them toward fully autonomous agricultural ecosystems.

4.1 Goal-Oriented Autonomous Agricultural Agents

Goal-oriented agentic systems form the core of autonomous agricultural decision-making, where agents operate based on explicitly defined objectives rather than predefined rules or static models

Haidemariam (2025); Hughes et al. (2025). These objectives may include maximizing crop yield, minimizing water consumption, optimizing labor efficiency, or ensuring sustainable resource utilization. Unlike traditional systems, which respond to immediate inputs, goal-oriented agents continuously evaluate their actions against long-term targets and adjust their strategies accordingly Lee and Park (2026).

At the architectural level, these systems integrate three fundamental components: perception, reasoning, and action Panda and Padhy (2025). Perception modules collect data from heterogeneous sources, including IoT sensors, UAV imagery, and satellite observations Poorvi et al. (2025). Reasoning modules, often powered by large-scale machine learning models or hybrid symbolic-neural systems, interpret this data in the context of predefined goals Liang et al. (2025). Action modules execute decisions through actuators such as automated irrigation systems, robotic platforms, or farm management interfaces Jiang et al. (2025); Hernández et al. (2025).

A defining characteristic of agentic systems is their ability to perform hierarchical task planning Dwivedi et al. (2026); Sapkota et al. (2025a). High-level goals are decomposed into sequences of subtasks, which are dynamically scheduled and executed based on environmental feedback Bandi et al. (2025). For example, an agent tasked with optimizing irrigation may first analyze soil moisture data and plant water status data, then predict future water requirements using weather forecasts, and finally generate an irrigation schedule that balances crop needs and resource constraints Swati et al. (2026); Srinivasu et al. (2025). This planning process is iterative and adaptive, allowing agents to refine their strategies as new information becomes available.

Autonomous decision loops further enhance system robustness. In these loops, agents continuously cycle through perception, reasoning, and action phases, updating their internal state and policies in response to environmental changes Garg (2025). This enables real-time adaptation to unexpected conditions such as sudden weather shifts, pest outbreaks, or equipment failures Panda and Padhy (2025); Salimi et al. (2026). Importantly, these loops are not purely reactive; they incorporate predictive modeling and scenario analysis to anticipate future states and proactively adjust decisions.

In robotic farming systems, goal-oriented planning plays a critical role in coordinating multiple operations Murad et al. (2026); Vishwakarma (2025). For instance, as depicted in Figure 11, a harvesting agent may plan routes for robotic arms, schedule tasks across multiple robots, and adapt strategies based on crop maturity and environmental constraints. Such systems demonstrate the transition from isolated automation to fully autonomous agricultural workflows, where agents manage complex sequences of interdependent tasks with minimal human intervention. Table 5 presents a comprehensive overview of the key features, functional modules, operational capacities, and practical roles of goal-oriented autonomous agricultural agents in modern farming systems.

4.2 Agentic AI for Crop Monitoring and Management

Agentic AI systems are increasingly being applied to the autonomous management of crop health and agricultural resources that integrates sensing, prediction, and planning capabilities to detect, diagnose, and respond to agricultural challenges in a proactive and adaptive manner KP and Gowrishankar (2025); Murad et al. (2026). Unlike traditional monitoring systems, which provide descriptive insights Fountas et al. (2015), agentic systems actively intervene in the environment to optimize outcomes Hughes et al. (2025).

Plant stress detection agents represent a key application area. These agents utilize multispectral imaging, computer vision models, and temporal analysis to identify early signs of various types of plant stresses caused by water and nutrient deficiency, disease, or pest infestation Srinivasu et al. (2025). By combining spatial and temporal data, they can distinguish between different types of stress factors and predict stress progression. Once a potential issue is identified, the agent initiates a decision-making process that may involve recommending targeted interventions, deploying robotic actuation, or adjusting environmental conditions to address them Bal et al. (2026); Vishwakarma (2025).

Smart irrigation agents exemplify the integration of predictive modeling and autonomous control as these agents continuously monitor soil moisture, plant water status, weather forecasts, and crop growth stages to determine optimal irrigation strategies Swati et al. (2026); Johnsen (2025). Rather than following fixed schedules, they dynamically adjust water delivery based on real-time conditions and long-term objectives. For example, during periods of anticipated rainfall, irrigation may be

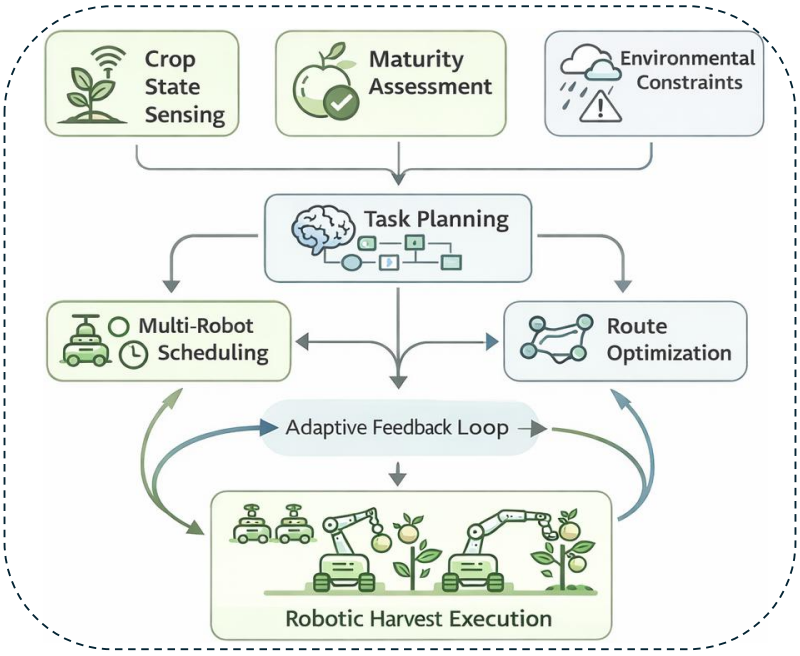


Fig. 11 System-level conceptual representation of a goal-oriented fruit harvesting agent in autonomous agriculture, integrating crop state sensing, maturity assessment, and environmental constraints into a centralized task planning agent. This example system coordinates multi-robot scheduling and route optimization to enable efficient robotic harvest execution. Continuous adaptive feedback loops update sensing and planning modules, ensuring dynamic response to changing field conditions. This framework highlights how intelligent agents orchestrate perception, decision-making, and coordinated action to achieve fully autonomous, scalable harvesting operations in precision agriculture.

reduced or delayed, conserving water resources while maintaining crop health Abdulameer et al. (2025); Cantonjos and Biswas (2026); Vishwakarma (2025).

Nutrient optimization agents further extend this paradigm by managing fertilization processes. These agents analyze soil composition, crop requirements, and historical nutrient application and yield data to generate precise nutrient application strategies AlZu'bi and Mughaid (2025b). By optimizing the timing, quantity, and spatial distribution of fertilizers, they reduce environmental impact while improving crop productivity Srinivasu et al. (2025). Integration with robotic or automated application systems enables precise execution of these strategies at the field level.

Pest monitoring and management systems also benefit from agentic AI approaches KP and Gowrishankar (2025); Srinivasu et al. (2025). These systems combine sensor data, remote sensing, and predictive analytics to track pest populations and forecast infestation risks Aziz et al. (2025); Chaudhary and Jabborova (2026). Agents can then coordinate interventions such as targeted pesticide application, biological control deployment, or environmental adjustments to mitigate pest impact.

A key advantage of agentic systems in this domain is their ability to integrate heterogeneous data sources and operate across multiple temporal scales San et al. (2026). Short-term decisions, such as immediate irrigation adjustments, are combined with long-term strategies, such as seasonal crop planning or disease prevention Wei et al. (2024); Hung and Yang (2021). This multi-scale reasoning capability enables more holistic and effective agricultural management.

Furthermore, these systems often incorporate feedback mechanisms that allow them to learn from past actions and outcomes. By analyzing the effectiveness of interventions, agents refine their models and improve future decision-making. This continuous learning process is essential for adapting to changing environmental conditions and evolving agricultural practices.

4.3 Agentic Farm Management Systems and Digital Twins

The emergence of digital twin technologies has further expanded the capabilities of agentic AI in agriculture, enabling the creation of virtual replicas of physical farm environments Olayinka et al. (2026); Sihare et al. (2026). These digital twins integrate real-time data, predictive models, and simulation frameworks to provide a comprehensive representation of agricultural systems Peladarinos et al. (2023). When combined with agentic AI, digital twins enable advanced planning, optimization, and decision-making capabilities Goyal et al. (2026).

Table 5 Core features, functional modules, operational capacities, and representative implementation characteristics of goal-oriented autonomous agricultural agents. The table summarizes how agentic agricultural systems extend beyond reactive automation by integrating perception, reasoning, hierarchical planning, adaptive control, and coordinated execution for long-horizon farm decision-making.

Dimension	Core Feature	Primary Function	Operational Capacity	Representative Agricultural Role
Goal formulation	Explicit objective specification	Defines high-level targets such as yield maximization, water minimization, labor reduction, or sustainability improvement	Supports long-horizon optimization instead of isolated short-term responses	Crop production planning, irrigation efficiency, resource-aware field operations
Perception layer	Multimodal environmental sensing	Collects heterogeneous data from IoT sensors, UAVs, satellites, robots, and weather services	Enables context-aware state estimation across spatially distributed field conditions	Soil moisture monitoring, canopy status tracking, crop maturity sensing, pest surveillance
State interpretation	Context-aware reasoning input	Converts raw data streams into actionable environmental states for downstream planning	Supports uncertainty-aware understanding of dynamic farm conditions	Detection of stress severity, maturity stage estimation, field accessibility assessment
Reasoning engine	Goal-conditioned inference	Interprets sensed data under predefined agronomic objectives using ML, symbolic logic, or hybrid models	Links observed field conditions with target-driven decisions rather than fixed rules	Deciding whether to irrigate, delay spraying, reprioritize harvesting, or reschedule robots
Knowledge integration	Agronomic knowledge grounding	Incorporates domain rules, constraints, thresholds, and management heuristics into decision-making	Improves consistency, interpretability, and safety of autonomous recommendations	Fertilizer timing logic, disease risk thresholds, machinery constraints, crop-specific rules
Hierarchical planning	Goal decomposition	Breaks high-level objectives into ordered sub-tasks, dependencies, and executable action chains using AI planning frameworks such as hierarchical task networks (HTN), reinforcement learning-based planners, and LLM-driven task decomposition models	Supports complex multi-step workflows under changing environmental conditions	Harvest planning, irrigation sequencing, scouting followed by targeted intervention
Task scheduling	Dynamic temporal coordination	Allocates timing and priority of subtasks based on current state, urgency, and resource availability	Enables adaptive execution under uncertain field conditions	Scheduling multiple harvest passes, irrigation windows, or pest inspection routines
Resource optimization	Constraint-aware allocation	Optimizes water, energy, labor, machinery time, and chemical inputs relative to goals	Reduces waste while maintaining agronomic performance	Precision irrigation, selective spraying, efficient robot deployment
Predictive modeling	Future-state anticipation	Uses forecasts and learned models including machine learning models, time-series forecasting, and process-based crop growth models (e.g., DSSAT, APSIM) to estimate crop needs, disease risk, weather impact, and yield outcomes	Enables proactive rather than purely reactive control	Forecast-driven irrigation, disease outbreak anticipation, yield-aware harvest timing
Autonomous decision loop	Closed-loop adaptation	Continuously cycles through perception, reasoning, planning, execution, and feedback updating	Maintains robust performance under nonstationary and uncertain environments	Real-time adaptation to rainfall events, equipment faults, or crop condition changes
Scenario evaluation	What-if analysis	Compares alternative operational strategies before action execution	Improves robustness by testing candidate policies under simulated or predicted conditions	Choosing between early harvest, delayed irrigation, or alternate robot routes
Action layer	Executable intervention control	Converts plans into operational commands for irrigation systems, robots, UAVs, and software interfaces	Bridges digital planning with physical farm execution	Actuating pumps, dispatching robots, triggering scouting flights, updating dashboards
Human interaction	Supervisory override and feedback	Allows growers or operators to review, modify, or approve autonomous recommendations	Supports human-in-the-loop governance and trust calibration	Farmer approval of irrigation plans, exception handling, mission reassignment
Multi-robot coordination	Distributed operational planning	Synchronizes multiple robots or devices to perform complementary tasks	Enables scalable autonomous workflows across large farms	Coordinated harvesting fleets, scouting UAV plus ground robot response, parallel field coverage
Robustness to uncertainty	Adaptive strategy revision	Updates policies when weather, crop state, market needs, or equipment status changes unexpectedly	Improves resilience under real-world variability	Replanning during storms, equipment malfunction recovery, shifting harvest priority
Temporal scope	Long-term decision orientation	Balances immediate interventions with medium- and long-term production objectives	Prevents short-term control behavior common in static automation systems	Season-long irrigation planning, phased nutrient management, multi-day robotic task allocation
Scalability	Farm-wide orchestration	Extends from plot-level control to coordinated field, greenhouse, or multi-farm management	Supports modular deployment across heterogeneous infrastructures	Greenhouse control, orchard block management, multi-zone farm optimization
Explainability potential	Traceable decision structure	Makes decisions interpretable through explicit goals, sub-tasks, and reasoning chains	Enhances trust, validation, and regulatory readiness	Explaining why harvest was advanced, why irrigation was delayed, or why a route changed
System-level outcome	Autonomous workflow management	Integrates perception, decision-making, and coordinated action into end-to-end agricultural operations	Enables scalable, adaptive, and minimally supervised farm management	Fully autonomous harvesting, adaptive irrigation ecosystems, coordinated robotic crop operations

In digital twin-based systems, agents operate within both physical and virtual environments. The virtual environment serves as a testing ground for evaluating different strategies, allowing agents to simulate potential outcomes before implementing actions in the real world Hu et al. (2023); San et al. (2026). For example, an agent may simulate different irrigation schedules or crop management strategies within the digital twin to identify optimal solutions under varying conditions Salwadkar (2024); Alves et al. (2023).

Farm simulation systems leverage these capabilities to model complex interactions between crops, soil, weather, and management practices Aderele et al. (2025). Agent-based simulations enable the representation of individual components such as plants, fields, or equipment as autonomous entities with distinct behaviors and interactions Mazzetto (2024); Abar et al. (2017). This allows for detailed analysis of system dynamics and the exploration of alternative management strategies.

Predictive agricultural modeling plays a central role in these systems, combining machine learning approaches with process-based crop growth models (e.g., DSSAT, APSIM) that explicitly represent plant development and soil–plant–atmosphere interactions Hernández Hernández et al. (2025); Jabed and Murad (2024). Emerging agentic AI frameworks increasingly embed such models within digital twin environments, enabling agents to simulate and evaluate management strategies under varying conditions prior to execution. This hybrid integration supports biologically informed and data-driven decision-making, where crop growth and yield predictions guide planting, irrigation, and harvest

strategies, while climate-aware simulations inform risk management and long-term planning Emi-Johnson et al. (2025); Delfani et al. (2024).

Agent-based farm orchestration represents the culmination of these capabilities, where multiple agents coordinate across sensing, planning/decision-making, and execution layers to manage entire farming operations Yang et al. (2026); Shadkam and Irannezhad (2025). In such systems, high-level goals such as maximizing profitability or sustainability are translated into coordinated actions across multiple subsystems Nugroho and Uehara (2023). This includes tasks such as scheduling irrigation, managing robotic operations, optimizing resource allocation, and responding to environmental changes.

Importantly, digital twin systems enable continuous feedback between physical and virtual environments Li and Yang (2025). Data from the real world updates the digital twin, while insights from simulations inform real-world actions San et al. (2026). This bidirectional interaction enhances system accuracy and enables continuous optimization of agricultural processes.

Despite their potential, these systems also present challenges, including computational complexity, data integration, and model accuracy Bariah and Debbah (2024); Mohanraj and Balaji (2025). Ensuring reliable synchronization between physical and virtual environments is critical for maintaining system performance. Additionally, integrating diverse data sources and models requires robust architectures and standardized interfaces.

5 Agentic Agriculture: Toward Fully Autonomous Farming Ecosystems

The convergence of artificial intelligence, robotics, sensing infrastructures, and multi-agent systems is giving rise to a new paradigm referred to as *Agentic Agriculture*. Unlike traditional precision agriculture systems that rely on isolated automation modules or decision-support tools, agentic agriculture as explained in Figure 12 envisions fully integrated ecosystems where autonomous agents perceive, reason, plan, and act in a coordinated and adaptive manner. These systems extend beyond reactive intelligence, incorporating goal-driven reasoning, long-term planning, and multi-agent orchestration to manage agricultural operations holistically.

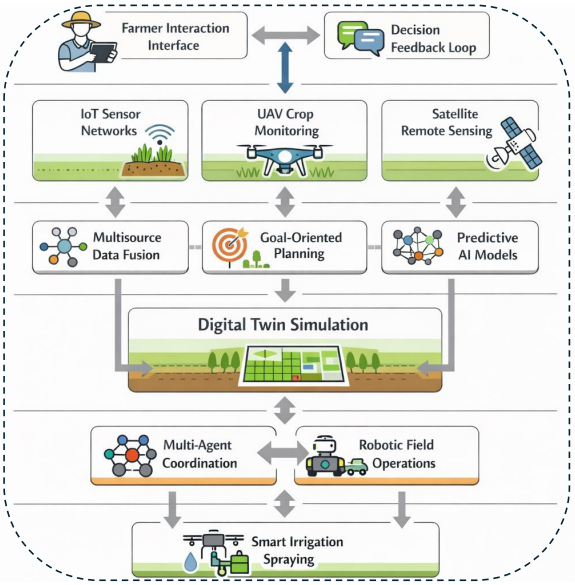


Fig. 12 This figure illustrates an integrated Agentic Agriculture Ecosystem that unifies sensing, intelligence, coordination, and action within a closed-loop framework. Multimodal perception from IoT, UAV, and satellite systems is fused and processed through predictive models, goal-oriented planning, and digital twin simulation. A multi-agent coordination layer translates decisions into automated/robotic field operations such as fruit picking, smart irrigation, and precision spraying. Continuous feedback from execution updates perception and planning, while farmer interaction provides oversight and adaptive guidance. The architecture enables real-time, data-driven, and autonomous agricultural decision-making with human-in-the-loop intelligence.

As illustrated in this Figure 12, the evolution toward agentic agriculture is grounded in the progressive development of AI agents, from early simulation-based decision models to modern multi-agent systems and, more recently, agentic AI architectures capable of autonomous decision-making. In this paradigm, farms are no longer viewed as static environments but as dynamic, cyber-physical systems where intelligent agents continuously interact among each other and with crops, soil, climate, and human stakeholders Wu and Zhong (2025); Holzinger et al. (2024). This shift enables the transformation of agriculture into a self-optimizing and resilient system capable of addressing global challenges such as changing weather patterns, food security, labor shortages, and environmental sustainability.

5.1 Human-AI Collaborative Agriculture

Despite the increasing autonomy of AI systems, human expertise remains indispensable in agricultural decision-making. Human-AI collaborative agriculture as depicted in Figure 13 represents a hybrid paradigm in which intelligent agents augment, rather than replace, human farmers. In such systems, AI agents act as cognitive assistants, providing insights, recommendations, and predictive analytics while allowing humans to retain oversight and strategic control.

As illustrated in Figure 13, a key feature of collaborative systems is the integration of human knowledge with data-driven intelligence. Farmers possess contextual understanding, experiential knowledge, and domain-specific intuition that are often difficult to encode explicitly into machine learning models Priya et al. (2025); Murad et al. (2026). However, as experienced farmers age and fewer individuals enter farming with comparable experiential backgrounds, there is a growing risk of knowledge loss Komarek et al. (2020); Meuwissen et al. (2001). In this context, AI-driven systems particularly those leveraging data-driven learning, digital twins, and knowledge-grounded models offer a promising pathway to capture, formalize, and transfer expert knowledge, thereby bridging the gap between traditional farming expertise and future autonomous agricultural systems. Agentic systems bridge this gap by incorporating human-in-the-loop mechanisms, where user feedback, preferences, and decisions are continuously integrated into the agent's reasoning process Hughes et al. (2025). This enables adaptive learning and personalization, ensuring that recommendations align with local farming practices and constraints.

Modern collaborative frameworks often employ conversational interfaces powered by large language models, enabling intuitive interaction between farmers and AI agents Tzachor et al. (2023); Gujjar and Kumar (2025). Through natural language queries, farmers can access complex analytics, explore alternative scenarios, and receive actionable recommendations NANDAMURU (2025); Reddy and Reddy (2025). For example, as illustrated in Figure 13, a farmer may query the system about optimal irrigation schedules under changing weather conditions, and the agent can respond by synthesizing sensor data, predictive models, and agronomic knowledge.

Explainability and trust are critical components of human-AI collaboration Romeo and Conti (2026). As AI systems become more autonomous, ensuring that their decisions are transparent and interpretable becomes essential for user acceptance.Agentic systems address this challenge by integrating explainability mechanisms that expose the data sources, assumptions, and reasoning processes underlying their recommendations Brohi et al. (2025); Garg (2025). Although such mechanisms do not completely resolve the intrinsic "black-box" nature of many AI models, they enhance transparency at the system level, thereby improving trust, accountability, and informed human decision-making.

Another important aspect is shared autonomy, where control is dynamically distributed between humans and agents Kiasari and Aly (2026). In routine operations, such as irrigation or pest monitoring, agents may operate autonomously, however, in high-stakes scenarios such as disease outbreaks or extreme weather events human intervention may be required to guide decision-making Saha (2026); Cantonjos and Biswas (2026). This flexible allocation of control enhances system robustness and ensures that critical decisions benefit from both human expertise and computational intelligence.

Collaborative systems also support multi-stakeholder interactions, enabling coordination among farmers, agronomists, policymakers, and supply chain actors. By acting as intermediaries, AI agents facilitate information exchange, optimize resource allocation, and align decision-making across different levels of the agricultural ecosystem. This is particularly important in large-scale farming operations and cooperative agricultural networks.

Overall, human-AI collaborative agriculture represents a transitional phase toward full autonomy, where intelligent agents enhance human capabilities while maintaining accountability, transparency,

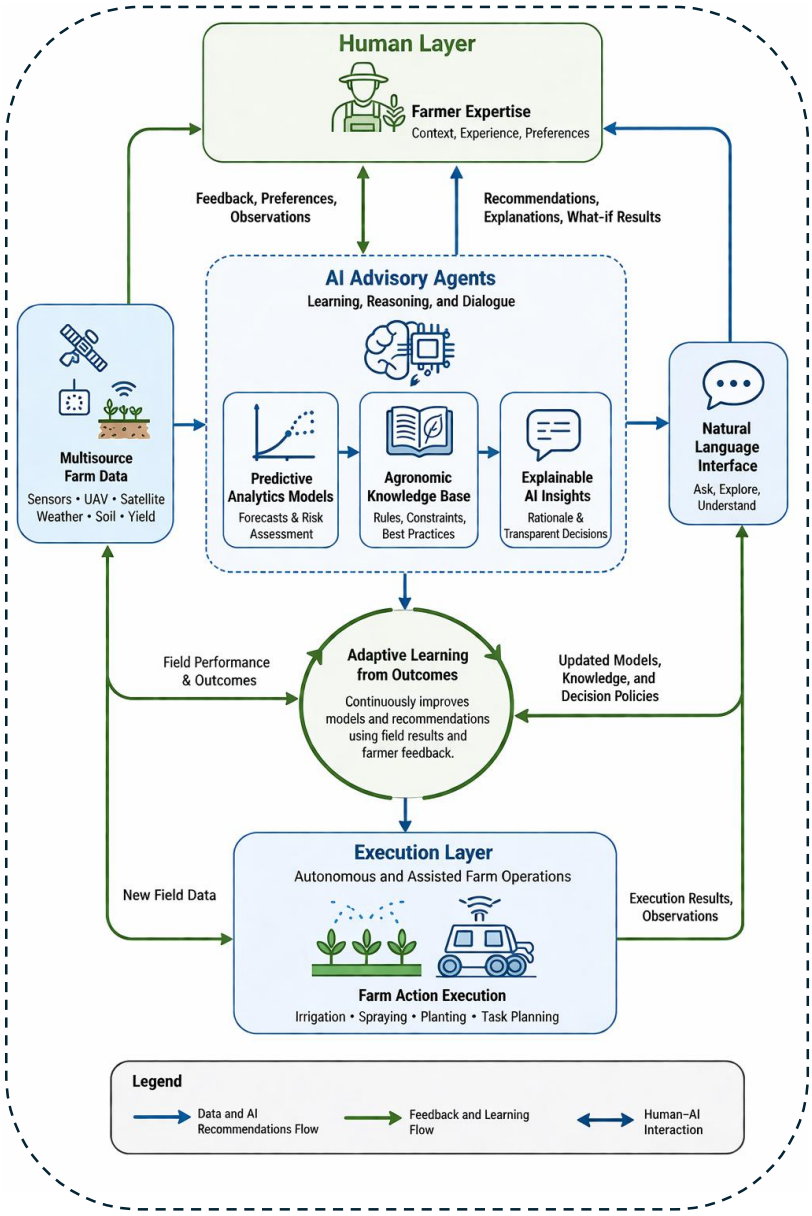


Fig. 13 Conceptual illustration of human-AI collaborative agriculture, where knowledgeable and experienced farmers interact with AI advisory agents through multi-source farm data and predictive analytics models. Agronomic knowledge bases support reasoning, while natural language interfaces enable intuitive communication. Explainable AI insights enhance transparency, and shared decision distributes authority between human and machine. Adaptive learning feedback continuously refines system behavior, and final decisions drive farm action execution. The framework highlights closed-loop collaboration, integrating human intuition and experience with data-driven intelligence for robust and context-aware agricultural decision-making.

and control. This paradigm ensures that technological advancements are aligned with practical agricultural needs and socio-economic considerations.

5.2 Agentic Agricultural Robotics and Autonomous Operations

Building upon collaborative frameworks, agentic agricultural robotics represents the realization of fully autonomous farming systems. In this paradigm, robotic agents equipped with advanced perception, reasoning, and actuation capabilities perform agricultural operations with minimal human intervention, while incorporating encoded human expertise through historical data, agronomic knowledge bases, and simulation-driven digital twins. This enables the transfer and reuse of experiential farming knowledge in autonomous systems without requiring real-time human input. By integrating sensing, decision-making, and execution within closed-loop architectures, these systems support continuous adaptation to dynamic and uncertain agricultural environments.

Autonomous robotic agents perform tasks such as crop monitoring, harvesting, spraying, weeding, and planting, using computer vision and sensor fusion to perceive crops, weeds, and plant health, tightly integrated with decision-making modules for optimal action selection.

A defining characteristic of agentic robotics is the integration of multi-agent coordination Yang et al. (2025); Habiba and Khan (2025). Rather than operating as isolated units, robotic agents collaborate to achieve collective objectives. For example, fleets of UAVs may coordinate to monitor large agricultural fields, while ground robots perform targeted interventions such as precision spraying or harvesting. Task allocation and coordination are managed through decentralized or semi-centralized mechanisms, enabling scalability and robustness.

Closed-loop autonomy is a critical feature of these systems. Robots continuously receive feedback from sensors and update their actions in real time, ensuring precise and efficient operation. For instance, a fruit harvesting robot may adjust its grasping strategy based on fruit position and ripeness Lin et al. (2026); Ao et al. (2025), while a spraying robot dynamically modulates application rates based on detected plant stress Wang et al. (2025); Jiao et al. (2025). These feedback loops enable high levels of precision and reduce resource wastage.

Agentic systems also incorporate long-term planning capabilities Haidemariam (2025), allowing robots to optimize operations over extended time horizons. This includes scheduling tasks, managing energy consumption, and coordinating activities across multiple robots Sapkota et al. (2025a). Integration with predictive models and digital twins further enhances planning capabilities, enabling robots to anticipate future conditions and adapt their strategies accordingly San et al. (2026).

Energy efficiency and operational sustainability are important considerations in autonomous farming systems. Agentic robots optimize resource usage by minimizing redundant actions, reducing chemical inputs, and improving operational efficiency. For example, precision spraying systems target only affected areas, reducing pesticide usage and environmental impact Davcev et al. (2026).

5.3 Technical Challenges, Safety, and Future Directions in Agentic Agriculture

The transition from traditional precision agriculture toward agentic agricultural systems introduces significant opportunities for autonomy, scalability, and efficiency, while simultaneously exposing critical technical, safety, and governance challenges. As outlined in this survey, agentic systems function as tightly coupled, closed-loop ecosystems integrating perception, reasoning, planning, and execution in dynamic agricultural environments; however, their practical deployment remains limited by challenges in reliability, uncertainty, safety, data governance, system integration, and the high cost and operational complexity associated with deploying and maintaining such systems at scale. Addressing these challenges requires not only robust engineering solutions but also a forward-looking research agenda that bridges autonomy with domain knowledge, human expertise, and responsible system design. Figure 14 shows the challenges and proposes potential solutions.

5.3.1 System Reliability and Robustness Under Uncertainty

Ensuring reliable performance of agentic agricultural systems remains a fundamental challenge due to the inherently stochastic and heterogeneous nature of agricultural environments. Unlike controlled industrial systems, farms exhibit variability driven by weather dynamics, biological processes, soil heterogeneity, and temporal changes across growing seasons. Autonomous agents rely on multimodal sensing inputs from IoT sensors, UAVs, and satellite platforms, which are often noisy, incomplete, or temporally inconsistent, leading to uncertainty in perception and downstream decision-making.

Moreover, models trained on limited datasets frequently fail to generalize across crops, regions, and management practices, limiting robustness in deployment scenarios Joshi et al. (2023); Aderele et al. (2025). Addressing these challenges requires integrating uncertainty-aware AI methods Zhang et al. (2023), including probabilistic modelingPugazhendi et al. (2026), Bayesian inferenceKrause and Bokinala (2023), and confidence-aware learningYang et al. (2025), to explicitly quantify and manage prediction uncertainty. In addition, redundancy through multimodal sensing and cross-validation of data streams can improve reliability Aniruddha et al. (2025).

A promising future direction involves the integration of digital twins for systematic validation and stress testing of agent behaviors under diverse environmental conditions San et al. (2026). Coupled with continual learning and domain adaptation, such frameworks enable agentic systems to evolve across seasons and operational contexts, improving long-term robustness and reliability.

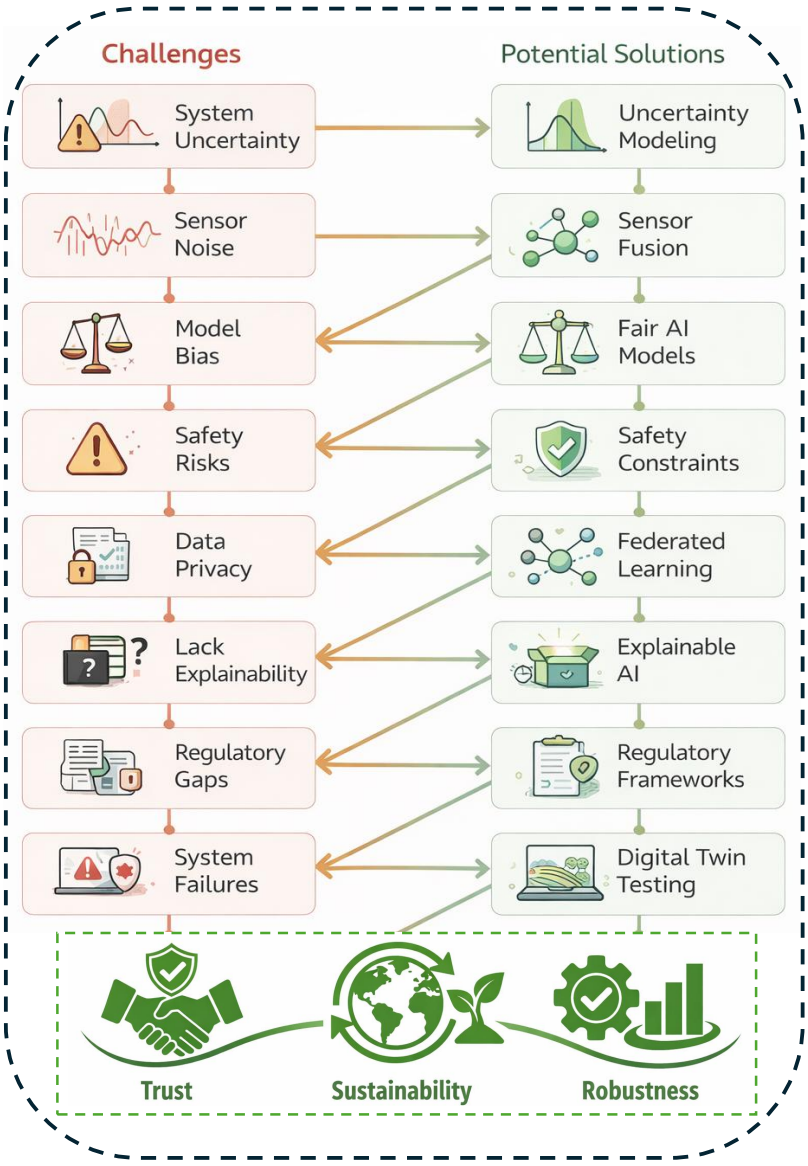


Fig. 14 Key challenges in agentic agriculture and corresponding strategies to address them. System uncertainty, sensor noise, and model bias degrade decision reliability, while safety risks, data privacy concerns, and lack of explainability hinder trustworthy deployment. Regulatory gaps and system failures further limit real-world adoption. These challenges are addressed through uncertainty modeling, sensor fusion, and fair AI models for robustness; safety constraints and federated learning for secure operation; explainable AI for transparency; regulatory frameworks for governance; and digital twin testing for safe validation of autonomous agricultural systems.

5.3.2 Safety and Human-Aware Autonomy in Agricultural Operations

Safety is a critical requirement for deploying autonomous systems in agriculture, particularly in environments where robots operate alongside humans, livestock, and sensitive crops. Unlike structured industrial settings, agricultural environments are unstructured and dynamic, increasing the risk of unintended interactions and system failures. Safety risks arise from perception errors, control failures, and coordination breakdowns among agents. For example, incorrect classification in spraying systems may lead to chemical misapplication, while navigation failures in robotic platforms can cause crop damage or human safety hazards. To mitigate these risks, safety-aware architectures must be embedded across system layers, including anomaly detection at the perception level, constraint-aware planning at the decision level, and fail-safe execution mechanisms.

In addition, runtime monitoring frameworks such as Trust, Risk, and Security Management (TRiSM) provide mechanisms for continuous risk assessment and policy enforcement Raza et al. (2026). Importantly, while full autonomy is a long-term objective, current systems must adopt

human-aware autonomy, where supervisory control and override mechanisms are retained. This perspective reconciles autonomy with human involvement, ensuring that systems remain controllable, interpretable, and aligned with real-world operational constraints.

5.3.3 Ethical, Socio-Technical, and Workforce Implications

The deployment of agentic AI in agriculture introduces broader socio-technical challenges related to fairness, transparency, accountability, and workforce transformation. Algorithmic bias remains a significant concern, as models trained on region-specific or high-input datasets may not generalize across diverse agricultural contexts, potentially disadvantaging smallholder farmers. Transparency and interpretability are equally important, as farmers must trust and understand system decisions, particularly in high-stakes scenarios such as pesticide application or crop management. The integration of explainable AI techniques is therefore essential to provide interpretable insights into decision processes and improve adoption.

Another critical dimension is workforce transformation Kalyanaraman et al. (2022). While agentic systems can address labor shortages, they may also displace traditional roles. Future systems should therefore emphasize human–AI collaboration, reskilling strategies, and inclusive deployment models that augment rather than replace human expertise, when appropriate. In this context, AI systems can also serve as repositories of accumulated agricultural knowledge, preserving experiential expertise from current generations of farmers and transferring it to future autonomous systems.

5.3.4 Data Governance, Privacy, and Interoperability

Agentic agriculture relies on large-scale data collection from distributed sensing systems, including environmental, operational, and economic data. This dependency raises significant concerns regarding data privacy, ownership, and governance. Farmers may be reluctant to adopt such systems without clear guarantees on data control and usage. To address these challenges, privacy-preserving learning approaches such as federated learningDembani et al. (2025), secure multi-party computationZhou et al. (2024), and differential privacy enabled collaborative model development without exposing raw dataGavai et al. (2025); Dembani et al. (2025). Additionally, blockchain-based frameworks have been proposed to support secure and transparent data sharing Kassanuk and Phasinam (2022); Nguyen et al. (2022). Equally important is the development of standardized data formats and interoperability protocols to avoid fragmentation across platforms. Without such standards, data silos can limit scalability and hinder the integration of multi-agent systems across agricultural ecosystems.

5.3.5 Regulatory Barriers and Deployment Constraints

The deployment of agentic systems is further limited by the lack of comprehensive regulatory frameworks addressing multi-agent autonomy in agriculture. Existing policies for UAVs and machinery do not fully capture the complexities of AI-driven, distributed decision-making systems. Key issues include liability in case of system failure, certification of autonomous systems, and compliance with environmental regulations. For instance, determining responsibility for actions taken by autonomous agents remains an open challenge. Addressing these issues requires the development of standardized validation and certification protocols, including safety benchmarks and performance evaluation frameworks. Collaboration between researchers, industry, and policymakers is essential to establish regulatory frameworks that support innovation while ensuring safety, accountability, and public trust Choudhury and Avvari (2025); Hughes et al. (2025).

5.3.6 Future Directions: Toward Trustworthy and Knowledge-Embedded Autonomy

Future research in agentic agriculture must move toward trustworthy, adaptive, and knowledge-embedded autonomous systems. This includes integrating explainability, uncertainty awareness, and human oversight directly into system architectures rather than treating them as add-on components. A key direction involves embedding agronomic knowledge and human expertise into agentic systems through digital twins, knowledge bases, and learning frameworks, enabling systems to operate autonomously while leveraging accumulated domain knowledge. This approach bridges the gap between human-centered decision-making and autonomous operation, reframing autonomy not as the absence of humans, but as the integration of human knowledge and some degree of oversight into system intelligence.

Ultimately, the future of agentic agriculture lies in developing systems that are not only autonomous but also trustworthy, adaptive, and aligned with agronomic, environmental, and societal objectives. By addressing the challenges outlined in this section, agentic systems can evolve into robust, scalable, and responsible agricultural solutions that support long-term sustainability and resilience.

6 Conclusion

This survey provided a structured synthesis of the evolution of intelligent agricultural systems, highlighting the transition from model-centric AI to integrated, goal-driven Agentic AI frameworks in precision agriculture. This shift reflects not only technological advancement but a broader architectural transformation toward systems that unify perception, reasoning, coordination, and action in dynamic environments.

To guide this analysis, four research questions were posed, addressing (RQ1) foundational AI agents, (RQ2) multi-agent collaboration, (RQ3) Agentic AI for autonomous decision-making, and (RQ4) future challenges and directions. In response to RQ1, the review showed that early AI agents enabled core agricultural tasks such as sensing, decision support, and robotic operations, supporting data-driven and site-specific management but remaining largely task-specific. For RQ2, multi-agent systems were found to extend these capabilities by enabling distributed farm intelligence, coordinated robotics, and adaptive management through IoT-agent architectures, swarm systems, and multi-agent learning. In addressing RQ3, Agentic AI emerged as a key paradigm shift, where agents evolve into goal-oriented, reasoning-driven systems capable of planning and long-horizon decision-making. The integration of digital twins, multimodal sensing, predictive modeling, and knowledge-based reasoning enables simulation-driven and proactive farm management strategies.

For RQ4, the survey identified persistent challenges in achieving reliable, scalable, and trustworthy deployment, including system robustness, interoperability, data governance, cost and complexity, and the need for standardized evaluation frameworks. The balance between autonomy and human-centered design further underscores the importance of embedding domain expertise and experiential knowledge into agentic systems. A key finding of this work is that the value of Agentic AI lies in integrating existing precision agriculture technologies into cohesive, closed-loop systems rather than replacing them. Emerging LLMs and VLMs further enhance decision-support capabilities by combining multimodal data with extensive agronomic knowledge.

Future research should focus on uncertainty-aware and knowledge-embedded architectures, scalable multi-agent coordination, and integration of data-driven and process-based models such as crop growth systems. Overall, agentic agriculture represents a promising pathway toward robust, adaptive, and sustainable farming systems capable of addressing complex global agricultural challenges.

Author contributions statement

Ranjan Sapkota: Conceptualization, Methodology, Literature Search, writing original draft, Vizualization, review and editing. **Manoj Karkee:** Review, critical feedback, Editing and Overall Funding to Supervisory.

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Declarations

The authors declare no conflicts of interest.

References

Abou Ali, M., Dornaika, F., Charafeddine, J.: Agentic ai: a comprehensive survey of architectures, applications, and future directions. Artificial Intelligence Review **59**(1), 11 (2025)

Aremu, M.B., Ahmed, G., Elferik, S., Saif, A.-W.A.: Autonomous mobile robot path planning techniques—a review: Metaheuristic and cognitive techniques. Robotics **15**(1), 23 (2026)

Assiri, A.S., Alsheref, F.k., Ghamdi, M.A., Abbassy, M.: Madrl-si: a multi-agent deep reinforcement learning-based swarm intelligence approach for coordinating multiple uavs in smart agriculture systems. International Journal of Information Technology, 1–7 (2025)

Abdulameer, L., Al-Khafaji, M.S., Al-Awadi, A.T., Al Maimuri, N.M., Al-Shammari, M., Al-Dujaili, A.N.: Artificial intelligence in climate-resilient water management: A systematic review of applications, challenges, and future directions. Water Conservation Science and Engineering **10**(1), 44 (2025)

Abro, G.E.M., Ali, Z.A., Masood, R.J.: Synergistic uav motion: A comprehensive review on advancing multi-agent coordination. ICCK Transactions on Sensing, Communication, and Control **1**(2), 72–88 (2024)

Aristeidis, K., Anastasios, G., Natalia, A., Christos, K.: Next-gen explainable ai (xai) for federated and distributed internet of things systems: A state-of-the-art survey. Future Internet **18**(2), 83 (2026)

Altalak, M., Ammad uddin, M., Alajmi, A., Rizg, A.: Smart agriculture applications using deep learning technologies: A survey. Applied Sciences **12**(12), 5919 (2022)

Arguenon, V., Bergues-Lagarde, A., Rosenberger, C., Bro, P., Smari, W.: Multi-agent based prototyping of agriculture robots. In: International Symposium on Collaborative Technologies and Systems (CTS'06), pp. 282–288 (2006). IEEE

Akkem, Y., Biswas, S.K., Varanasi, A.: Streamlit-based enhancing crop recommendation systems with advanced explainable artificial intelligence for smart farming. Neural Computing and Applications **36**(32), 20011–20025 (2024)

Albrecht, S.V., Christianos, F., Schäfer, L.: Multi-agent Reinforcement Learning: Foundations and Modern Approaches. MIT Press, ??? (2024)

Adikari, S.: Using a dynamic smarm of intelligent agents for advising farmers-agroagent. In: 2010 The 2nd International Conference on Computer and Automation Engineering (ICCAE), vol. 1, pp. 177–181 (2010). IEEE

Ajagekar, A., Decardi-Nelson, B., You, F.: Energy management for demand response in networked greenhouses with multi-agent deep reinforcement learning. Applied Energy **355**, 122349 (2024)

Ali, Z.A., Deng, D., Shaikh, M.K., Hasan, R., Khan, M.A.: Ai-based uav swarms for monitoring and disease identification of brassica plants using machine learning: A review. Computer Systems Science & Engineering **48**(1) (2024)

Akintan, O.A., Gebremedhin, K.G., Uyeh, D.D.: Linking animal feed formulation to milk quantity, quality, and animal health through data-driven decision-making. Animals **15**(2), 162 (2025)

Afrin, M., Jin, J., Rahman, A., Rahman, A., Wan, J., Hossain, E.: Resource allocation and service provisioning in multi-agent cloud robotics: A comprehensive survey. IEEE Communications Surveys & Tutorials **23**(2), 842–870 (2021)

Ao, J., Ji, W., Yu, X., Ruan, C., Xu, B.: End-effectors for fruit and vegetable harvesting robots: A review of key technologies, challenges, and future prospects. Agronomy **15**(11), 2650 (2025)

Anchekov, M.I., Kurashev, Z.K.: Multi-agent modeling in plant biology. News of the Kabardino-Balkarian Scientific Center of the Russian Academy of Sciences **27**(5), 26–33 (2025)

Aarif KO, M., Alam, A., Hotak, Y.: Smart sensor technologies shaping the future of precision agriculture: Recent advances and future outlooks. Journal of Sensors **2025**(1), 2460098 (2025)

Akerkar, R.: Ai agents: From perception to action. In: Artificial Intelligence: Transcending Traditional Paradigms, pp. 303–355. Springer, ??? (2026)

Ahmad, S., Kaushik, R., Ghatuary, R., Kotiyal, A., Jarial, S., Kumar, R.: Utilizing iot and ai for soil health monitoring and enhancement in sustainable agriculture. In: IoT and Advanced Intelligence Computation for Smart Agriculture, pp. 110–125. CRC Press, ??? (2025)

Akintuyi, O.B.: Adaptive ai in precision agriculture: a review: investigating the use of self-learning algorithms in optimizing farm operations based on real-time data. Research Journal of Multidisciplinary Studies **7**(02), 016–030 (2024)

Albahar, M.: A survey on deep learning and its impact on agriculture: challenges and opportunities. Agriculture **13**(3), 540 (2023)

Álvarez-López, C., González-Briones, A., Li, T.: Explainable ai and multi-agent systems for energy management in iot-edge environments: A state of the art review. Electronics **15**(2), 385 (2026)

Alotibi, Y.S.: A socio-technical agent-based simulation model for predicting smart agriculture adoption dynamics. Scientific Reports **15**(1), 43536 (2025)

Alqudsi, Y., Makaraci, M.: Exploring advancements and emerging trends in robotic swarm coordination and control of swarm flying robots: A review. Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science **239**(1), 180–204 (2025)

AlZu'bi, S., Mughaid, A.: The digital farmer agents in precise and sustainable agricultural ecosystems. In: 2025 3rd International Conference on Foundation and Large Language Models (FLLM), pp. 202–209 (2025). IEEE

Ahmad, T., Morel, A., Cheng, N., Palaniappan, K., Calyam, P., Sun, K., Pan, J.: Future uav/drone systems for intelligent active surveillance and monitoring. ACM Computing Surveys **58**(2), 1–37 (2025)

Alves, R.G., Maia, R.F., Lima, F.: Development of a digital twin for smart farming: Irrigation management system for water saving. Journal of Cleaner Production **388**, 135920 (2023)

Ali, Z., Muhammad, A., Lee, N., Waqar, M., Lee, S.W.: Artificial intelligence for sustainable agriculture: A comprehensive review of ai-driven technologies in crop production. Sustainability **17**(5), 2281 (2025)

Aryan, K., Mishra, S., Patel, S.K., Kaushik, S., Singh, K.: Ai-powered integrated platform for farmer support: real-time disease diagnosis, precision irrigation advisory, and expert consultation services. Journal of Scientific Innovation and Advanced Research (JSIAR) **1**(2), 222–229 (2025)

Anjomshoae, S., Najjar, A., Calvaresi, D., Främling, K.: Explainable agents and robots: Results from a systematic literature review. In: 18th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2019), Montreal, Canada, May 13–17, 2019, vol. 2, pp. 1078–1088 (2019). International Foundation for Autonomous Agents and Multiagent Systems

Alva, L., Pandey, B.: Agentic ai systems in the age of generative models: architectures, cloud scalability, and real-world applications. Artificial Intelligence Review (2026)

Agilandeeswari, L., Prabukumar, M., Radhesyam, V., Phaneendra, K.L.B., Farhan, A.: Crop classification for agricultural applications in hyperspectral remote sensing images. Applied Sciences **12**(3), 1670 (2022)

Adinarayana, S., Raju, M.G., Srirangam, D.P., Prasad, D.S., Kumar, M.R., veesam, S.b.: Enhancing resource management in precision farming through ai-based irrigation optimization. How Machine Learning is Innovating Today's World: A Concise Technical Guide, 221–251 (2024)

Aziz, D., Rafiq, S., Saini, P., Ahad, I., Gonal, B., Rehman, S.A., Rashid, S., Saini, P., Rohela, G.K., Aalum, K., *et al.*: Remote sensing and artificial intelligence: revolutionizing pest management in agriculture. Frontiers in Sustainable Food Systems **9**, 1551460 (2025)

Aderele, M.O., Srivastava, A.K., Butterbach-Bahl, K., Rahimi, J.: Integrating machine learning with agroecosystem modelling: Current state and future challenges. European Journal of Agronomy **168**, 127610 (2025)

Ahmed, I., Syed, M.A., Maaruf, M., Khalid, M.: Distributed computing in multi-agent systems: a survey of decentralized machine learning approaches. Computing **107**(1), 2 (2025)

Abar, S., Theodoropoulos, G.K., Lemarinier, P., O'Hare, G.M.: Agent based modelling and simulation tools: A review of the state-of-art software. Computer Science Review **24**, 13–33 (2017)

Arsenoaia, V.N., Topa, D.C., Ratu, R.N., Tenu, I.: From sensing to intervention: A critical review of agricultural drones for precision agriculture, data-driven decision making, and sustainable intensification. Agronomy **16**(5), 564 (2026)

Aniruddha, P., Valiyaparambil, A.S., Sornalakshmi, K.: Evaluating multimodal fusion strategies for resilient agricultural sensing systems. In: International Conference on Data Science and Applications, pp. 249–259 (2025). Springer

Anbari, M.J., Zarghami, M., Nadiri, A.-A.: An uncertain agent-based model for socio-ecological simulation of groundwater use in irrigation: A case study of lake urmia basin, iran. Agricultural water management **249**, 106796 (2021)

Babenko, Y.: Trends and innovations in modern agriculture: from fluorescent sensors to decision support systems. Academia Engineering **3**(1) (2026)

Bayar, J., Ali, N., Cao, Z., Ren, Y., Dong, Y.: Artificial intelligence of things (aiot) for precision agriculture: Applications in smart irrigation, nutrient and pest management. Smart Agricultural Technology, 101629 (2025)

Badage, A.: Crop disease detection using machine learning: Indian agriculture. Int. Res. J. Eng. Technol **5**(9), 866–869 (2018)

Balaska, V., Adamidou, Z., Vryzas, Z., Gasteratos, A.: Sustainable crop protection via robotics and artificial intelligence solutions. Machines **11**(8), 774 (2023)

Boubin, J., Burley, C., Han, P., Li, B., Porter, B., Stewart, C.: Marble: Multi-agent reinforcement learning at the edge for digital agriculture. In: 2022 IEEE/ACM 7th Symposium on Edge Computing (SEC), pp. 68–81 (2022). IEEE

Barreteau, O., Bousquet, F., Millier, C., Weber, J.: Suitability of multi-agent simulations to study irrigated system viability: application to case studies in the senegal river valley. Agricultural systems **80**(3), 255–275 (2004)

Barnaud, C., Bousquet, F., Trebuil, G.: Multi-agent simulations to explore rules for rural credit in a highland farming community of northern thailand. Ecological Economics **66**(4), 615–627 (2008)

Bariah, L., Debbah, M.: The interplay of ai and digital twin: Bridging the gap between data-driven and model-driven approaches. IEEE Wireless Communications **31**(3), 219–225 (2024)

Beni, G.: Swarm intelligence. In: Complex Social and Behavioral Systems: Game Theory and Agent-based Models, pp. 791–818. Springer, ??? (2020)

Berger, T.: Agent-based spatial models applied to agriculture: a simulation tool for technology

diffusion, resource use changes and policy analysis. Agricultural economics **25**(2-3), 245–260 (2001)

Bal, S., Gunsola, D., Kukreti, R., Ballesteros, L.M.S., Acosta, N.K.S., Korgaonkar, S.N., Boutaj, H.: Data-driven pest management: Harnessing ai for precision control. In: Advancements in Entomology: Bridging Forensic Science and Sustainable Agriculture, pp. 381–421. Springer, ??? (2026)

Bandi, A., Kongari, B., Naguru, R., Pasnoor, S., Vilipala, S.V.: The rise of agentic ai: A review of definitions, frameworks, architectures, applications, evaluation metrics, and challenges. Future Internet **17**(9), 404 (2025)

Blackmore, S.: Precision farming: an introduction. Outlook on agriculture **23**(4), 275–280 (1994)

Betalo, M.L., Leng, S., Abishu, H.N., Dharejo, F.A., Seid, A.M., Erbad, A., Naqvi, R.A., Zhou, L., Guizani, M.: Multi-agent deep reinforcement learning-based task scheduling and resource sharing for o-ran-empowered multi-uav-assisted wireless sensor networks. IEEE Transactions on Vehicular Technology **73**(7), 9247–9261 (2023)

Bousquet, F., Le Page, C.: Multi-agent simulations and ecosystem management: a review. Ecological modelling **176**(3-4), 313–332 (2004)

Badillo-Márquez, A.E., Aguilar-Lasserre, A.A., Miranda-Ackerman, M.A., Sandoval-González, O.O., Villanueva-Vásquez, D., Posada-Gómez, R.: An agent-based model-driven decision support system for assessment of agricultural vulnerability of sugarcane facing climatic change. Mathematics **9**(23), 3061 (2021)

Brohi, S., Mastoi, Q.-u.-a., Jhanjhi, N., Pillai, T.R.: A research landscape of agentic ai and large language models: Applications, challenges and future directions. Algorithms **18**(8), 499 (2025)

Banerjee, S., Palsani, D., Mondal, A.C.: Nutritional content detection using vision transformers-an intelligent approach. Int. J. Innov. Res. Eng. Manag **11**(6) (2024)

Bossens, D.M., Ramchurn, S., Tarapore, D.: Resilient robot teams: a review integrating decentralised control, change-detection, and learning. Current Robotics Reports **3**(3), 85–95 (2022)

Berger, T., Schreinemachers, P.: Agent-based land use models for collaborative learning about agri-environmental policies. In: New Perspectives on Agri-environmental Policies, pp. 157–169. Routledge, ??? (2009)

Berger, T., Troost, C.: Agent-based modelling of climate adaptation and mitigation options in agriculture. Journal of Agricultural Economics **65**(2), 323–348 (2014)

Choudhury, A., Avvari, V.: Regulatory frameworks for autonomous ai: Balancing innovation and safety. In: The Power of Agentic AI: Redefining Human Life and Decision-Making: In Industry 6.0, pp. 273–289. Springer, ??? (2025)

Castelfranchi, C.: Modelling social action for ai agents. Artificial intelligence **103**(1-2), 157–182 (1998)

Cantonjos, N.A.P., Biswas, A.: Agroaskai: A multi-agentic ai framework for supporting smallholder farmers' enquiries globally. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 40, pp. 40217–40223 (2026)

Chreim, A., Belarouci, A., Merzouki, R.: Ai-agent-based modeling for supervision a system of systems for mushroom harvesting. In: 2023 18th Annual System of Systems Engineering Conference (SoSe), pp. 1–6 (2023). IEEE

Chevalier, A., Copot, C., De Keyser, R., Hernandez, A., Ionescu, C.: A multi agent system for precision agriculture. In: Handling Uncertainty and Networked Structure in Robot Control, pp. 361–386. Springer, ??? (2016)

Chaudhary, M., Jabborova, D.: Smart sensing technologies: Monitoring insect pests with ai. In: Advancements in Entomology: Bridging Forensic Science and Sustainable Agriculture, pp. 423–440. Springer, ??? (2026)

Chowhan, R.S., Kumari, S., Karel, A.: E-green revolution: intentional pollination using intelligent agent based autonomous drone-pollinator-bees. Int. J. Pure Appl. Biosci. **5**, 1139–1142 (2017)

Cavaliere, D., Senatore, S.: A multi-agent knowledge-enhanced model for decision-supporting agro-forestry systems. In: 2021 IEEE Symposium Series on Computational Intelligence (SSCI), pp. 01–08 (2021). IEEE

Cortès, U., Sànchez-Marrè, M., Ceccaroni, L., R-Roda, I., Poch, M.: Artificial intelligence and environmental decision support systems. Applied intelligence **13**(1), 77–91 (2000)

Chen, J., Seng, K.P., Smith, J., Ang, L.-M.: Situation awareness in ai-based technologies and multimodal systems: Architectures, challenges and applications. IEEE Access **12**, 88779–88818 (2024)

Castella, J.-C., Trung, T.N., Boissau, S.: Participatory simulation of land-use changes in the northern mountains of vietnam: the combined use of an agent-based model, a role-playing game, and a geographic information system. Ecology and Society **10**(1) (2005)

Chan, A., Wei, K., Huang, S., Rajkumar, N., Perrier, E., Lazar, S., Hadfield, G.K., Anderljung, M.: Infrastructure for ai agents. arXiv preprint arXiv:2501.10114 (2025)

Celis, J., Xiao, X., Wagle, P., Adler, P.R., White, P.: A review of yield forecasting techniques and their impact on sustainable agriculture. Transformation Towards Circular Food Systems, 139–168 (2024)

Carbo, J., Zubelzu, S., Molina, J.: An agents-based tool for simulating farmers operation in irrigation districts. Computers and Electronics in Agriculture **245**, 111518 (2026)

Dazzi, P.: The internet of ai agents (iaia): A new frontier in networked and distributed intelligence. International Journal of Networked and Distributed Computing **13**(1), 16 (2025)

Dutta, S.K., Bhutia, B., Misra, T., Mishra, V., Singh, S., Patel, V.: Application and prospects of artificial intelligence (ai)-based technologies in fruit production systems. Applied Fruit Science **67**(1), 16 (2025)

Dao, S.D., Gostar, A.K., Tennakoon, R., Chuah, W.Q., Fellows, P., Bab-Hadiashar, A.: Modernizing livestock operations: Review of smart feedlot technologies and their impact. Authorea Preprints (2025)

Dwivedi, Y.K., Helal, M.Y., Elgendy, I.A., Alahmad, R., Walton, P., Suh, A., Singh, V., Jeon, I.: Agentic ai systems: What it is and isn't. Global Business and Organizational Excellence **45**(3), 253–263 (2026)

Dhal, S.B., Kar, D.: Transforming agricultural productivity with ai-driven forecasting: Innovations in food security and supply chain optimization. Forecasting **6**(4), 925–951 (2024)

Dembani, R., Karvelas, I., Akbar, N.A., Rizou, S., Tegolo, D., Fountas, S.: Agricultural data privacy and federated learning: A review of challenges and opportunities. Computers and Electronics in Agriculture **232**, 110048 (2025)

Davcev, D., Kalajdziski, S., Dimitrovski, I., Kitanovski, I., Mitreski, K.: Agentic ai-based iot precision agriculture framework—our vision and challenges. AgriEngineering **8**(4), 147 (2026)

Dhanaselvam, P.S., Karthikeyan, B., Kumar, P., Salama, H., Nasri, F., Kavitha, K.: Artificial intelligence in sensor technology. In: Adaptive AI in Sensor Informatics, pp. 1–23. Elsevier, ??? (2026)

Dusadeerungsikul, P.O., Nof, S.Y.: Precision agriculture with ai-based responsive monitoring algorithm. International Journal of Production Economics **271**, 109204 (2024)

Dong, M., Ota, K., Yang, L.T., Chang, S., Zhu, H., Zhou, Z.: Mobile agent-based energy-aware and user-centric data collection in wireless sensor networks. Computer networks **74**, 58–70 (2014)

De Silva, R., Cielniak, G., Wang, G., Gao, J.: Deep learning-based crop row detection for infield navigation of agri-robots. Journal of field robotics **41**(7), 2299–2321 (2024)

Delfani, P., Thuraga, V., Banerjee, B., Chawade, A.: Integrative approaches in modern agriculture: Iot, ml and ai for disease forecasting amidst climate change. Precision Agriculture **25**(5), 2589–2613 (2024)

EMARA, E.I., Al-SAEED, A.M., HAMED, L.M.: Ai-driven decision support system enhances productivity, water use efficiency, and soil sustainability of strategic crops in sandy soils. Agricultural Water Management **325**, 110198 (2026)

Emi-Johnson, O., Fasanya, O., Adeniyi, A.: Explainable ai for pesticide decision-making: Enhancing trust in datadriven crop protection models. International Journal of Computer Applications Technology and Research **14**(01), 147–161 (2025)

El Jarroudi, M., Kouadio, L., Delfosse, P., Bock, C.H., Mahlein, A.-K., Fettweis, X., Mercatoris, B., Adams, F., Lenné, J.M., Hamdioui, S.: Leveraging edge artificial intelligence for sustainable agriculture. Nature Sustainability **7**(7), 846–854 (2024)

Essakkat, K., Wu, L., Atallah, S.S., Khanna, M.: Spatial-dynamic adoption of ai weeding robots: Insights from a choice experiment and an agent-based model. UMN (2025)

Eiffert, S., Wallace, N.D., Kong, H., Pirmarzdashti, N., Sukkarieh, S.: Resource and response aware path planning for long-term autonomy of ground robots in agriculture. Field Robotics **2**, 1–33 (2022)

El Yasmine, A.S.L., Ghani, B.A., Trentesaux, D., Bouziane, B.: Supply chain management using multi-agent systems in the agri-food industry. In: Service Orientation in Holonic and Multi-Agent Manufacturing and Robotics, pp. 145–155. Springer, ??? (2014)

Fountas, S., Carli, G., Sørensen, C.G., Tsiropoulos, Z., Cavalaris, C., Vatsanidou, A., Liakos, B., Canavari, M., Wiebensohn, J., Tisserye, B.a.: Farm management information systems: Current situation and future perspectives. Computers and electronics in agriculture **115**, 40–50 (2015)

Ficili, I., Giacobbe, M., Tricomi, G., Puliafito, A.: From sensors to data intelligence: Leveraging iot, cloud, and edge computing with ai. Sensors **25**(6), 1763 (2025)

FUKATSU, T., HIRAFUJI, M., KIURA, T.: A distributed agent system for managing a web-based sensor network with field servers. In: Computers in Agriculture and Natural Resources, 23-25 July 2006, Orlando Florida, p. 223 (2006). American Society of Agricultural and Biological Engineers

Fukatsu, T., Hirafuji, M., Kiura, T., *et al.*: Agent system for operating web-based sensor nodes via the internet. Journal of Robotics and Mechatronics **18**(2), 186 (2006)

Fountas, S., Malounas, I., Athanasakos, L., Avgoustakis, I., Espejo-Garcia, B.: Ai-assisted vision for agricultural robots. AgriEngineering **4**(3), 674–694 (2022)

Fuentes-Peñailillo, F., Gutter, K., Vega, R., Silva, G.C.: Transformative technologies in digital agriculture: Leveraging internet of things, remote sensing, and artificial intelligence for smart crop management. Journal of Sensor and Actuator Networks **13**(4), 39 (2024)

Franklin, S.: Autonomous agents as embodied ai. Cybernetics & Systems **28**(6), 499–520 (1997)

Fang, M., Tan, Z., Tang, Y., Chen, W., Huang, H., Dananjayan, S., He, Y., Luo, S.: Pest-conformer: A hybrid cnn-transformer architecture for large-scale multi-class crop pest recognition. Expert

Systems with Applications **255**, 124833 (2024)

Feng, Z., Zhang, R., Qin, X., Xing, C., Zhao, N., Niyato, D., Al-Dhahir, N.: Toward intelligent and secure satellite communications: Agentic ai for covert semantics. IEEE Wireless Communications (2026)

Gupta, S., Arora, S., Qamar, S.: Federated learning in smart farming: Applications and challenges. Convergence of AI, Federated Learning, and Blockchain for Sustainable Development, 121–146 (2025)

Garg, V.: Designing the mind: How agentic frameworks are shaping the future of ai behavior. Journal of Computer Science and Technology Studies **7**(5), 182–193 (2025)

González-Briones, A., Castellanos-Garzón, J.A., Mezquita-Martín, Y., Prieto, J., Corchado, J.M.: A multi-agent system framework for autonomous crop irrigation. In: 2019 2nd International Conference on Computer Applications & Information Security (ICCAIS), pp. 1–6 (2019). IEEE

Gavai, A.K., Bouzembrak, Y., Xhani, D., Sedrakyan, G., Meuwissen, M.P., Souza, R.G.-S., Marvin, H.J., Hillegersberg, J.: Agricultural data privacy: Emerging platforms & strategies. Food and Humanity **4**, 100542 (2025)

Gutierrez Cejudo, J., Enguix Andres, F., Lujak, M., Carrascosa Casamayor, C., Fernandez, A., Hernandez Lopez, L.: Towards agrirobot digital twins: Agri-ro5—a multi-agent architecture for dynamic fleet simulation. Electronics **13**(1), 80 (2023)

Gutiérrez-Cejudo, J., Lujak, M., Fernández, A.: Agrobots architecture and agrobots-sim simulator for dynamic agri-robot coordination. In: International Conference on Practical Applications of Agents and Multi-Agent Systems, pp. 5–17 (2023). Springer

Gronauer, S., Diepold, K.: Multi-agent deep reinforcement learning: a survey. Artificial Intelligence Review **55**(2), 895–943 (2022)

Gonzalez-de-Santos, P., Fernández, R., Sepúlveda, D., Navas, E., Emmi, L., Armada, M.: Field robots for intelligent farms—inhering features from industry. Agronomy **10**(11), 1638 (2020)

Gabriel Filho, L.R., Cremasco, C.P., Putti, F.F., Chacur, M.G.: Application of fuzzy logic for the evaluation of livestock slaughtering. Engenharia Agrícola **31**(4), 813–825 (2011)

Gao, Y., Jiang, Q., Wang, M., Dong, X.: Advances in path-planning algorithms for agricultural robots. Journal of Field Robotics **43**(1), 89–145 (2026)

Gujjar, J., Kumar, H.: Agri friendly conversational ai chatbot using open source framework. Indian Journal of Science and Technology **18**(8), 580–585 (2025)

Guo, W., Li, W., Zhong, Y., Lodewijks, G., Shen, W.: Agent-based negotiation framework for agricultural supply chain supported by third party logistics. In: 2016 IEEE 20th International Conference on Computer Supported Cooperative Work in Design (CSCWD), pp. 584–589 (2016). IEEE

Griepentrog, H.W., Stein, A.: Comparison of robot concepts for new sustainable crop production systems. Smart Agricultural Technology **8**, 100499 (2024)

Goyal, I., Sai, S., Pabbati, M., Chamola, V., Ranjan, V.: Empowering digital twins with agentic ai: Applications, case studies, and limitations. IEEE Communications Standards Magazine (2026)

Gupta, S.K.: Ai-augmented sustainability management in agri-food supply chain: rethinking decision-making in the era of algorithms. Journal of Agribusiness in Developing and Emerging Economies, 1–16 (2026)

Gao, X., Wang, Y., Liu, B., Zhou, X., Zhang, R., Wang, J., Niyato, D., Kim, D.I., Jamalipour, A., Yuen, C., et al.: Agentic satellite-augmented low-altitude economy and terrestrial networks: A survey on generative approaches. IEEE Communications Surveys & Tutorials (2026)

Haidemariam, T.: From the logic of coordination to goal-directed reasoning: the agentic turn in artificial intelligence. Frontiers in Artificial Intelligence **8**, 1728738 (2025)

Happe, K.: Agricultural Policies and Farm Structures: Agent-based Modelling and Application to EU-policy Reform vol. 30. Studies on the Agricultural and Food Sector in Central and Eastern Europe, ??? (2004)

Hughes, L., Dwivedi, Y.K., Malik, T., Shawosh, M., Albashrawi, M.A., Jeon, I., Dutot, V., Appanderanda, M., Crick, T., De', R., *et al.*: Ai agents and agentic systems: A multi-expert analysis. Journal of Computer Information Systems **65**(4), 489–517 (2025)

Holzinger, A., Fister, I., Kaul, H.-P., Asseng, S.: Human-centered ai in smart farming: toward agriculture 5.0. IEEe Access **12**, 62199–62214 (2024)

Hernández Hernández, G.C., Gómez Gómez, J., Jiménez-Cabas, J.: Predictive models based on artificial intelligence to estimate crop yield: A literature review. Agriculture **15**(23), 2438 (2025)

Habiba, M., Khan, N.I.: Revisiting gossip protocols: A vision for emergent coordination in agentic multi-agent systems. arXiv preprint arXiv:2508.01531 (2025)

Happe, K., Kellermann, K., Balmann, A.: Agent-based analysis of agricultural policies: an illustration of the agricultural policy simulator agripolis, its adaptation and behavior. Ecology and society **11**(1) (2006)

Hu, X., Li, S., Huang, T., Tang, B., Huai, R., Chen, L.: How simulation helps autonomous driving: A survey of sim2real, digital twins, and parallel intelligence. IEEE Transactions on Intelligent Vehicles **9**(1), 593–612 (2023)

Han, H., Liu, Z., Li, J., Zeng, Z.: Challenges in remote sensing based climate and crop monitoring: navigating the complexities using ai. Journal of cloud computing **13**(1), 1–14 (2024)

Hanson, J., Liebig, M., Merrill, S., Tanaka, D., Krupinsky, J., Stott, D.: Dynamic cropping systems: increasing adaptability amid an uncertain future. Agronomy Journal **99**(4), 939–943 (2007)

Hernández, H.A., Mondragón, I.F., González, S.R., Pedraza, L.F.: Reconfigurable agricultural robotics: Control strategies, communication, and applications. Computers and Electronics in Agriculture **234**, 110161 (2025)

Heideker, A., Ottolini, D., Zyrianoff, I., Neto, A.T., Cinotti, T.S., Kamienski, C.: Iot-based measurement for smart agriculture. In: 2020 IEEE International Workshop on Metrology for Agriculture and Forestry (MetroAgriFor), pp. 68–72 (2020). IEEE

Hamadani, H., Rashid, S.M., Parrah, J., Khan, A., Dar, K., Ganie, A., Gazal, A., Dar, R., Ali, A.: Traditional farming practices and its consequences. In: Microbiota and Biofertilizers, Vol 2: Ecofriendly Tools for Reclamation of Degraded Soil Environs, pp. 119–128. Springer, ??? (2021)

Hosseini, S., Seilani, H.: The role of agentic ai in shaping a smart future: A systematic review. Array **26**, 100399 (2025)

Hu, M.-C., Tsang, K.-M.: Ai agent system for agricultural water resources management. In: IoT and AI in Agriculture: Strategies to Develop Smart Agricultural Space for In-field and In-house Crop Production, pp. 287–289. Springer, ??? (2026)

Huang, K.: Ai agents in healthcare. In: Agentic AI: Theories and Practices, pp. 303–321. Springer, ??? (2025)

Hung, F., Yang, Y.E.: Assessing adaptive irrigation impacts on water scarcity in nonstationary environments—a multi-agent reinforcement learning approach. Water Resources Research **57**(9), 2020–029262 (2021)

Hu, G., You, F.: Renewable energy-powered semi-closed greenhouse for sustainable crop production using model predictive control and machine learning for energy management. Renewable and Sustainable Energy Reviews **168**, 112790 (2022)

He, L., Yan, Y., Liu, J., Zhao, C.: Intelligent decision-making for field crop production. Tsinghua Science and Technology **31**(3), 1393–1410 (2026)

Indira, P., Arafat, I.S., Karthikeyan, R., Selvarajan, S., Balachandran, P.K.: Fabrication and investigation of agricultural monitoring system with iot & ai. SN Applied Sciences **5**(12), 322 (2023)

Ionescu, Ş., Delcea, C., Chiriţă, N., Nica, I.: Exploring the use of artificial intelligence in agent-based modeling applications: A bibliometric study. Algorithms **17**(1), 21 (2024)

Ishaaya, I., Kontsedalov, S., Mazirov, D., Horowitz, A.R.: Biorational agents: mechanism and importance in ipm and irm programs for controlling agricultural pests. Meded Rijksuniv Gent Fak Landbouwkd Toegep Biol Wet **66**(2a), 363–374 (2001)

Ishaq, R.A.F., Zhou, G., Jing, G., Shah, S.R.A., Ali, A., Imran, M., Jiang, H.: Geospatial robust wheat yield prediction using machine learning and integrated crop growth model and time-series satellite data. Remote Sensing **17**(7), 1140 (2025)

Jaggavarapu, M.K.R.: The evolution of agentic ai: architecture and workflows for autonomous systems. Journal Of Multidisciplinary **5**(7), 418–427 (2025)

Janssen, M.A.: Agent-based modelling. Modelling in ecological economics **155**(1), 172–181 (2005)

Jimenez, A.-F., Cardenas, P.-F., Jimenez, F., Ruiz-Canales, A., López, A.: A cyber-physical intelligent agent for irrigation scheduling in horticultural crops. Computers and electronics in agriculture **178**, 105777 (2020)

Jia, H., Chen, W., Su, Z., Sun, Y., Qian, Z., Huang, L.: Ai-driven cooperative control for autonomous tractors and implements: A comprehensive review. AgriEngineering **7**(11), 394 (2025)

Jha, K., Doshi, A., Patel, P., Shah, M.: A comprehensive review on automation in agriculture using artificial intelligence. Artificial Intelligence in Agriculture **2**, 1–12 (2019)

Joshi, A., Guevara, D., Earles, M.: Standardizing and centralizing datasets for efficient training of agricultural deep learning models. Plant Phenomics **5**, 0084 (2023)

Jayarathna, H., Hettige, B.: Wmac: Web-based multi-agent solution for agriculture community. SRI LANKA ASSOCIATION FOR ARTIFICIAL INTELLIGENCE, 25 (2013)

Ji, W., Huang, X., Wang, S., He, X.: A comprehensive review of the research of the "eye–brain–hand" harvesting system in smart agriculture. Agronomy **13**(9), 2237 (2023)

Jepsen, M.R., Leisz, S., Rasmussen, K., Jakobsen, J., MØller-Jensen, L., Christiansen, L.: Agent-based modelling of shifting cultivation field patterns, vietnam. International Journal of Geographical Information Science **20**(9), 1067–1085 (2006)

Jabed, M.A., Murad, M.A.A.: Crop yield prediction in agriculture: A comprehensive review of machine learning and deep learning approaches, with insights for future research and sustainability. Heliyon **10**(24) (2024)

Johnsen, M.: Agentic AI. Maria Johnsen, ??? (2025)

Joshi, S.: A comprehensive review of gen ai agents: Applications and frameworks in finance, investments and risk domains. International Journal of Innovative Science and Research Technology, 1339–1355 (2025)

Jia, L., Pei, Y.: Recent advances in multi-agent reinforcement learning for intelligent automation and

control of water environment systems. Machines **13**(6), 503 (2025)

Jamal, A., Raza, H., Erdenetsogt, T., Singh, A., Farooq, M., Kabeer, M.M., Aslam, M.S.: Ai and data analytics for precision agriculture: Current progress and future directions. JATAED: Journal of Appropriate Technology for Agriculture, Environment, and Development **2**(2), 36–46 (2025)

Jiao, Y., Sun, Z., Jin, Y., Cui, L., Zhang, X., Wang, S., Zhang, S., Chang, C., Ding, S., Xue, X.: Current status and future prospects of key technologies in variable-rate spray. Agriculture; Basel **15**(20) (2025)

Javed, K., Smagghe, G., Kang, Y.-Q., Wang, Q., Wang, Y.: Artificial intelligence in the mass production of natural enemies for biological control in modern agriculture. Pest management science **81**(12), 7577–7592 (2025)

Juroszek, P., Tiedemann, A.: Potential strategies and future requirements for plant disease management under a changing climate. Plant Pathology **60**(1), 100–112 (2011)

Jiang, L., Xu, B., Husnain, N., Wang, Q.: Overview of agricultural machinery automation technology for sustainable agriculture. Agronomy **15**(6), 1471 (2025)

Kiasari, M., Aly, H.: Agentic artificial intelligence for smart grids: A comprehensive review of autonomous, safe, and explainable control frameworks. Energies **19**(3), 617 (2026)

Kamel, A.: An agent-based framework for agricultural knowledge based systems. IFAC Proceedings Volumes **37**(2), 6–10 (2004)

Kannan, S.: Transforming Agriculture for the Digital Age: Integrating Artificial Intelligence, Cloud Computing, and Big Data Solutions for Sustainable and Smart Farming Systems. Deep Science Publishing, ??? (2025)

Kannan, P., Abishek, J., Pazhanivelan, S., Firnass, M.M.R.A.: Farming from above: the role of uavs in modern agricultural practices. In: Digital Farming and Smart Agriculture for Sustainable Future, pp. 235–254. Chapman and Hall/CRC, ??? (2025)

Krause, P.J., Bokinala, V.: A tutorial on data mining for bayesian networks, with a specific focus on iot for agriculture. Internet of Things **22**, 100738 (2023)

Katikaridis, D., Benos, L., Busato, P., Kateris, D., Papageorgiou, E., Karras, G., Bochtis, D.: Experimental comparative analysis of centralized vs. decentralized coordination of aerial–ground robotic teams for agricultural operations. Robotics **14**(9), 119 (2025)

Khriji, S., Benbelgacem, Y., Chéour, R., Houssaini, D.E., Kanoun, O.: Design and implementation of a cloud-based event-driven architecture for real-time data processing in wireless sensor networks. The Journal of Supercomputing **78**(3), 3374–3401 (2022)

Kalyanaraman, A., Burnett, M., Fern, A., Khot, L., Viers, J.: Special report: The agaid ai institute for transforming workforce and decision support in agriculture. Computers and Electronics in Agriculture **197**, 106944 (2022)

Kaye-Blake, W., Li, F.Y., Martin, A., McDermott, A., Rains, S., Sinclair, S., Kira, A.: Multi-agent simulation models in agriculture: a review of their construction and uses. Lincoln University. Agribusiness and Economics Research Unit (2010)

Kalyani, Y., Collier, R.: A systematic survey on the role of cloud, fog, and edge computing combination in smart agriculture. Sensors **21**(17), 5922 (2021)

Karetsos, S., Costopoulou, C., Lambrou, M., Zorkadis, V., Authority, H.D.P.: The use of intelligent agents in the e-agribusiness environment. HAICTA, Athens, 19–26 (2002)

Komarek, A.M., De Pinto, A., Smith, V.H.: A review of types of risks in agriculture: What we know and what we need to know. Agricultural systems **178**, 102738 (2020)

Khemakhem, F., Ellouzi, H., Ltifi, H., Ayed, M.B.: Agent-based intelligent decision support systems: a systematic review. IEEE Transactions on Cognitive and Developmental Systems **14**(1), 20–34 (2020)

KP, A.R., Gowrishankar, S.: Agentic ai for pathogen-based plant disease detection. In: 2025 6th International Conference on Artificial Intelligence, Robotics and Control (AIRC), pp. 323–328 (2025). IEEE

Khan, M.N.: Edge ai based on real-time robotic decision-making in resource-constrained environment. International Journal of Advanced and Innovative Research (IJAIR) **1**(2), 19–26 (2025)

Khaliq, A., Khan, A., Jan, S., Umair, M., Gulshair, A., Ali, A., Shah, U.A.: Ai-driven smart agriculture: An integrated approach for soil analysis, irrigation, and crop-fertilizer recommendations. IEEE Access (2025)

Kazanskiy, N., Khabibullin, R., Nikonorov, A., Khonina, S.: A comprehensive review of remote sensing and artificial intelligence integration: advances, applications, and challenges. Sensors **25**(19), 5965 (2025)

Khatri, P., Kumar, P., Shakya, K.S., Kirlas, M.C., Tiwari, K.K.: Understanding the intertwined nature of rising multiple risks in modern agriculture and food system. Environment, Development and Sustainability **26**(9), 24107–24150 (2024)

Karunathilake, E., Le, A.T., Heo, S., Chung, Y.S., Mansoor, S.: The path to smart farming: Innovations and opportunities in precision agriculture. Agriculture **13**(8), 1593 (2023)

Kumari, K., Mirzakhani Nafchi, A., Mirzaee, S., Abdalla, A.: Ai-driven future farming: achieving climate-smart and sustainable agriculture. AgriEngineering **7**(3), 89 (2025)

Kassanuk, T., Phasinam, K.: Design of blockchain based smart agriculture framework to ensure safety and security. Materials Today: Proceedings **51**, 2313–2316 (2022)

Kamilaris, A., Prenafeta-Boldú, F.X.: Deep learning in agriculture: A survey. Computers and electronics in agriculture **147**, 70–90 (2018)

Kandamali, D.F., Porter, W.M., Porter, E., McLemore, A., Rains, G.C.: Cottonbot: An ai-driven cotton farming assistant and irrigation advisor using llm-rag and agentic ai tools. Smart Agricultural Technology, 101640 (2025)

Kale, V.P., Rao, J.M.: Multi agent approach for climate resilient fertilizer recommendations. In: 2025 International Conference on Future Technologies (ICFT), pp. 1–8 (2025). IEEE

Kaushik, S., Singh, K.: Ai-driven smart irrigation and resource optimization for sustainable precision agriculture. Europe **250**, 70 (2025)

Kumar, G.K., Sankar, P.: Artificial intelligence powered satellite communications and sentinel satellite constellations: An overview and future perspectives. SN Computer Science **6**(6), 735 (2025)

Khan, Z., Shen, Y., Liu, H.: Objectdetection in agriculture: A comprehensive review of methods, applications, challenges, and future directions. Agriculture **15**(13), 1351 (2025)

Kuandykov, A., Uskenbayeva, R., Im Cho, Y., Kozhamzharova, D., Baimuratov, O., Chinibayev, Y., Karimzhan, N.: Multi-agent based anti-locust territory protection system. Procedia Computer Science **56**, 477–483 (2015)

Kumar, N., Wei, X., Zhang, H.: Agentic artificial intelligence as a new frontier in information systems: Promise, peril, and research opportunities. Information & Management, 104317 (2026)

Laveglia, S., Altieri, G., Genovese, F., Matera, A., Di Renzo, G.C.: Advances in sustainable crop

management: integrating precision agriculture and proximal sensing. AgriEngineering **6**(3), 3084–3120 (2024)

Liu, J., Du, Y., Yang, K., Wu, J., Wang, Y., Hu, X., Wang, Z., Liu, Y., Sun, P., Boukerche, A., et al.: Edge-cloud collaborative computing on distributed intelligence and model optimization: A survey. IEEE Communications Surveys & Tutorials (2026)

Lopez-Jimenez, J., Quijano, N., Vande Wouwer, A.: Agent-based sensor location strategy for smart irrigation of large crop fields. Computers and Electronics in Agriculture **214**, 108282 (2023)

Lochan, K., Khan, A., Elsayed, I., Suthar, B., Seneviratne, L., Hussain, I.: Advancements in precision spraying of agricultural robots: A comprehensive review. IEEE Access **12**, 129447–129483 (2024)

Lytridis, C., Kaburlasos, V.G., Pachidis, T., Manios, M., Vrochidou, E., Kalampokas, T., Chatzistamatis, S.: An overview of cooperative robotics in agriculture. Agronomy **11**(9), 1818 (2021)

Liu, Z., Lei, H., Lei, L., Sheng, H.: Spatial prediction of total nitrogen in soil surface layer based on machine learning. Sustainability **14**(19), 11998 (2022)

Lavendelis, E., Liekna, A., Nikitenko, A., Grabovskis, A., Grundspenkis, J.: Multi-agent robotic system architecture for effective task allocation and management. Recent Researches in Communications, Electronics, Signal Processing & Automatic, 22–24 (2012)

Lee, Y., Park, E.: Toward sustainable agentic ai systems: A survey of architectures and methodologies. Sustainable Development (2026)

Lin, T., Sun, F., Li, X., Guo, X., Ying, J., Wu, H., Li, H.: A review of key technologies and recent advances in intelligent fruit-picking robots. Horticulturae **12**(2), 158 (2026)

Lujak, M., Sklar, E., Semet, F.: On multi-agent coordination of agri-robot fleets. In: Eleventh International Workshop on Agents in Traffic and Transportation (2020)

Li, Z., Wu, W., Guo, Y., Sun, J., Han, Q.-L.: Embodied multi-agent systems: A review. IEEE/CAA Journal of Automatica Sinica **12**(6), 1095–1116 (2025)

Liu, J.-J., Wu, H., Riaz, I.: Advanced technologies for smart fertilizer management in agriculture: A review. IEEE Access (2025)

Liang, B., Wang, Y., Tong, C.: Ai reasoning in deep learning era: From symbolic ai to neural–symbolic ai. Mathematics **13**(11), 1707 (2025)

Li, J., Yang, S.X.: Digital twins to embodied artificial intelligence: Review and perspective. Intelligence & Robotics **5**(1), 202–227 (2025)

Liu, S., Zhu, Y.: Agent-based wheat simulation model cooperation research. In: International Conference on Computer and Computing Technologies in Agriculture, pp. 1281–1289 (2008). Springer

Murad, M., Ahmed, M., din, N.u., Shahid, M.F., Siddiqui, S., Byers, D., Tanveer, M.H., Voicu, R.C.: Agentic ai framework to automate traditional farming for smart agriculture. AgriEngineering **8**(1), 8 (2026)

Maharramova, A., Abbasova, A., Imanova, S., Hakhverdiyev, T., Rustamovna, A.K.: Artificial intelligence (ai) agents. Reliability: Theory & Applications **20**(SI 10 (88)), 166–175 (2025)

Matthews, R.: The people and landscape model (palm): Towards full integration of human decision-making and biophysical simulation models. Ecological modelling **194**(4), 329–343 (2006)

Mazzetto, S.: Interdisciplinary perspectives on agent-based modeling in the architecture, engineering, and construction industry: A comprehensive review. Buildings **14**(11), 3480 (2024)

Mohanraj, R., Balaji, S.N.: Digital twin technology: A comprehensive review of modeling, applications, challenges and future directions in complex system integration. Archives of Computational Methods in Engineering, 1–26 (2025)

McCown, R.L.: Changing systems for supporting farmers' decisions: problems, paradigms, and prospects. Agricultural systems **74**(1), 179–220 (2002)

Mallik, S., Chakraborty, A., Podder, K., Talukdar, S., Rahman, A., Mishra, U.: Enhancing soil moisture prediction with explainable ai: Integrating iot and multi-sensor remote sensing data through soft computing. Applied Soft Computing **180**, 113406 (2025)

Miller, T., Durlik, I., Kostecka, E., Kozlovska, P., Łobodzińska, A., Sokołowska, S., Nowy, A.: Integrating artificial intelligence agents with the internet of things for enhanced environmental monitoring: applications in water quality and climate data. Electronics **14**(4), 696 (2025)

Miller, T., Durlik, I., Łobodzińska, A., Dorobczyński, L., Jasionowski, R.: Ai in context: Harnessing domain knowledge for smarter machine learning. Applied Sciences **14**(24), 11612 (2024)

Misra, S., Ghosh, A.: Agriculture paradigm shift: a journey from traditional to modern agriculture. In: Biodiversity and Bioeconomy, pp. 113–141. Elsevier, ??? (2024)

Mgendi, G.: Unlocking the potential of precision agriculture for sustainable farming. Discover Agriculture **2**(1), 87 (2024)

Macmillan, W., Huang, H.: An agent-based simulation model of a primitive agricultural society. Geoforum **39**(2), 643–658 (2008)

Moore, A.D., Holzworth, D.P., Herrmann, N.I., Brown, H.E., Voil, P.G., Snow, V.O., Zurcher, E.J., Huth, N.I.: Modelling the manager: Representing rule-based management in farming systems simulation models. Environmental Modelling & Software **62**, 399–410 (2014)

Meuwissen, M., Hardaker, J., Huirne, R., Dijkhuizen, A.: Sharing risks in agriculture; principles and empirical results. NJAS-Wageningen Journal of Life Sciences **49**(4), 343–356 (2001)

Mikula, K., Izydorczyk, G., Skrzypczak, D., Mironiuk, M., Moustakas, K., Witek-Krowiak, A., Chojnacka, K.: Controlled release micronutrient fertilizers for precision agriculture–a review. Science of the Total Environment **712**, 136365 (2020)

Ming, R., Jiang, R., Luo, H., Lai, T., Guo, E., Zhou, Z.: Comparative analysis of different uav swarm control methods on unmanned farms. Agronomy **13**(10), 2499 (2023)

Mishra, H., Kayusi, F., Adamopoulos, I., Olayinka, O.T.: Ai-driven irrigation and water management systems. In: Robotics and Intelligent Machines in Smart Agriculture, pp. 270–309. CRC Press, ??? (2026)

McKinion, J., Lemmon, H.: Expert systems for agriculture. Computers and Electronics in Agriculture **1**(1), 31–40 (1985)

Malley, D., Martin, P., Ben-Dor, E.: Application in analysis of soils. Near-infrared spectroscopy in agriculture **44**, 729–784 (2004)

Miller, T., Mikiciuk, G., Durlik, I., Mikiciuk, M., Łobodzińska, A., Śnieg, M.: The iot and ai in agriculture: The time is now—a systematic review of smart sensing technologies. Sensors **25**(12), 3583 (2025)

McMahan, B., Moore, E., Ramage, D., Hampson, S., Arcas, B.A.: Communication-efficient learning of deep networks from decentralized data. In: Artificial Intelligence and Statistics, pp. 1273–1282 (2017). Pmlr

Marwah, N., Singh, V.K., Kashyap, G.S., Wazir, S.: An analysis of the robustness of uav agriculture field coverage using multi-agent reinforcement learning. International Journal of Information

Technology **15**(4), 2317–2327 (2023)

Mansour, M., Sathyanarayanan, K.K., Sauerteig, P., Streif, S.: Adaptive robust greenhouse climate control: Combining deep reinforcement learning and economic optimization. Smart Agricultural Technology, 101327 (2025)

Meshram, A.T., Vanalkar, A.V., Kalambe, K.B., Badar, A.M.: Pesticide spraying robot for precision agriculture: A categorical literature review and future trends. Journal of Field Robotics **39**(2), 153–171 (2022)

McEnroe, P., Wang, S., Liyanage, M.: A survey on the convergence of edge computing and ai for uavs: Opportunities and challenges. IEEE Internet of Things Journal **9**(17), 15435–15459 (2022)

Nielsen, K., Appel, J., Demazeau, Y.: Applying ai to cooperating agricultural robots. In: IFIP International Conference on Artificial Intelligence Applications and Innovations, pp. 262–270 (2006). Springer

Nagireddy, S.K.: Sensor-driven autonomy in agriculture: A multi-modal approach to precision farming. Journal of Computer Science and Technology Studies **7**(7), 979–986 (2025)

Nair, A.S.: A hub-ci model for networked telerobotics in collaborative monitoring of agricultural greenhouses. Master's thesis, Purdue University (2019)

NANDAMURU, A.P.: Ai recommended agricultural assistant chatbot system. Journal of Nonlinear Analysis and Optimization **16**(1) (2025)

Ng, T.L., Eheart, J.W., Cai, X., Braden, J.B.: An agent-based model of farmer decision-making and water quality impacts at the watershed scale under markets for carbon allowances and a second-generation biofuel crop. Water Resources Research **47**(9) (2011)

Neethirajan, S.: Digital twins for cows and chickens: From hype cycles to hard evidence in precision livestock farming. Agriculture **16**(2), 166 (2026)

Nguyen, L.V.: Swarm intelligence-based multi-robotics: A comprehensive review. AppliedMath **4**(4), 1192–1210 (2024)

Niazi, M., Hussain, A.: Agent-based computing from multi-agent systems to agent-based models: a visual survey. Scientometrics **89**(2), 479–499 (2011)

Nguyen, D.-H., Huynh-Tuong, N., Pham, H.-A.: A blockchain-based framework for developing traceability applications towards sustainable agriculture in vietnam. Security and Communication Networks **2022**(1), 1834873 (2022)

Nagini, S., Kanth, T.R., Kiranmayee, B.: Agriculture yield prediction using predictive analytic techniques. In: 2016 2nd International Conference on Contemporary Computing and Informatics (IC3I), pp. 783–788 (2016). IEEE

Nyéki, A., Neményi, M.: Crop yield prediction in precision agriculture. MDPI (2022)

Neo, D.C.J., Ong, M.M.X., Lee, Y.Y., Teo, E.J., Ong, Q., Tanoto, H., Xu, J., Ong, K.S., Suresh, V.: Shaping and tuning lighting conditions in controlled environment agriculture: A review. ACS Agricultural Science & Technology **2**(1), 3–16 (2022)

Niaz, H.U., Qadeer, Q.B., Niaz, H., Mansib, H., Awais, M., Khan, H.: Artificial intelligence assisted autonomous unmanned aerial vehicles (uavs) and aerial drones based on machine vision for enhancing remote sensing of precision crop health monitoring. The Asian Bulletin of Big Data Management **5**(4), 155–177 (2025)

Nebot Roglá, P., et al.: Agent-based architecture for multirobot cooperative tasks: Design and applications. Universitat Jaume I (2008)

Nfor, K.A., Theodore Armand, T.P., Ismaylovna, K.P., Joo, M.-I., Kim, H.-C.: An explainable cnn and vision transformer-based approach for real-time food recognition. Nutrients **17**(2), 362 (2025)

Naqvi, S.M.Z.A., Tahir, M.N., Raghavan, V., Awais, M., Hu, J., Said, Y., Othman, N.A., Ashurov, M., Khan, M.I.: Ai-enhanced iot sensors for real-time crop monitoring: an era towards self-monitored agriculture. Telecommunication Systems **88**(3), 100 (2025)

Nugroho, S., Uehara, T.: Systematic review of agent-based and system dynamics models for social-ecological system case studies. Systems **11**(11), 530 (2023)

Olson, D., Anderson, J.: Review on unmanned aerial vehicles, remote sensors, imagery processing, and their applications in agriculture. Agronomy Journal **113**(2), 971–992 (2021)

Ok, A.O., Akar, O., Gungor, O.: Evaluation of random forest method for agricultural crop classification. European Journal of Remote Sensing **45**(1), 421–432 (2012)

Obeidat, M.A., Abdallah, J., Hamadneh, T., Qawaqneh, H., Mansour, A.M.: Enhancing agricultural operations through ai-driven agent communication in smart farming systems. Ingenierie des Systemes d'Information **29**(3), 917 (2024)

Ogbu, D.: Agentic ai in computer vision domain-recent advances and prospects. International Journal of Research Publication and Reviews **4**(12), 5102–5120 (2023)

Onteddu, A.R., Kundavaram, R.R., Kamisetty, A., Gummadi, J.C.S., Manikyala, A.: Enhancing agricultural efficiency with robotics and ai-powered autonomous farming systems. Malaysian Journal of Medical and Biological Research **12**(1), 7–22 (2025)

Olatinwo, D.D., Myburgh, H.C., De Freitas, A., Abu-Mahfouz, A.M.: A review of smart crop technologies for resource constrained environments: Leveraging multimodal data fusion, edge-to-cloud computing, and iot virtualization. Journal of Sensor and Actuator Networks **14**(5), 99 (2025)

Oliveira, L.F., Moreira, A.P., Silva, M.F.: Advances in agriculture robotics: A state-of-the-art review and challenges ahead. Robotics **10**(2), 52 (2021)

Oyeboade, J., Olagoke-Komolafe, O.: Implementing innovative data-driven solutions for sustainable agricultural development and productivity. International Journal of Multidisciplinary Futuristic Development **4**(1), 24–31 (2023)

Oprea, M.: Agent-based modelling of multi-robot systems. In: IOP Conference Series: Materials Science and Engineering, vol. 444, p. 052026 (2018). IOP Publishing

Ota, J.: Multi-agent robot systems as distributed autonomous systems. Advanced engineering informatics **20**(1), 59–70 (2006)

Olayinka, A.S., Ukhurebor, K.E., Aigbe, U.O.: Digital Twins, Artificial Intelligence, and Internet of Things for a Sustainable Food Production in Agriculture: Transformative Technologies. CRC Press, ??? (2026)

Polwaththa, K., Amarasinghe, A., Nandasena, G.: A review of innovative fertilization strategies in precision agriculture. Open Access Research Journal of Science and Technology **12**(2), 49–57 (2024)

Pulido, J.S., Bocco, G.: The traditional farming system of a mexican indigenous community: The case of nuevo san juan parangaricutiro, michoacán, mexico. Geoderma **111**(3-4), 249–265 (2003)

Pugazhendi, P., Badgujar, C.M., Ganapathy, M.R., Arumugam, M.: Uncertainty-aware deep learning for sugarcane leaf disease detection using monte carlo dropout and mobilenetv3. AgriEngineering **8**(1), 31 (2026)

Pinheiro, P.O., Collobert, R.: From image-level to pixel-level labeling with convolutional networks. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 1713–1721

(2015)

Phang, S.K., Chiang, T.H.A., Happonen, A., Chang, M.M.L.: From satellite to uav-based remote sensing: A review on precision agriculture. Ieee Access **11**, 127057–127076 (2023)

Papadopoulos, T., Evangelidis, K., Kaskalis, T.H., Evangelidis, G.: The metaverse is geospatial: a system model architecture integrating spatial computing, digital twins, and virtual worlds. ISPRS International Journal of Geo-Information **14**(3), 126 (2025)

Polanía, E.C.M., Fernandes, H.R., Albiero, D., Garcia, A.P.: Control system for the navigation of the agricultural robots: A review. Journal of Field Robotics (2026)

Prajapati, H.A., Kadam, D., Aitwar, S.S., Jagtap, P.D., Singh, D., Nandeha, N., Mukherjee, D.: Application of robotics, artificial intelligence and deep learning in modern agriculture technology: a review. International Journal of Plant & Soil Science **35**(23), 106–116 (2023)

Poorvi, J., Kalita, A., Gurusamy, M.: Reliable and efficient data collection in uav based iot networks. IEEE Communications Surveys & Tutorials (2025)

Pavlovic, M., Koumboulis, F.N., Tzamtzi, M.P., Rozman, C.: Role of automation agents in agribusiness decision support systems. Agrociencia **42**(8), 913–923 (2008)

Pomar, J., López, V., Pomar, C.: Agent-based simulation framework for virtual prototyping of advanced livestock precision feeding systems. Computers and electronics in agriculture **78**(1), 88–97 (2011)

Parrott, L., Lacroix, R., Wade, K.M.: Design considerations for the implementation of multi-agent systems in the dairy industry. Computers and electronics in agriculture **38**(2), 79–98 (2003)

Poole, D.L., Mackworth, A.K.: Artificial Intelligence: Foundations of Computational Agents. Cambridge University Press, ??? (2010)

Parker, D.C., Manson, S.M., Janssen, M.A., Hoffmann, M.J., Deadman, P.: Multi-agent systems for the simulation of land-use and land-cover change: a review. Annals of the association of American Geographers **93**(2), 314–337 (2003)

Posadas, B.B., Ogunyiola, A., Niewolny, K.: Socially responsible ai assurance in precision agriculture for farmers and policymakers. In: AI Assurance, pp. 473–499. Elsevier, ??? (2023)

Panda, S., Padhy, A.: Agentic ai in distributed systems: Toward autonomous, goal-oriented decision-making in multi-agent environments. In: 2025 5th International Conference on Mobile Networks and Wireless Communications (ICMNWC), pp. 1–6 (2025). IEEE

Peladarinos, N., Piromalis, D., Cheimaras, V., Tserepas, E., Munteanu, R.A., Papageorgas, P.: Enhancing smart agriculture by implementing digital twins: A comprehensive review. Sensors **23**(16), 7128 (2023)

Prasanna, T., Pandit, V., Ravali, C., Shyamsunder, B., Chatterjee, S., Tariq, A.: Precision agriculture and resource optimization. In: Harnessing AI to Reshape the Future of Agriculture, pp. 411–436. Springer, ??? (2026)

Priya, P.K., Reethika, A., Sathyamoorthy, M., Dhanaraj, R.K.: Domain-specific ai algorithms and models in decision-making: An overview. Ethical Decision-Making Using Artificial Intelligence: Challenges, Solutions and Applications, 27–53 (2025)

Perini, A., Susi, A.: Developing a decision support system for integrated production in agriculture. Environmental Modelling & Software **19**(9), 821–829 (2004)

Pérez-Salazar, M.d.R., Aguilar-Lasserre, A.A., Cedillo-Campos, M.G., Posada-Gómez, R., Moral-Argumedo, M.J., Hernández-González, J.C.: An agent-based model driven decision support system for reactive aggregate production scheduling in the green coffee supply chain. Applied Sciences

9(22), 4903 (2019)

Park, J., Samarakoon, S., Elgabli, A., Kim, J., Bennis, M., Kim, S.-L., Debbah, M.: Communication-efficient and distributed learning over wireless networks: Principles and applications. Proceedings of the IEEE **109**(5), 796–819 (2021)

Pargo, T.A., Shirazi, M.A., Fadai, D.: Smart and efficient iot-based irrigation system design: utilizing a hybrid agent-based and system dynamics approach. arXiv preprint arXiv:2502.18298 (2025)

Priya, N.K., Sivanarayana, G., Rahul, M.S.: Integrating digital tools to revolutionize agricultural learning. New Frontiers in Agricultural Extension Strategies **1** (2025)

Patel, A., Shukla, C., Trivedi, A., Balasaheb, K.S., Sinha, M.: Smart farming: Utilization of robotics, drones, remote sensing, gis, ai, and iot tools in agricultural operations and water management. In: Integrated Land and Water Resource Management for Sustainable Agriculture Volume 1, pp. 127–151. Springer, ??? (2025)

Panigrahi, S., Ting, K.: Artificial Intelligence for Biology and Agriculture. Springer, ??? (2012)

Pierre, N., Viviane, I.I., Lambert, U., Shadrack, I., Erneste, B., Schadrack, N., Alexis, N., Francois, K., Theogene, H.: Ai based real-time weather condition prediction with optimized agricultural resources. European Journal of Technology **7**(2), 36–49 (2023)

Pentjuss, A., Zacepins, A., Gailums, A., et al.: Improving precision agriculture methods with multiagent systems in latvian agricultural field. Engineering for rural development, Latvia, Jelgava, 109–114 (2011)

Qu, C., Boubin, J., Gafurov, D., Zhou, J., Aloysius, N., Nguyen, H., Calyam, P.: Uav swarms in smart agriculture: Experiences and opportunities. In: 2022 IEEE 18th International Conference on e-Science (e-Science), pp. 148–158 (2022). IEEE

Qadir, M., Boers, T.M., Schubert, S., Ghafoor, A., Murtaza, G.: Agricultural water management in water-starved countries: challenges and opportunities. Agricultural water management **62**(3), 165–185 (2003)

Qin, Y.-M., Tu, Y.-H., Li, T., Ni, Y., Wang, R.-F., Wang, H.: Deep learning for sustainable agriculture: A systematic review on applications in lettuce cultivation. Sustainability **17**(7), 3190 (2025)

Qiao, Y., Valente, J., Su, D., Zhang, Z., He, D.: AI, sensors and robotics in plant phenotyping and precision agriculture. Frontiers (2022)

Rasheed, A.A., Abdullah, M.N., Al-Araji, A.S.: A review of multi-agent mobile robot systems applications. International Journal of Electrical and Computer Engineering **12**(4), 3517–3529 (2022)

Rafea, A.: Expert system applications: Agriculture. Central Laboratory for Expert Systems, Giza Egypt, 1–4 (2009)

Romeo, G., Conti, D.: Exploring automation bias in human–ai collaboration: a review and implications for explainable ai. AI & SOCIETY **41**(1), 259–278 (2026)

Ravaioli, G., Domingos, T., Teixeira, R.F.: A framework for data-driven agent-based modelling of agricultural land use. Land **12**(4), 756 (2023)

Roberts-Elliott, L., Das, G.P., Millard, A.G.: Agent-based simulation of multi-robot soil compaction mapping. In: Annual Conference Towards Autonomous Robotic Systems, pp. 251–265 (2022). Springer

Rahman, T.U., Faraz, A., Nawaz, T., Saud, S., Fahad, S., Harrison, M.T.: Towards sustainable solutions: climate change and food security in a globalized world. Food and Energy Security **14**(5),

70126 (2025)

Ruiz-Garcia, L., Steinberger, G., Rothmund, M.: A model and prototype implementation for tracking and tracing agricultural batch products along the food chain. Food control **21**(2), 112–121 (2010)

Rangarajan, S., Harshad, A., Dhinesh, A., et al.: Comparing and analysing the effectiveness of multi-agent reinforcement learning (marl) algorithms for simplistic coordination in row cultivation applications. Smart Agricultural Technology, 101842 (2026)

Rimada, Y.: Adaptive machine learning model for optimizing tractor path planning and fuel efficiency in large farms. Journal of Environmental Sustainability, Climate Resilience, and Agro-Ecosystems, 16–23 (2025)

Roy, C.S., Krishnan, V.G., Ganganala, S., Mukherjee, A., Joseph, D., Gupta, G.: Development of task-based robots for artificially intelligent farm systems. In: Robotics and Intelligent Machines in Smart Agriculture, pp. 89–103. CRC Press, ??? (2025)

Ren, C., Kim, D.-K., Jeong, D.: A survey of deep learning in agriculture: techniques and their applications. Journal of Information Processing Systems **16**(5), 1015–1033 (2020)

Raman, R.K., Kumar, A., Sarkar, S., Yadav, A.K., Mukherjee, A., Meena, R.S., Kumar, U., Singh, D., Das, S., Kumar, R., *et al.*: Reconnoitering precision agriculture and resource management: a comprehensive review from an extension standpoint on artificial intelligence and machine learning. Indian Research Journal of Extension Education **24**(1), 108–123 (2024)

Romera, A., Morris, S., Hodgson, J., Stirling, W., Woodward, S.: A model for simulating rule-based management of cow–calf systems. Computers and electronics in agriculture **42**(2), 67–86 (2004)

Radhakrishnan, S., Prasanth, A.: Cognitive intelligence and distributed computing systems applications in smart farming. Computer Vision in Smart Agriculture and Crop Management, 159–186 (2025)

Reddy, A.M., Reddy, N.R.: Design and evaluation of a natural language-based chatbot for smart farming advisory services. Fringe Multi-Engineering Proceedings (FMEP, ISSN: 3107-6149) **1**(3), 53–65 (2025)

Roberts, D.P., Short Jr, N.M., Sill, J., Lakshman, D.K., Hu, X., Buser, M.: Precision agriculture and geospatial techniques for sustainable disease control. Indian Phytopathology **74**(2), 287–305 (2021)

Raza, S., Sapkota, R., Karkee, M., Emmanouilidis, C.: Trism for agentic ai: A review of trust, risk, and security management in llm-based agentic multi-agent systems. AI Open (2026)

Ramaiah, M., Settu, P., Ravi, V.: Artificial intelligence techniques enabled soil moisture estimation frameworks using remote sensing satellite images: Challenges and future directions-review. Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery **15**(3), 70032 (2025)

Ramani, D.R., Sujitha, B.B., Tangade, S.: Smart environmental monitoring systems: Iot and sensor-based advancements. Environmental Monitoring Using Artificial Intelligence, 45–60 (2025)

Saha, S.: AI for Food Security: Protecting Agriculture, Infrastructure, and Supply Chains.: AI for Food Security: Protecting Agriculture, Infrastructure, and Supply Chains. Geh press, ??? (2026)

Salwadkar, M.: Digital twin–integrated smart irrigation system for real-time water optimization in arid agriculture. Journal of Environmental Sustainability, Climate Resilience, and Agro-Ecosystems, 23–29 (2024)

Shaffer, M., Brodahl, M.: Rule-based management for simulation in agricultural decision support systems. Computers and Electronics in Agriculture **21**(2), 135–152 (1998)

Schreinemachers, P., Berger, T.: Land use decisions in developing countries and their representation

in multi-agent systems. Journal of land use science **1**(1), 29–44 (2006)

Schreinemachers, P., Berger, T.: An agent-based simulation model of human–environment interactions in agricultural systems. Environmental Modelling & Software **26**(7), 845–859 (2011)

Singh, R.K., Berkvens, R., Weyn, M.: Agrifusion: An architecture for iot and emerging technologies based on a precision agriculture survey. IEEE Access **9**, 136253–136283 (2021)

Srivastav, A.K., Das, P.: Edge computing and ai in agricultural iot. In: Biotechnology and IoT in Agriculture and Food Production: Green Innovation, pp. 187–200. Springer, ??? (2025)

Sarma, W., Dey, S., Tiwari, S.: Autonomous iot: Ai-driven edge computing to power intelligent decision-making. International Journal of AI, BigData, Computational and Management Studies **3**(2), 52–61 (2022)

Seralathan, P., Edward, A.S.: Reinforcement learning based dynamic vegetation index formulation for rice crop stress detection using satellite and mobile imagery. Scientific Reports (2025)

Salimi, N., Ferdosian, M., Mahjouri, N.: An adaptive agent-based framework for simulating farmers' behavior in the face of drought: Application of numerical weather predictions. Water Resources Management **40**(1), 40 (2026)

Swati, N.P., Gupta, S.V., Duddela, N.S., Parvathy, L.R.: Agentic ai-driven autonomous decision support system for smart agriculture. Scientific Reports (2026)

Srivastava, A.K., Goering, C.E., Rohrbach, R.P., Buckmaster, D.R.: Engineering principles of agricultural machines. American society of agricultural engineers St. Joseph, Michigan (1993)

Shams, M.Y., Gamel, S.A., Talaat, F.M.: Enhancing crop recommendation systems with explainable artificial intelligence: a study on agricultural decision-making. Neural Computing and Applications **36**(11), 5695–5714 (2024)

SS, V.C., Hareendran, A., Albaaji, G.F.: Precision farming for sustainability: An agricultural intelligence model. Computers and Electronics in Agriculture **226**, 109386 (2024)

Sassenrath, G.F., Heilman, P., Luschei, E., Bennett, G.L., Fitzgerald, G., Klesius, P., Tracy, W., Williford, J.R., Zimba, P.V.: Technology, complexity and change in agricultural production systems. Renewable Agriculture and Food Systems **23**(4), 285–295 (2008)

Shukla, S., Hassan, M.F., Tran, D.C., Akbar, R., Paputungan, I.V., Khan, M.K.: Improving latency in internet-of-things and cloud computing for real-time data transmission: a systematic literature review (slr). Cluster Computing **26**(5), 2657–2680 (2023)

Shadkam, E., Irannezhad, E.: A comprehensive review of simulation optimization methods in agricultural supply chains and transition towards an agent-based intelligent digital framework for agriculture 4.0. Engineering applications of artificial intelligence **143**, 109930 (2025)

Sihare, M., Khule, S., Arora, R., Dixit, N., Dubey, G., Sharma, Y.K.: Cultivating the future of agriculture where digital twin meets artificial intelligence. In: Transforming Industries, Empowering Societies, pp. 137–148. Elsevier, ??? (2026)

Steckel, T., Kersting, T., Nüßer, W.: Towards supporting mobile business processes in non-deterministic agricultural environments by using agent-based technologies. KI-Künstliche Intelligenz **27**(4), 359–362 (2013)

Sun, R., Nolan, J., Kulshreshtha, S.: Agent-based modeling of policy induced agri-environmental technology adoption. SN Business & Economics **2**(8), 101 (2022)

Schreinemachers, P., Potchanasin, C., Berger, T., Roygrong, S.: Agent-based modeling for ex ante assessment of tree crop innovations: litchis in northern thailand. Agricultural Economics **41**(6), 519–536 (2010)

Srinivasu, P.N., Pavate, A., JayaLakshmi, G., Shafi, J., Choi, J., Ijaz, M.F.: Agentic ai for smart and sustainable precision agriculture. Frontiers in Plant Science **16**, 1706428 (2025)

Sapkota, R., Qureshi, R., Hadi, M.U., Hassan, S.Z., Sadak, F., Shoman, M., Sajjad, M., Dharejo, F.A., Paudel, A., Li, J., et al.: Multi-modal llms in agriculture: A comprehensive review. IEEE Transactions on Automation Science and Engineering (2025)

San, O., Rasheed, A., Bozdemir, E., Deng, J.: The evolution of digital twins from reactive to agentic systems. Nature Computational Science **6**(1), 6–10 (2026)

Sapkota, R., Roumeliotis, K.I., Karkee, M.: Ai agents vs. agentic ai: A conceptual taxonomy, applications and challenges. Information Fusion, 103599 (2025)

Sapkota, R., Roumeliotis, K.I., Karkee, M.: Uavs meet agentic ai: A multidomain survey of autonomous aerial intelligence and agentic uavs. arXiv preprint arXiv:2506.08045 (2025)

Senapaty, M.K., Ray, A., Padhy, N.: A decision support system for crop recommendation using machine learning classification algorithms. Agriculture **14**(8), 1256 (2024)

Sengupta, R., Sieber, R.: Geospatial agents, agents everywhere... Transactions in GIS **11**(4), 483–506 (2007)

Singh, A., Sharma, A.: A framework for semantics and agent based personalized information retrieval in agriculture. In: 2015 2nd International Conference on Computing for Sustainable Global Development (INDIACom), pp. 929–931 (2015). IEEE

Santoso, A., Surya, Y.: Maximizing decision efficiency with edge-based ai systems: advanced strategies for real-time processing, scalability, and autonomous intelligence in distributed environments. Quarterly Journal of Emerging Technologies and Innovations **9**(2), 104–132 (2024)

Sharma, K., Shivandu, S.K.: Integrating artificial intelligence and internet of things (iot) for enhanced crop monitoring and management in precision agriculture. Sensors International **5**, 100292 (2024)

Santos, L., Santos, F.N., Oliveira, P.M., Shinde, P.: Deep learning applications in agriculture: A short review. In: Iberian Robotics Conference, pp. 139–151 (2019). Springer

Sarma, S.K., Singh, K.R., Singh, A.: An expert system for diagnosis of diseases in rice plant. International Journal of Artificial Intelligence **1**(1), 26–31 (2010)

Tey, Y.S., Brindal, M., Wong, S.Y., Ardiansyah, Ibragimov, A., Yusop, M.R.: Evolution of precision agricultural technologies: a patent network analysis. Precision Agriculture **25**(1), 376–395 (2024)

Traversari, S., Cacini, S., Galieni, A., Nesi, B., Nicastro, N., Pane, C.: Precision agriculture digital technologies for sustainable fungal disease management of ornamental plants. Sustainability **13**(7), 3707 (2021)

Tang, Y., Chen, M., Wang, C., Luo, L., Li, J., Lian, G., Zou, X.: Recognition and localization methods for vision-based fruit picking robots: A review. Frontiers in plant science **11**, 510 (2020)

Tzachor, A., Devare, M., King, B., Avin, S., Ó hÉigeartaigh, S.: Responsible artificial intelligence in agriculture requires systemic understanding of risks and externalities. Nature Machine Intelligence **4**(2), 104–109 (2022)

Tzachor, A., Devare, M., Richards, C., Pypers, P., Ghosh, A., Koo, J., Johal, S., King, B.: Large language models and agricultural extension services. Nature food **4**(11), 941–948 (2023)

Tugrul, B., Elfatimi, E., Eryigit, R.: Convolutional neural networks in detection of plant leaf diseases: A review. Agriculture **12**(8), 1192 (2022)

Tapia, D.I., Fraile, J.A., Rodríguez, S., Alonso, R.S., Corchado, J.M.: Integrating hardware agents into an enhanced multi-agent architecture for ambient intelligence systems. Information Sciences

222, 47–65 (2013)

Tweedale, J., Ichalkaranje, N., Sioutis, C., Jarvis, B., Consoli, A., Phillips-Wren, G.: Innovations in multi-agent systems. Journal of Network and Computer Applications **30**(3), 1089–1115 (2007)

Thakur, P.S., Khanna, P., Sheorey, T., Ojha, A.: Vision transformer for plant disease detection: Plantvit. In: International Conference on Computer Vision and Image Processing, pp. 501–511 (2021). Springer

Titirmare, S., Margal, P.B., Gupta, S., Kumar, D.: Ai-powered predictive analytics for crop yield optimization. In: Agriculture 4.0, pp. 89–110. CRC Press, ??? (2024)

Tariq, M.U., Saqib, S.M., Mazhar, T., Khan, M.A., Shahzad, T., Hamam, H.: Edge-enabled smart agriculture framework: Integrating iot, lightweight deep learning, and agentic ai for context-aware farming. Results in Engineering, 107342 (2025)

Tantalaki, N., Souravlas, S., Roumeliotis, M.: Data-driven decision making in precision agriculture: The rise of big data in agricultural systems. Journal of agricultural & food information **20**(4), 344–380 (2019)

Tripathi, P.K., Singh, C.K., Singh, R., Deshmukh, A.K.: A farmer-centric agricultural decision support system for market dynamics in a volatile agricultural supply chain. Benchmarking: an international journal **30**(10), 3925–3952 (2023)

Tejada, J.C., Toro-Ossaba, A., Lopez-Gonzalez, A., Hernandez-Martinez, E.G., Sanin-Villa, D.: A review of multi-robot systems and soft robotics: challenges and opportunities. Sensors **25**(5), 1353 (2025)

Upadhyay, A., Chandel, N.S., Singh, K.P., Chakraborty, S.K., Nandede, B.M., Kumar, M., Subeesh, A., Upendar, K., Salem, A., Elbeltagi, A.: Deep learning and computer vision in plant disease detection: a comprehensive review of techniques, models, and trends in precision agriculture. Artificial Intelligence Review **58**(3), 92 (2025)

Uma, M., Harish, B.V., Hariharan, K., Kaviyaraj, R.: An agentic explainable ai system for agricultural crop production prediction and suitability mapping. In: Intelligent and Sustainable Systems, pp. 325–331. CRC Press, ??? (2026)

Ugwu, O.P.-C., Ogenyi, F.C., Alum, E.U., Eze, V.H.U., Basajja, M., Ugwu, J.N., Ugwu, C.N., Ejemot-Nwadiaro, R.I., Okon, M.B., Egba, S.I., *et al.*: Implementing artificial intelligence and machine learning algorithms for optimized crop management: a systematic review on data-driven approach to enhancing resource use and agricultural sustainability. Cogent Food & Agriculture **11**(1), 2569982 (2025)

Vos, R., Bellù, L.G.: Global trends and challenges to food and agriculture into the 21st century. Sustainable food and agriculture, 11–30 (2019)

Villarrubia, G., De Paz, J.F., De La Iglesia, D.H., Bajo, J.: Combining multi-agent systems and wireless sensor networks for monitoring crop irrigation. Sensors **17**(8), 1775 (2017)

Vishwakarma, S.K.: Agentic ai–driven decision-support framework for climate-responsive agricultural adaptation using reinforcement learning. International Journal of Applied Mathematics **38**(6s), 293–312 (2025)

Vimalajeewa, D., Kulatunga, C., Berry, D.P., Balasubramaniam, S.: A service-based joint model used for distributed learning: Application for smart agriculture. IEEE Transactions on Emerging Topics in Computing **10**(2), 838–854 (2021)

Vougioukas, S.G.: Agricultural robotics. Annual review of control, robotics, and autonomous systems **2**(1), 365–392 (2019)

Velusamy, P., Rajendran, S., Mahendran, R.K., Naseer, S., Shafiq, M., Choi, J.-G.: Unmanned aerial vehicles (uav) in precision agriculture: Applications and challenges. Energies **15**(1), 217 (2021)

Vinyals, M., Sabbadin, R., Couture, S., Sadou, L., Thomopoulos, R., Chapuis, K., Lesquoy, B., Taillandier, P.: Toward ai-designed innovation diffusion policies using agent-based simulations and reinforcement learning: The case of digital tool adoption in agriculture. Frontiers in Applied Mathematics and Statistics **9**, 1000785 (2023)

Verma, K.K., Song, X.-P., Kumari, A., Jagadesh, M., Singh, S.K., Bhatt, R., Singh, M., Seth, C.S., Li, Y.-R.: Climate change adaptation: challenges for agricultural sustainability. Plant, Cell & Environment **48**(4), 2522–2533 (2025)

Vijayakumar, S., Shanmugapriya, P., Saravanane, P., Ramesh, T., Murugaiyan, V., Ilakkiya, S.: Precision weed control using unmanned aerial vehicles and robots: Assessing feasibility, bottlenecks, and recommendations for scaling. NDT **3**(2), 10 (2025)

Vijay, S., Tamizhselvi, A., *et al.*: A next-generation cyber-physical architecture with multi-agent ai, iot sensing, and autonomous robotic swarms. In: 2025 6th International Conference on IoT Based Control Networks and Intelligent Systems (ICICNIS), pp. 299–305 (2025). IEEE

Valbuena, D., Verburg, P.H., Veldkamp, A., Bregt, A.K., Ligtenberg, A.: Effects of farmers' decisions on the landscape structure of a dutch rural region: An agent-based approach. Landscape and Urban Planning **97**(2), 98–110 (2010)

Wadoux, A.M.-C.: Artificial intelligence in soil science. European Journal of Soil Science **76**(2), 70080 (2025)

Waqar, M., Bhatti, I., Khan, A.H.: Leveraging machine learning algorithms for autonomous robotics in real-time operations. International Journal of Advanced Engineering Technologies and Innovations **4**(1), 1–24 (2024)

Wong, A., Bäck, T., Kononova, A.V., Plaat, A.: Deep multiagent reinforcement learning: Challenges and directions. Artificial Intelligence Review **56**(6), 5023–5056 (2023)

Wang, H., Chen, X., Wu, Q., Yu, Q., Hu, X., Zheng, Z., Bouguettaya, A.: Integrating reinforcement learning with multi-agent techniques for adaptive service composition. ACM Transactions on Autonomous and Adaptive Systems (TAAS) **12**(2), 1–42 (2017)

Wang, J., Chen, M., Zhou, J., Li, P.: Data communication mechanism for greenhouse environment monitoring and control: An agent-based iot system. Information Processing in Agriculture **7**(3), 444–455 (2020)

Wen, W.: A knowledge-based intelligent electronic commerce system for selling agricultural products. Computers and electronics in agriculture **57**(1), 33–46 (2007)

Wei, Y., Han, C., Yu, Z.: An environment safety monitoring system for agricultural production based on artificial intelligence, cloud computing and big data networks. Journal of Cloud Computing **12**(1), 83 (2023)

Wu, Z., He, M., Zhou, G., Yan, X., Zhang, J., Liu, Z., Tan, X., Zhang, H.: Tvim: Text-enhanced uav framework with vegetation-index masking for precision pine wilt localization. IEEE Transactions on Geoscience and Remote Sensing (2026)

Wang, Y., Jia, W., Ou, M., Wang, X., Dong, X.: A review of orchard canopy perception technologies for variable-rate spraying. Sensors **25**(16), 4898 (2025)

Wang, X., Li, Q., Jia, W.: Cognitive edge computing: A comprehensive survey on optimizing large models and ai agents for pervasive deployment. arXiv preprint arXiv:2501.03265 (2025)

Wang, C., Liu, S., Wang, Y., Xiong, J., Zhang, Z., Zhao, B., Luo, L., Lin, G., He, P.: Application of

convolutional neural network-based detection methods in fresh fruit production: a comprehensive review. Frontiers in plant science **13**, 868745 (2022)

Waqas, M., Naseem, A., Humphries, U.W., Hlaing, P.T., Dechpichai, P., Wangwongchai, A.: Applications of machine learning and deep learning in agriculture: A comprehensive review. Green Technologies and Sustainability **3**(3), 100199 (2025)

Wijayathunga, L., Rassau, A., Chai, D.: Challenges and solutions for autonomous ground robot scene understanding and navigation in unstructured outdoor environments: A review. Applied Sciences **13**(17), 9877 (2023)

Wei, H., Xu, W., Kang, B., Eisner, R., Muleke, A., Rodriguez, D., deVoil, P., Sadras, V., Monjardino, M., Harrison, M.T.: Irrigation with artificial intelligence: Problems, premises, promises. Human-Centric Intelligent Systems **4**(2), 187–205 (2024)

Wu, P., Zhong, Y.: Artificial intelligence in sustainable agriculture: Towards a socio-technical roadmap. Smart Agricultural Technology, 101578 (2025)

Xiang, C., Feng, C., Xie, X., Shi, B., Lu, H., Lv, Y., Yang, M., Niu, Z.: Multi-sensor fusion and cooperative perception for autonomous driving: A review. IEEE Intelligent Transportation Systems Magazine **15**(5), 36–58 (2023)

Xing, Y., Liu, X., Wang, X.: Integrating uavs, satellite remote sensing, and machine learning in precision agriculture: pathways to sustainable food production, resource efficiency, and scalable innovation. Frontiers in Agronomy **7**, 1670380 (2025)

Yue, E., *et al.*: Application of the agent in agricultural expert system inspection software. Agricultural Sciences in China **7**(1), 117–122 (2008)

Yang, C.: Remote sensing and precision agriculture technologies for crop disease detection and management with a practical application example. Engineering **6**(5), 528–532 (2020)

Yang, Z., Chen, Y., Zhou, X., Yan, J., Song, D., Liu, Y., Li, Y., Zhang, Y., Zhou, P., Chen, H., et al.: Agentic robot: A brain-inspired framework for vision-language-action models in embodied agents. arXiv preprint arXiv:2505.23450 (2025)

Ye, X., Deng, Z., Shi, Y., Shen, W.: Toward energy-efficient routing of multiple agvs with multi-agent reinforcement learning. Sensors **23**(12), 5615 (2023)

Yellanki, S.K.: Autonomous agentic ai for dynamic travel experience management: From discovery to posttrip feedback without human intervention. Advances in Consumer Research **2**(3), 558–568 (2025)

Yamashita, R., Hoshino, S.: Development of an agent-based model for estimation of agricultural land preservation in rural japan. Agricultural systems **164**, 264–276 (2018)

Ye, K., Hu, G., Tong, Z., Xu, Y., Zheng, J.: Key intelligent pesticide prescription spraying technologies for the control of pests, diseases, and weeds: A review. Agriculture **15**(1), 81 (2025)

Yang, H., Jin, Y., Jiang, L., Lu, J., Wen, G.: Ai roles in 4r crop pest management—a review. Agronomy **15**(7), 1629 (2025)

Yang, G., Liu, C., Li, G., Chen, H., Chen, K., Wang, Y., Hu, X.: Diffknet-tl: Maize phenology monitoring with confidence-aware constrained diffusion model based on uav platform. Computers and Electronics in Agriculture **239**, 110936 (2025)

Yang, L., Li, P., Qian, S., Quan, H., Miao, J., Liu, M., Hu, Y., Memetimin, E.: Path planning technique for mobile robots: A review. Machines **11**(10), 980 (2023)

Yang, Z.-X., Li, Y., Wang, R.-F., Hu, P., Su, W.-H.: Deep learning in multimodal fusion for sustainable plant care: A comprehensive review. Sustainability **17**(12), 5255 (2025)

Yu, P., Teng, F., Zhu, W., Shen, C., Chen, Z., Song, J.: Cloud–edge–device collaborative computing in smart agriculture: architectures, applications, and future perspectives. Frontiers in Plant Science **16**, 1668545 (2025)

Yang, B., Zhang, Y., Chen, Y., Feng, L., Xu, X., Aierken, N., Li, S.: Agriagent: Contract-driven planning and capability-aware tool orchestration in real-world agriculture. arXiv preprint arXiv:2601.08308 (2026)

Zaman, Q.: Precision agriculture: Evolution, insights and emerging trends. Elsevier (2023)

Ziervogel, G., Bithell, M., Washington, R., Downing, T.: Agent-based social simulation: a method for assessing the impact of seasonal climate forecast applications among smallholder farmers. Agricultural systems **83**(1), 1–26 (2005)

Zhang, Q., Chen, D., Mahajan, Y., Chen, R., Ha, D.S., Cho, J.-H.: Attack-resistant, energy-adaptive monitoring for smart farms: Uncertainty-aware deep reinforcement learning approach. IEEE Internet of Things Journal **10**(16), 14254–14268 (2023)

Zhao, Y., Gong, L., Huang, Y., Liu, C.: A review of key techniques of vision-based control for harvesting robot. Computers and Electronics in Agriculture **127**, 311–323 (2016)

Zhang, W.: Ai-driven assessment of plant adaptation to climate change: The web tool based on physiological, biological, morphological, and genetic indicators of plant species. Computational Ecology and Software **16**(2), 99–153 (2026)

Zhu, H., Lin, C., Liu, G., Wang, D., Qin, S., Li, A., Xu, J.-L., He, Y.: Intelligent agriculture: Deep learning in uav-based remote sensing imagery for crop diseases and pests detection. Frontiers in Plant Science **15**, 1435016 (2024)

Zhu, Y., Li, S., Liu, S., Xu, Y.: Agent-based cooperative analysis and assistant decision-making method for regional agricultural economic information. In: 2009 Sixth International Conference on Fuzzy Systems and Knowledge Discovery, vol. 7, pp. 317–321 (2009). IEEE

Zhu, Y., Li, S., Liu, S., Yue, E.: Design of agent-based agricultural product quality control system. In: International Conference on Computer and Computing Technologies in Agriculture, pp. 476–486 (2010). Springer

Zhang, M., Liu, C., Li, Z., Yin, B.: From convolutional networks to vision transformers: evolution of deep learning in agricultural pest and disease identification. Agronomy **15**(5), 1079 (2025)

Zhang, S., Lai, C., Zhao, J., Wang, J.: Big data and ai-powered modeling: A pathway to sustainable precision animal nutrition. Advanced Science **12**(41), 07564 (2025)

Zhou, Z., Majeed, Y., Naranjo, G.D., Gambacorta, E.M.: Assessment for crop water stress with infrared thermal imagery in precision agriculture: A review and future prospects for deep learning applications. Computers and Electronics in Agriculture **182**, 106019 (2021)

Zeng, Z., Mahmood, T., Wang, Y., Rehman, A., Mujahid, M.A.: Ai-driven smart agriculture using hybrid transformer-cnn for real time disease detection in sustainable farming. Scientific Reports **15**(1), 25408 (2025)

Zhou, I., Tofigh, F., Piccardi, M., Abolhasan, M., Franklin, D., Lipman, J.: Secure multi-party computation for machine learning: A survey. IEEE Access **12**, 53881–53899 (2024)

Zuo, A.: The rise of autonomous ai agents: Automating complex tasks. International Journal of Artificial Intelligence for Science (IJAI4S) **1**(2) (2025)

Zhou, H., Wang, X., Au, W., Kang, H., Chen, C.: Intelligent robots for fruit harvesting: Recent developments and future challenges. Precision Agriculture **23**(5), 1856–1907 (2022)

Zhang, Z., Zhu, L.: A review on unmanned aerial vehicle remote sensing: Platforms, sensors, data

processing methods, and applications. drones **7**(6), 398 (2023)

Zhang, X., Zheng, X., Gao, Z., Fan, Y., Zhou, K., Zhang, W., Chang, X.: Development of a reinforcement learning-based intelligent irrigation decision-making model. Agronomy **15**(10), 2416 (2025)

Zhu, Y., Zhang, J., Li, S., Xue, L., Liu, S., Xue, Y.: Agriculture and forestry economy decision support system based on agent. New Zealand Journal of Agricultural Research **50**(5), 1339–1346 (2007)